Supplementary Information for *Beyond the Original Sin: Lay Evaluations of Creative Contribution in AI-Assisted Artworks*

David G. Kamper[1,2,*], Alice Lin[1] and Keith J. Holyoak[1]

[1]Department of Psychology, University of California, Los Angeles, Los Angeles, CA, USA
[2]University of Chicago Law School, Chicago, IL, USA
[*]Corresponding author: davidgkamper@ucla.edu

This document has four parts. Section 1 gives Supplementary Tables 1 and 2: every registered test with its status and verdict, and the AI label's effect on second works. Section 2 holds Supplementary Notes 1 to 7, in the order in which the main text's Results cite them. Section 3, the Supplementary Methods, gives the detail behind the main text's Methods. Sections 4 to 11 give the registrations, then each study's materials, procedure, sample and remaining results, then the text measures shared by all five studies and the exploratory cross-study analyses. Each study's section begins by naming the notes and tables that hold its other results.

Contents

1 Registered tests and their verdicts **2**

2 Supplementary Notes **6**
Supplementary Note 1: Study 1: hypotheses 3 and 4 6
Supplementary Note 2: Study 1: robustness checks and the belief check 8
Supplementary Note 3: Study 3: stated time, the gap at each time and the cell means 10
Supplementary Note 4: The image check 12
Supplementary Note 5: Study 5: the label's effect at each stated editing time 14
Supplementary Note 6: The pooled analysis of second works 16
Supplementary Note 7: What the explanations referred to 20

3 Supplementary Methods **22**

4 Registrations **30**

5 Study 1 **36**

6 Study 2 **51**

7 Study 3 **57**

8 Study 4 **60**

9 Study 5 **65**

10 Text measures shared by all five studies **70**

11 Exploratory cross-study analyses **74**

1 Registered tests and their verdicts

A registration is a dated record of a study's hypotheses and analysis plan. All five studies were registered on the Open Science Framework (OSF), an online archive for study plans, on 8 October 2026, after their data had been collected; these registrations have nothing to do with copyright registration. Supplementary Table 1 lists every registered test of the five studies with its status, its verdict and what had been seen before its registration, and Section 4 defines each status. Supplementary Table 2 gives the AI label's effect on ratings of second works, study by study, measure by measure and level by level, with the interval and the tests behind each verdict.

Supplementary Table 1: Registered tests and their verdicts. One row per registered test; measures that reached the same verdict share a row. Verdicts are worded by each registration's decision rule.

Hypothesis or test	Measure	Status	Registered verdict	Known before registration
Originals: the same AI-made images, rated for creativity and effort				
Study 1				
H1: the AI label lowers ratings	Creativity; effort	Confirmatory	Supported: AI lower on both; both survive Holm's correction	Both drops: t-tests, talk
H2: the drop is larger on effort	Effort minus creativity	Confirmatory	Supported: survives Holm's correction; the ordinal model agrees	Larger effort drop: talk
H3: the same penalty, figure or not	Creativity	Own rule	Inconclusive: cannot separate no effect from a small one	Rule set after analysis
	Effort	Own rule	Supported: not significant, and equivalent within ± 0.20	As above
H4: spread between vs within people	Creativity; effort	Confirmatory	Not supported as registered: met on effort, not on creativity	Quantity set after analysis
Study 3				
Label (F test)	Creativity; effort	Confirmatory	Supported: significant after Holm's correction, AI lower	Reported in the talk
Stated time (F test)	Creativity; effort	Confirmatory	Supported after Holm's correction; 100 hours above under 1 hour	Reported in the talk
Label by time (F test)	Creativity; effort	Confirmatory	Does not survive Holm's correction	Talk: on effort only
100-hour gap: equivalence, decision	Creativity; effort	Confirmatory	Not equivalent. Decision: the gap persists	Written knowing the talk
100 hours minus under 1 hour	Creativity; effort	Secondary	Positive on both ratings (95% intervals above zero)	–
Second works: hand edits of a labeled original, rated for transformativeness, essence change and creativity				
Study 2				
H1: ratings rise with the level	All three	Confirmatory	Holds on every measure: both steps positive and significant	Seen: large effects

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Supplementary Table 1, continued

Hypothesis or test	Measure	Status	Registered verdict	Known before registration
H2: averaged label effect, ±0.224	Transformativeness	Confirmatory	Small or absent: inside ±0.224	Bound set after Study 1
	Essence change	Confirmatory	Small or absent: inside ±0.224	As above
	Creativity	Confirmatory	Not shown to be small (interval inside; Holm $p = .068$)	Seen: a creativity effect
H3: label effect, None minus Dramatic	All three	Confirmatory	Not significant	Seen: none significant
Study 4				
H1: ratings rise with the level	All three	Confirmatory	Supported on every measure	Seen: large effects
H2: averaged label effect, ±0.224	Transformativeness	Confirmatory	Not shown inside the bound (interval inside; Holm $p = .064$)	Seen: label not significant
	Essence change	Confirmatory	Inside the bound (Holm $p = .020$); not shown in the refit	As above
	Creativity	Confirmatory	Not shown inside the bound (the estimate runs the other way)	As above
H3: hours on the original (F test)	All three	Confirmatory	Not significant after Holm's correction	Seen: a creativity effect
H3: those hours by level (F test)	All three	Confirmatory	Not significant after Holm's correction	–
Label by those hours (F test)	All three	Secondary	Not significant (no correction applies)	–
Those hours: 100 minus under 1	All three	Secondary	Inside the bound on all three measures	–
Label effect at each level (Supplementary Table 2)	All three	Secondary	Inside the bound or not shown inside it, by measure and level	Seen: on essence, by level
Refit without flagged participants	All three	Secondary	1 of 15 verdicts changes: essence change (Holm $p = .0501$)	–
Study 5				
Editing hours (F test)	All three	Confirmatory	Not significant after Holm's correction	–
Editing hours: 100 minus under 1	All three	Secondary	Absent or small: inside the bound	–
Label by editing hours (F test)	Transformativeness	Confirmatory	Not significant after Holm's correction; not shown to be absent	–
	Essence change	Confirmatory	Not significant after Holm's correction; not shown to be absent	–
	Creativity	Confirmatory	Significant after Holm's correction (Holm $p = .048$); 1 of 6 passed	–

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Supplementary Table 1, continued

Hypothesis or test	Measure	Status	Registered verdict	Known before registration
Label effect at each level (Supplementary Table 2)	All three	Secondary	None: absent or small (all three). Slight: significant, not inside the bound (all three). Dramatic: transformativeness as at Slight; essence change absent or small; creativity inconclusive	–
Studies 1–5				
Text checks on the explanations	Explanations	Secondary	Verdicts in the note below	Study 1: partly known

Notes to Supplementary Table 1.

- **Studies** (analyzed participants; collection): 1 (111; 25 February to 3 March 2025); 2 (235; 23 November to 5 December 2024); 3 (85; 4 to 9 March 2025); 4 (117; 27 January to 2 February 2025); 5 (223; 21 October to 2 November 2025).
- **Registered** means written into a registration, not fixed before collection. All five were filed on OSF on 8 October 2026, after collection: Study 1's after its analyses had been run, the others before the analyses reported here.
- **Seen before registration**: April 2025 *t*-tests (Study 1); analyses of variance run by February 2025 and reported in the 2025 proceedings of the Cognitive Science Society (CogSci 2025) [1] (Studies 2 and 4) or run in April 2025 (Study 3); a September 2025 talk on Studies 1 to 4 ('talk'). For Study 5 no rating had been summarized by condition, and both bounds were fixed before any Study 5 model was fitted.
- **Status**. *Confirmatory*: decided by the registration's rule, usually after Holm's correction, which raises each *p*-value in a set of tests so that the chance of any false positive in the set stays at .05. *Secondary*: outside that set, and uncorrected unless stated. *Own rule*: Study 1's H3, a significance test read with an equivalence test; its power on creativity was 0.35, so that failed equivalence test is uninformative.
- An **equivalence test** asks whether an effect is shown to lie inside a bound. It passes when the 90% confidence interval lies wholly inside, which means small, not zero. An F test asks whether a factor changes the mean rating.
- Bounds in points on 1-to-5 scales: ±0.20 (Study 1's H3); ±0.224, half of Study 1's creativity penalty of 0.448 (label effects and Study 3's gap); ±0.219, the size of Study 3's creativity time effect (editing hours).
- **Checks**. Ordinal refits, which treat the answers as ordered categories, agreed with every verdict they re-tested in Studies 2 to 5, except Study 2's creativity refit, which did not converge (not completed). Study 5's analysis with items rescaled from 0 to 1 gave the same verdicts. Study 3's required refit with effects varying by person and image was singular (a variance at the edge of its range), so that requirement was waived. Study 1's H4 needs both ratings; its between-person test survives in the Holm set, which does not mean H4 passed. Flagged: Study 4's 7 Human-label participants who wrote that a machine made the work.
- **Text checks** on the explanations, each written after its rating. Word counts recover the rating in every study: they predict it for people left out of the fit better than a shuffled baseline. They recover the label in Study 1, for Study 3's effort and, barely, in Study 5, and are not shown to do so elsewhere. They add to the rating in telling the labels apart in Studies 1 and 2 and are inconclusive in Studies 3 to 5. Greater own-word reuse under the AI label is supported for Study 1's effort only.

Supplementary Table 2: The AI label's effect on ratings of second works. Each estimate is the Human-label minus AI-label difference in the model's predicted mean rating, in points on the 1-to-5 scale; a positive value means the second work was rated lower when its original carried the AI label. Studies 2 and 4 registered the effect averaged over levels as a confirmatory test; Study 4 also registered it at each level, and Study 5 registered it only at each level, both as secondary tests. Each row is a registered test of whether the effect lies inside ±0.224 points.

Measure	Level	Estimate	90% interval	Two-sided test	Equivalence test	Registered verdict
Study 2, averaged over levels (H2, confirmatory; equivalence p after Holm's correction)						
Transformativeness	Averaged	0.06	−0.02 to 0.14	$p = .249$[a]	$p = .003$	Small or absent: inside ±0.224
Essence change	Averaged	0.07	−0.01 to 0.16	$p = .161$[a]	$p = .008$	Small or absent: inside ±0.224
Creativity	Averaged	0.11	0.03 to 0.20	$p = .026$[a]	$p = .068$	Not shown to be small
Study 4, averaged over levels and over the hours stated for the original (H2, confirmatory; equivalence p after Holm's correction)						
Transformativeness	Averaged	0.04	−0.08 to 0.17	$p = .550$[b]	$p = .064$	Not shown inside the bound
Essence change	Averaged	0.02	−0.10 to 0.14	$p = .778$[b]	$p = .020$	Inside the bound[c]
Creativity	Averaged	−0.11	−0.27 to 0.04	$p = .229$[b]	$p = .803$	Not shown inside the bound
Study 4, each level, averaged over the hours stated for the original (secondary; uncorrected)						
Transformativeness	None	0.01	−0.13 to 0.16	$p = .882$	$p = .009$	Inside the bound
	Slight	0.20	0.05 to 0.35	$p = .024$	$p = .399$	Not shown inside the bound
	Dramatic	−0.08	−0.23 to 0.06	$p = .345$	$p = .051$	Not shown inside the bound
Essence change	None	−0.05	−0.18 to 0.09	$p = .583$	$p = .017$	Inside the bound
	Slight	0.19	0.05 to 0.33	$p = .024$	$p = .337$	Not shown inside the bound
	Dramatic	−0.08	−0.22 to 0.05	$p = .318$	$p = .046$	Inside the bound
Creativity	None	−0.13	−0.30 to 0.04	$p = .199$	$p = .188$	Not shown inside the bound
	Slight	−0.03	−0.20 to 0.14	$p = .764$	$p = .031$	Inside the bound
	Dramatic	−0.17	−0.34 to 0.00	$p = .096$	$p = .308$	Not shown inside the bound

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Supplementary Table 2, continued

Measure	Level	Estimate	90% interval	Two-sided test	Equivalence test	Registered verdict
Study 5, each level, averaged over editing hours (secondary; each test judged at .05 without correction)						
Transformativeness	None	−0.04	−0.15 to 0.07	$p = .594$	$p = .002$	Absent or small: inside the bound
	Slight	0.13	0.02 to 0.24	$p = .049$	$p = .083$	Significant, not inside the bound
	Dramatic	0.17	0.06 to 0.28	$p = .009$	$p = .204$	Significant, not inside the bound
Essence change	None	0.02	−0.08 to 0.12	$p = .752$	$p < .001$	Absent or small: inside the bound
	Slight	0.17	0.07 to 0.28	$p = .006$	$p = .211$	Significant, not inside the bound
	Dramatic	0.10	0.00 to 0.21	$p = .104$	$p = .027$	Absent or small: inside the bound
Creativity	None	0.09	−0.02 to 0.20	$p = .171$	$p = .022$	Absent or small: inside the bound
	Slight	0.22	0.11 to 0.33	$p = .001$	$p = .452$	Significant, not inside the bound
	Dramatic	0.12	0.01 to 0.23	$p = .080$	$p = .052$	Inconclusive: neither significant nor equivalent[d]

Notes to Supplementary Table 2.

- *Equivalence test*: two one-sided tests of whether the difference lies inside ±0.224 points, half of Study 1's label penalty on the creativity of originals (0.448; 95% confidence interval 0.19 to 0.70). It passes when the 90% interval lies wholly inside the bound. A pass means the effect is small, not that it is zero; a fail means only that it is not shown to be small.
- *Two-sided test*: whether the difference is zero.
- *Holm's correction* raises each p-value in a set of tests so that the chance of any false positive in the set stays at .05.
- None is the original shown again unchanged, where ratings sit near the floor of the scale. Editing hours are the hours stated for the hand edit (Study 5).
- Study 4's registration gives no wording for its tests at each level; 'inside the bound' marks a passed equivalence test.
- [a]Description only; Study 2's verdict rests on the equivalence test.
- [b]The label's F test, a term outside Study 4's registered set of tests (it asks whether the label changes the mean rating), shown as context.
- [c]Not shown in the registered refit without the 7 Human-label participants who wrote that a machine made the work (Holm $p = .0501$).
- [d]The interval's upper end is 0.225, just outside the bound.
- In Study 5, 'significant' means a two-sided p below .05. Study 5's estimates average over editing hours, across which the creativity penalty differed (Supplementary Table 1).

2 Supplementary Notes

Supplementary Note 1: Study 1: hypotheses 3 and 4

This note reports in full Study 1's third and fourth hypotheses (H3 and H4). The main text's Results on Study 1 give only their verdicts, in one sentence.

Study 1's 111 participants rated ten AI-made images, five showing a human or humanoid figure and five not, for creativity and for effort on 1-to-5 scales. A label told each participant that the artist, Alex, had made the images either with design and sketching tools (the Human label; 56 participants) or by prompting an AI system (the AI label; 55 participants). Each

participant saw one label and rated every image. We call the mean rating under the Human label minus the mean under the AI label the label's penalty; a positive penalty means that the AI label lowered the rating. Each estimate comes from a linear mixed model, a regression that gives each participant and each image its own baseline, fitted to each rating separately. Study 1 was registered on the Open Science Framework (OSF), an online archive for study plans, on 8 October 2026, after its data had been collected and analyzed. H3's bound and decision table, and the quantity that decides H4, were therefore set with the estimates already known. Supplementary Table 1 lists both verdicts beside the other registered tests.

H3: the same penalty for images with and without a human figure. H3 predicted that the penalty would not differ between the two kinds of image. It is a registered hypothesis decided outside Study 1's set of confirmatory tests, by its own rule rather than by a corrected p-value: a significance test read with an equivalence test. The significance test asks whether the difference in penalty between the two kinds of image is distinguishable from zero. Its p-value is the chance of a difference at least this large if there were none, and a difference is called significant when p is below .05. An equivalence test asks whether an effect is shown to lie inside a stated bound. Judged alone, as H3's is, it passes when the effect's 90% confidence interval (CI), the interval that matches this test, lies wholly inside the bound, which shows that the effect is small, not that it is zero.[2,3] H3's bound was ± 0.20 points. Read together, the two tests decide H3: a difference that is not significant and lies inside the bound supports H3, and one that is not significant but is not shown inside the bound leaves H3 inconclusive, because the design cannot then separate no difference from a small one.

On effort, H3 was supported: the difference in penalty was not significant ($p = .833$) and lay inside the bound (-0.02 points; 90% CI -0.18 to 0.14; equivalence $p = .030$). The bound was set after the analyses, so this pass is weaker evidence than a pass against a bound fixed in advance. On creativity, H3 was inconclusive (-0.08 points; 90% CI -0.24 to 0.08; $p = .393$), because the interval crossed the bound. The equivalence test's power, its chance of passing if the true difference were zero, was only 0.35, so that failure says little. By the registered rule, a failed equivalence test is uninformative whenever this power is below 80%, since the failure then reflects the width of the interval this design produces, not the difference itself. Power on effort was much the same (0.36), but a pass stands whatever the power, because the equivalence test limits the chance of wrongly declaring a difference small. Were the true difference a tenth of a point, the test would pass with probability 0.22 on creativity and 0.21 on effort. These power figures come from simulated data and describe the design, not the observed data. They are a registered description: a result that the registration asks us to report, with no test.

Both verdicts rest on the primary model. Each verdict was also to hold in a fuller model, one that further lets the label's effect vary from image to image and the effect of image kind vary from person to person, unless that model was singular. A singular fit estimates a variance or correlation at the edge of its possible range, a sign that the data cannot support that many quantities. The fuller model was singular on both ratings, so under the registered fallback the verdicts rest on the primary model alone. The singular fuller model gave the same readings (90% CI -0.18 to 0.14 on effort, -0.27 to 0.11 on creativity).

H4: more spread between people under the AI label. H4 predicted that, under the AI label, ratings would spread out more between people and less across one person's ratings of different images. It is a registered confirmatory hypothesis: Study 1's registration names it among the study's confirmatory tests. Its registered criterion, on each rating, was an AI-to-Human ratio of the variance (the spread) between people with a 95% CI wholly above 1, and a ratio within people with a 95% CI wholly below 1. A 95% CI is the range of values the data do not rule out at the 95% level; here each came from recomputing the ratio after drawing participants again at random, 1,000 times, within each label group. The criterion was met on

effort: the ratio was 3.88 between people (2.21 to 6.73) and 0.63 within people (0.48 to 0.84). It was not met on creativity. There the ratio between people was 2.04 (1.18 to 3.57), but the ratio within people, 0.82, had an interval that reached 1.02. Because H4 required both ratings, it was not supported as registered.

Study 1's four confirmatory tests were judged together after Holm's correction,[4] which raises each p-value in a set of tests so that the chance of calling any result in the set significant by mistake stays at .05. H4's row in that set survived the correction (Holm $p = .026$), but that row does not decide H4. It is only the likelihood-ratio test of the spread between people on creativity, the rating with the weaker result, which the registration placed in the set; the test compares a model in which that spread differs by label with one in which it does not. Passing it is not passing H4. Three further points limit the reading. The two ratios come from fits with different structures, and the fit for the spread within people gives images no baseline of their own, so that spread also carries differences between images. In simulated data with ratios equal to those observed, the test of the spread between people would reject in 59.9% of samples on creativity and 99.3% on effort, so the creativity half was the harder to detect (registered description of what the design can resolve). And the quantity that decides H4 was chosen after the analyses had been run.

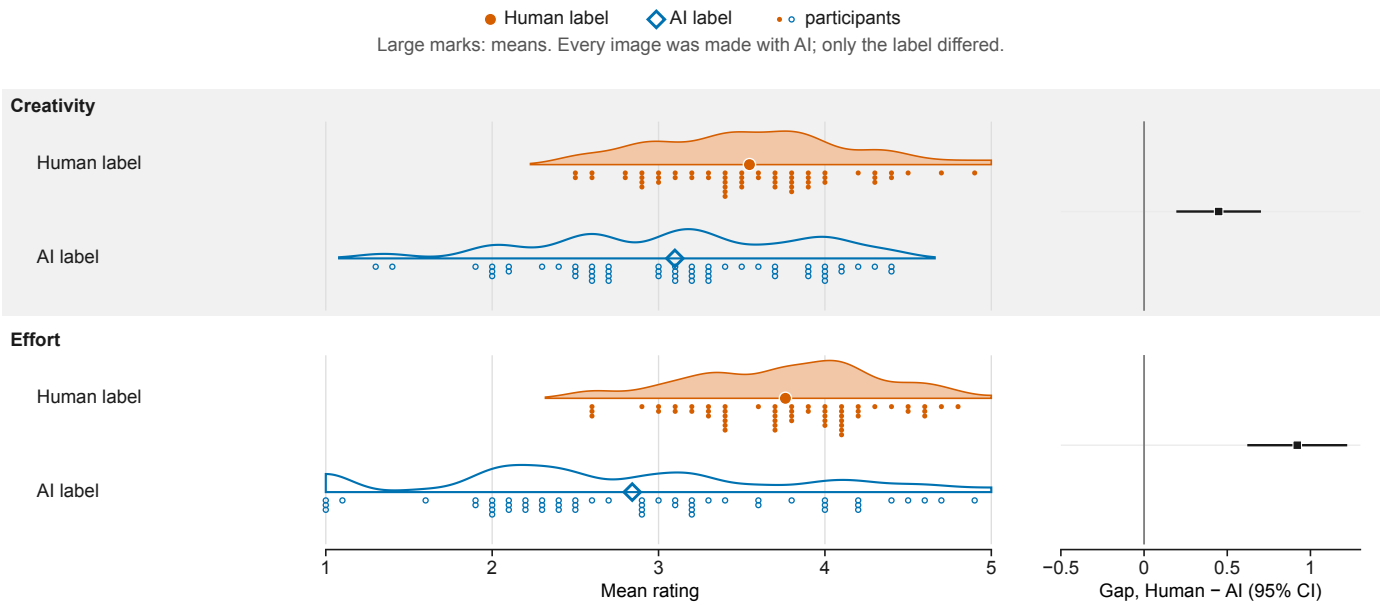
As description that the registration places outside the H4 criterion, more AI-label participants gave the same rating to seven or more of the ten images. On effort, 23 of 55 AI-label participants did so, against 12 of 56 Human-label participants; on creativity, 13 against 4. These counts describe the ratings and test nothing. Supplementary Fig. 1 shows each participant's mean rating under each label; that these means spread more under the AI label is likewise description, not a test of H4.

Supplementary Note 2: Study 1: robustness checks and the belief check

This note gives the checks on Study 1's label penalty and on what Human-label participants wrote about machines, which the main text summarizes in its Results on Study 1.

Study 1's 111 participants rated the same ten AI-made images for creativity and for effort on 1-to-5 scales. A label told each participant that the artist, Alex, had made them either with design and sketching tools (the Human label) or by prompting an AI system (the AI label). The label's penalty is the mean rating under the Human label minus the mean under the AI label, so a positive penalty means that the AI label lowered the rating. Its primary estimate comes from a linear mixed model, a regression that gives each participant and each image its own baseline: 0.45 points on creativity and 0.92 on effort, the registered confirmatory test of Study 1's first hypothesis (H1). All but one of the analyses below are named in Study 1's registration, as checks on that estimate or as counts to report, which makes them registered secondary analyses: reported, but not decisions about a hypothesis. The exception, the comparison with random groups, is exploratory description, as the registration names it. None replaces the primary estimate. Study 1 was registered on 8 October 2026, after its analyses had been run.

Robustness of the penalty. A registered robustness check, descriptive by registration, refit the penalty 18 ways for each rating. It varied the model type (for example a linear model, or an ordinal model that treats the five answers as ordered categories rather than evenly spaced numbers), its baselines (which differences between participants and between images the model allows for), its background variables (such as age and gender) and whether ratings were first averaged per person. In all 36 versions the AI label still lowered the rating. Among the versions that give the penalty in rating points (all but the ordinal ones), the penalty ran from 0.44 to 0.48 points on creativity and from 0.89 to 0.92 on effort. Descriptive by registration means that neither the middle of these estimates nor the share of versions with a significant penalty counts as evidence for a hypothesis.



Supplementary Figure 1: Study 1: each participant's mean rating under each label. Each small dot is one participant's mean rating of the ten originals, filled under the Human label (56 participants) and open under the AI label (55), stacked where values repeat. The outline above the dots, a half-violin, is a smoothed curve of how they spread. The large marks are the two label means, which here equal the means predicted by Study 1's statistical model. The right column is the gap between them, the Human-label mean minus the AI-label mean, with its 95% confidence interval, as in main-text Fig. 2a; these gaps are the registered confirmatory tests of H1, that the AI label lowers both ratings. The two groups' dots overlap, so a gap is a difference between group means, not a difference that every participant shows. The dots also spread more under the AI label, more on effort than on creativity; this spread is descriptive. H4, the registered confirmatory test that the AI label spreads ratings more between people and less within each person, met its criterion on effort only, so it was not supported as registered.

The other registered checks agreed. The ordinal model on its own gave a penalty on both ratings ($p < .001$ on creativity and $p < .001$ on effort, where a p-value is the chance of a penalty at least this large if the label had no effect). Its estimates are on another scale, so their size cannot be compared with the rating-point penalties. Adding age, gender and formal artistic training as background variables gave penalties of 0.44 and 0.90 points. Refitting the model once without each participant, and once without each image, moved the penalty by at most 0.04 points on creativity and 0.04 on effort, and never changed its sign. One effort item, on the first image with a human figure, labeled its top two answers differently from the others; dropping it moved the effort penalty by 0.007 points and the difference between the effort and creativity penalties (Study 1's H2) by 0.007. That last check was settled after the estimates had been read. The two items also differed in wording. Only the creativity item carried a definition ('the presence of at least a minimal original, intellectual contribution to the creation of the work'), and it pointed raters toward an intellectual contribution, which both descriptions gave to Alex ('Alex had a new idea'); the effort item carried none. Part of the difference between the two penalties may therefore come from the wording of the items.

The belief check. Every image was AI-made, so the Human label was a cover story, and no question asked participants whether they believed it. Instead, the registered belief check counts what Human-label participants wrote in the short explanation required for every rating, using fixed word lists that a script applied without seeing the label or the rating. Of the 56 Human-label participants, 28 (50.0%; 95% confidence interval, CI, 37.5% to 64.3%) mentioned machines, prompting or generating at least once, and 25 (44.6%; 95% CI 32.1% to 57.1%) wrote at least once that a work is or looks machine-made. A 95% CI is the range of values the data do not rule out at the 95% level; these come from drawing participants again at random 2,000 times. Both are counts of statements, not of disbelief. A participant who doubted the label may have written nothing about it, and one who wrote that a work looks AI-generated may still have accepted the description. The rule that flags a statement that a work is or looks machine-made was settled in September and October 2026, after results by label had been read.

Disbelief in the cover story would make Human-label ratings more like AI-label ones, shrinking the penalty, not creating it. In a registered refit without those 25 participants, the penalty grew to 0.55 points on creativity and 1.17 on effort. The refit is a sensitivity check and does not replace the full-sample estimate.

Dropping low raters from the Human-label group enlarges a penalty by arithmetic alone. To gauge that, an analysis that the registration names as exploratory description dropped 2,000 random groups of 25 Human-label participants whose mean effort rating matched the excluded group's within a fixed tolerance. It compared Cohen's d, the penalty divided by the pooled standard deviation (a measure of spread) of participants' mean ratings, before and after each removal. On creativity, d rose from 0.66 in the full sample to 0.80 after the exclusion, inside the middle 95% of the random removals (0.63 to 0.91), so the change is what removing any such group produces. On effort, d rose from 1.15 to 1.40, above every random removal (middle 95%: 1.16 to 1.31). The registered reading of the effort result is only that the change is not compatible with that reference. The matching held the mean effort rating alone, so the comparison cannot say which other difference between the groups moved the estimate.

Supplementary Note 3: Study 3: stated time, the gap at each time and the cell means

This note gives the Study 3 tests and estimates behind the main text's Results section on stated hours.

Study 3's 85 new participants each rated nine of the ten Study 1 images for creativity and for effort on 1-to-5 scales. A label told each participant that the artist, Alex, had made the images either with design and sketching tools (the Human label; 43 participants) or by prompting an

AI system (the AI label; 42 participants). Each image came with a stated time that Alex had spent on it, under 1 hour, 10 hours or 100 hours, three images at each time. The label varied between participants and the stated time within them. Under the Human label Alex spent those hours drawing; under the AI label he spent them revising prompts, so under the AI label more time also meant more control over the image, and the two cannot be told apart. Study 3's registration calls the Human-label mean minus the AI-label mean the gap; a positive gap means that the AI label lowered the rating. Estimates come from the registered linear mixed model, a regression that gives each participant and each image its own baseline, fitted to each rating separately. Study 3 was registered on 8 October 2026, after its data had been collected in March 2025, after the authors had analyzed them in April 2025, and after a talk in September 2025 had reported effects of label and time (Supplementary Table 1).

Six F tests were registered as confirmatory: label, stated time and their interaction, each on creativity and on effort. They were judged together after Holm's correction,[4] which raises each p-value in a set of tests so that the chance of calling any result in the set significant by mistake stays at .05. A p-value is the chance of a result at least this large if the factor had no effect. Two registered secondary quantities are reported as estimates, not decisions: the contrast between 100 hours and under 1 hour, and the gap at each time.

Label and stated time. The four F tests for label and for time, each on creativity and on effort, ask whether a factor changes the mean rating, and all four survived Holm's correction (registered confirmatory; supported). The label changed creativity ratings ($F(1, 82.9) = 7.65$, Holm $p = .021$) and effort ratings ($F(1, 83) = 8.63$, Holm $p = .017$). Stated time changed creativity ratings ($F(2, 667.8) = 6.02$, Holm $p = .013$) and effort ratings ($F(2, 665.1) = 47.55$, Holm $p < .001$). An F test gives no direction. The gaps in Supplementary Table 3 show the AI label lower at every time. Averaged over labels, the mean rating at 100 hours minus the mean at under 1 hour was 0.22 points on creativity (95% confidence interval, CI, the range of differences the data do not rule out at the 95% level: 0.07 to 0.36) and 0.75 on effort (0.60 to 0.90; registered secondary). Most of the creativity rise came at 100 hours: under both labels, the predicted creativity mean at 10 hours sat at about its level at under 1 hour. The predicted effort means rose at each step under both labels (Supplementary Table 3). No test of the rise within each label, or of equal rises under the two labels, was registered.

Supplementary Table 3: Study 3: predicted mean ratings and the gap at each stated time. Means are the registered linear mixed model's predicted means for each combination of label and stated time, on 1-to-5 scales (descriptive). The gap is the Human-label mean minus the AI-label mean, with its 95% confidence interval (registered description: estimates the registration names, with no test of their own). Gaps are computed before rounding, so a gap can differ in its last digit from the difference of the rounded means. Each participant rated three images at each time; Human label $N = 43$, AI label $N = 42$.

Rating		Under 1 hour	10 hours	100 hours
Creativity	Human label	3.43	3.42	3.65
	AI label	3.08	3.07	3.29
	Gap	0.35 (0.05 to 0.65)	0.35 (0.05 to 0.65)	0.36 (0.06 to 0.67)
Effort	Human label	3.24	3.41	3.94
	AI label	2.65	3.14	3.45
	Gap	0.58 (0.23 to 0.93)	0.26 (−0.08 to 0.61)	0.49 (0.14 to 0.84)

Whether the gap changed with time. The label-by-time interaction asks whether the gap changed with time. On neither rating did it survive Holm's correction (registered confirmatory):

on creativity, $F(2, 667.6) = 0.01$ and Holm $p = .994$; on effort, $F(2, 664.5) = 2.24$ and Holm $p = .214$. In the ordinal check, a registered secondary analysis that treats the five answers as ordered categories rather than evenly spaced numbers, the effort interaction came nearer the threshold (uncorrected $p = .028$, Holm $p = .057$), but still fell short of significance after correction. On every term the ordinal check reached the same verdict as the linear model. The September 2025 talk had also reported an effort interaction from an earlier analysis that still included participants the registration later excluded; the registered model does not reproduce it. A nonsignificant interaction does not show that the gap stayed the same, and no test of equal gaps across times was registered.

The registered decision on both ratings, that the gap persists at 100 hours, is reported in the main text. Part of that decision, also registered confirmatory, is an equivalence test, which asks whether the 100-hour gap is shown to lie inside ±0.224 points (half of Study 1's creativity penalty). The rule asked that this test be repeated in a fuller model that lets the time effect vary from person to person and the label's effect vary from image to image. It waived that requirement if the fuller model was singular, meaning that it estimated a variance or correlation at the edge of its possible range, and on both ratings it was. The fuller model's 90% CIs for the 100-hour gap (0.10 to 0.64 on creativity, 0.19 to 0.79 on effort) also reach past the bound, so the waiver changes no verdict.

The gap at each time. As registered description, the creativity gap was 0.35, 0.35 and 0.36 points at under 1, 10 and 100 hours, and the effort gap was 0.58, 0.26 and 0.49 (Supplementary Table 3; main-text Fig. 2a plots both). Each 95% CI lay above zero except that of the effort gap at 10 hours. By the registration, a gap significant at one time and not at another decides nothing about whether time closes the gap.

Each effort gap was smaller than Study 1's effort penalty of 0.92 points, a comparison across studies that is untested. The two studies differ in their participants, in whether a time was stated and in the wording of the Human label, which in Study 3 named the drawing tools (iPad apps and Adobe Photoshop); any of these, or chance, could make the difference. Study 3 registered no test of effort against creativity, and with time stated its two gaps were close: at 10 hours the effort gap was the smaller. The larger drop on effort therefore rests on Study 1.

Comparisons without a test. On effort, the AI-label mean at 100 hours (3.45) sat above the Human-label mean at under 1 hour (3.24); on creativity it sat below (3.29 against 3.43). These two cells differ in both label and stated time, and no test compares them, so the comparison is descriptive.

Supplementary Note 4: The image check

This note describes the check of the stimulus images that the main text's Results summarize in one sentence, in the section on how ratings of second works follow the level of modification.

Studies 2, 4 and 5 showed an original beside three second works that Alex was said to have made from it: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy one. Participants rated each second work's transformativeness, essence change and creativity. The design takes three things about the pictures for granted: that the ten originals differ from one another, that each artwork's Slight version is closer to it than its Dramatic version is, and that every edited version, however much it was changed, stays closer to its own original than to any of the other nine. The third matters most, because a question about how far a second work moved from its original rests on a pairing that the pictures must support. No registration names this check. It was specified on 10 October 2026, after every registered result was known, so it is exploratory and changes no registered verdict; its comparison with the ratings is descriptive, with no test.

What was compared. The check used the image files the surveys displayed: the ten originals and their ten Slight and ten Dramatic versions. They were taken from a printout of the Study 5 survey, a PDF file printed from the survey software that holds each image as the survey served it, and the survey files of Studies 2, 4 and 5 name the same image files. The None second work is the original's own file, with identical pixels, so its distance from the original is zero. Distances came from three measures that notice different things. Two are image-recognition networks, programs trained on large collections of pictures that turn any image into a long list of numbers, so that any two images lie at a distance from each other.[5,6] DINOv2 learned, without labels, to give matching numbers to different views of the same photograph, so its numbers describe layout, shapes and content. CLIP learned to match pictures with their captions, so its numbers reflect what a picture would be described as. For both, the distance between two images is 1 minus the cosine similarity of their lists, a standard measure of how closely two lists of numbers point the same way, so identical images lie at zero. The third measure uses no network. It compares the mix of colors in two images, taken as the share of pixels that fall into each cell of a fixed grid of colors, and is zero when two images have the same mix. Each network saw each image in two ways: the whole picture resized to a square, the primary input, and the network's own standard crop, which keeps the central square and drops the edges of a picture that is not square. The rule for passing each check was written into the code before any distance was computed, and no person or language model judged any image.

The three checks. On both networks and with both inputs, all three checks passed. The two closest originals lay farther apart than any original lay from its own Slight version. The Slight version was closer to its original than the Dramatic version for every artwork (10 artworks on DINOv2 and 10 on CLIP). These steps are ordered within each artwork but overlap across artworks: the Dramatic version of one artwork can lie nearer its original than the Slight version of another lies to its own. Every edited image lay closest to its own original (20 images on DINOv2 and 20 on CLIP). The two inputs gave the same verdict on every check.

The color measure failed all three checks. Some Slight edits moved an image's colors farther than the two most similar originals lie apart, so the first check failed. It put the Slight version closer than the Dramatic version for 9 artworks, not all ten, and put only half of the twenty edited images (10) closest to their own original. Because this measure ignores where colors lie in a picture, its failures show that the Slight edits changed palettes substantially and that some edited images have a palette nearer another artwork's. They do not show that an edited image resembles another artwork. The color measure counts a Slight edit as a much larger share of a Dramatic edit than either network does.

The Slight versions were also cropped or resized. The registrations describe the Slight edits as adjustments of color and contrast, and the CogSci 2025 paper, our proceedings paper on Studies 2 and 4, describes them as adjustments of hue, coloration and contrast.[1] But 7 of the ten Slight versions also differ from their original in pixel size and in the ratio of width to height, so those edits cropped or resized the picture as well (main-text Fig. 1a shows a Slight version that was not). The descriptions participants read never said how much had been changed. Supplementary Note 6 lists this cropping among the possible reasons for the peak in the label's penalty at Slight.

Distance and ratings, description only. The check also set each edited image's distance from its original beside its mean rating in Studies 2, 4 and 5. Each image's mean was computed within each study and then averaged over the three, so each study counts once whatever its size. Across the twenty edited images (nineteen for transformativeness, because one Slight item used a six-point scale and was set aside), images farther from their original were rated higher, strongly so for every distance measure and every rating, by Spearman correlations, ordinary

correlations computed on the ranks of the values. Most of that relation comes from the level of modification itself: every Dramatic image was rated above every Slight image, and the Dramatic images are as a group also farther from their originals. Within a level, across its ten images, the correlations were mostly small and changed with the measure, the input and the label group. With so few images these correlations describe the stimuli and test nothing, and no p-value is reported. The ratings are therefore shown to follow the level of modification; they are not shown to track the amount of visible change within a level.

Limits. Neither network was trained on judgments of art or of artistic change, so neither distance measures what participants saw or judged; the checks ask whether the pictures are arranged as the design assumes, as two networks with different training describe them. The pixels are those of the printout. The survey files show that Studies 2, 4 and 5 named the same images, but they cannot show that the survey platform served the same bytes on every date each study ran, and no copy of the live survey pages was kept. That the None second work showed the original rests on the survey file, the printout and the ratings near the floor of the scale at None.

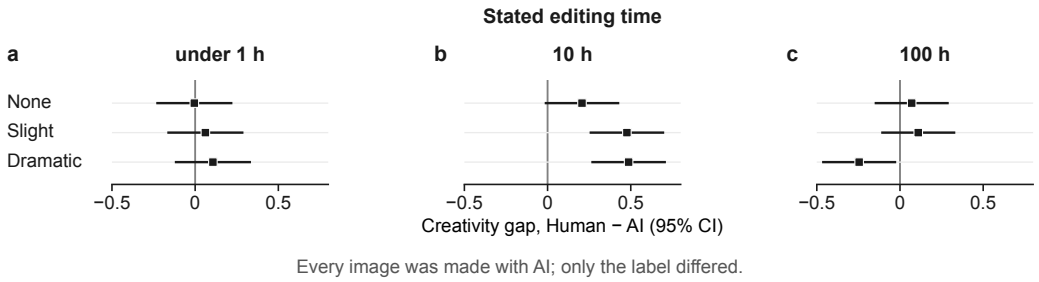
Code and images. The image check's folder in the analysis code (main text, Code availability) holds the script that extracts the images, the analysis notebook, the images themselves with a checksum for each (a fingerprint of a file's exact bytes), the pinned versions of both networks, every table and two figures. Its README explains how to rerun the check from a fresh copy. The networks run on one processor thread with a deterministic setting, so a rerun on the same kind of computer gives the same numbers.

Supplementary Note 5: Study 5: the label's effect at each stated editing time

This note gives, cell by cell, the exploratory breakdown behind Study 5's one significant registered confirmatory test: the AI label's creativity penalty differed across stated editing times. The main text reports that test in its Results section on stated editing hours.

Design. In Study 5, 223 participants each saw six original images, all generated with an AI image generator. With each original came three second works that Alex was said to have made from it with his own design and sketching tools: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy one. Every participant saw all three levels. Two factors varied between participants, so each person saw one level of each for the whole session. The label said that Alex had made the originals either with design and sketching tools (the Human label) or by prompting an AI system (the AI label); because every original was AI-generated, the Human label was a cover story. The stated editing time said that Alex spent under 1 hour, 10 hours or 100 hours making each second work. The six groups held 37, 37 and 38 participants under the Human label and 35, 38 and 38 under the AI label, at under 1, 10 and 100 hours (descriptive). The sentence that stated the editing time was the same at every level, so at None the description participants read said that Alex had spent the stated hours on an unchanged copy. A time for the original was also shown on every trial but was not recorded, so it enters no analysis. Participants rated each second work's transformativeness, essence change and creativity on 1-to-5 scales. As in Studies 2 and 4, the model's predicted means rose with the level of modification; on creativity under the Human label they were 1.21, 1.92 and 3.32 at None, Slight and Dramatic (descriptive). Study 5's registration was filed on 8 October 2026, after data collection but before any of its ratings had been averaged or compared by label.

The penalty. Each estimate below is the label's penalty: the predicted mean rating under the Human label minus the predicted mean under the AI label, in points on the 1-to-5 scale,



Supplementary Figure 2: Exploratory: Study 5's creativity gap at each stated editing time. Each square is the gap in creativity ratings of the second work, the Human-label mean minus the AI-label mean, from Study 5's registered statistical model fitted again to give a gap at each editing time and level, with its 95% confidence interval, not corrected for the nine comparisons. Right of zero, the AI label lowered the rating. Editing time varied between participants, so each panel compares two separate groups of 35 to 38 people, one per label. None is the original shown again unchanged. The registered test behind this breakdown, whether the label's creativity gap differed across editing times, was significant after Holm's correction (Holm $p = .048$). The breakdown was specified on 9 October 2026, after the registered results were known, and it changes no registered verdict.

from Study 5's registered linear mixed model, a regression that gives each participant and each image its own baseline. A positive penalty means that the AI label lowered the rating, and a negative one that it raised it. A p-value is the chance of a difference at least this large if there were no effect, and a result is called significant when p is below .05.

The registered test. Study 5's registration named 6 confirmatory tests, fixed with their decision rule before the analysis: the effect of editing time and the label-by-editing-time interaction, which asks whether the label's penalty differs across stated editing times, on each of the three ratings. They were judged together after Holm's correction, which raises each p-value in a set of tests so that the chance of calling any result in the set significant by mistake stays at .05. Only 1 of them passed: the label's creativity penalty differed across stated editing times ($F(2, 216.8) = 4.93$, uncorrected $p = .008$, Holm $p = .048$). Two checks that the registration named as secondary agreed. An ordinal model, which treats the five answers as ordered categories rather than evenly spaced numbers, found the same interaction ($\chi^2 = 9.65$, Holm $p = .048$) and nothing else among the six tests. The registered second analysis, with every item rescaled from 0 to 1, gave the same verdicts ($F = 4.93$, Holm $p = .048$). The interaction was not significant on the other two measures (Holm $p > .999$ on transformativeness and $p > .999$ on essence change). Averaged over the two labels, stated editing hours did not detectably change ratings on any measure (registered confirmatory; reported in the main text). The interaction concerns the penalty, not the mean rating, so the two results do not conflict. That null cannot be set against Study 3's effect of stated time on originals as a finding, because in Study 3 each participant saw every stated time and in Study 5 only one.

The shape of the interaction. The registration sketched one shape: if editing counts as Alex's own contribution, the penalty should shrink as the hours rise. The data took another. An exploratory breakdown, specified on 9 October 2026 after the registered results were known, estimated the creativity penalty at each editing time and level from Study 5's registered model, fitted again for this purpose (Supplementary Fig. 2). It carries no verdict, and its p-values are not corrected for the nine comparisons. At under 1 hour, the penalty was small at every level: 0.00, 0.06 and 0.11 points at None, Slight and Dramatic. At 10 hours it was largest at every level: 0.21, 0.48 and 0.49 (uncorrected $p < .001$ at Slight and $p < .001$ at Dramatic). At Slight

and Dramatic these are about as large as Study 1's creativity penalty on originals, 0.45 points, a comparison across studies rather than a test. At 100 hours the penalty was small at None and Slight (0.07 and 0.11) but reversed at Dramatic (-0.25, uncorrected $p = .031$). The model term that lets this editing-time pattern differ across levels was also significant on creativity ($F(4, 3769.6) = 5.47$, uncorrected $p < .001$); it is exploratory, because the registration names no test of it. The same breakdown on the other two measures, whose registered interactions were not significant, also put the largest penalty in the 10-hour groups at every level; at Slight and Dramatic it was 0.27 and 0.38 on transformativeness and 0.27 and 0.33 on essence change.

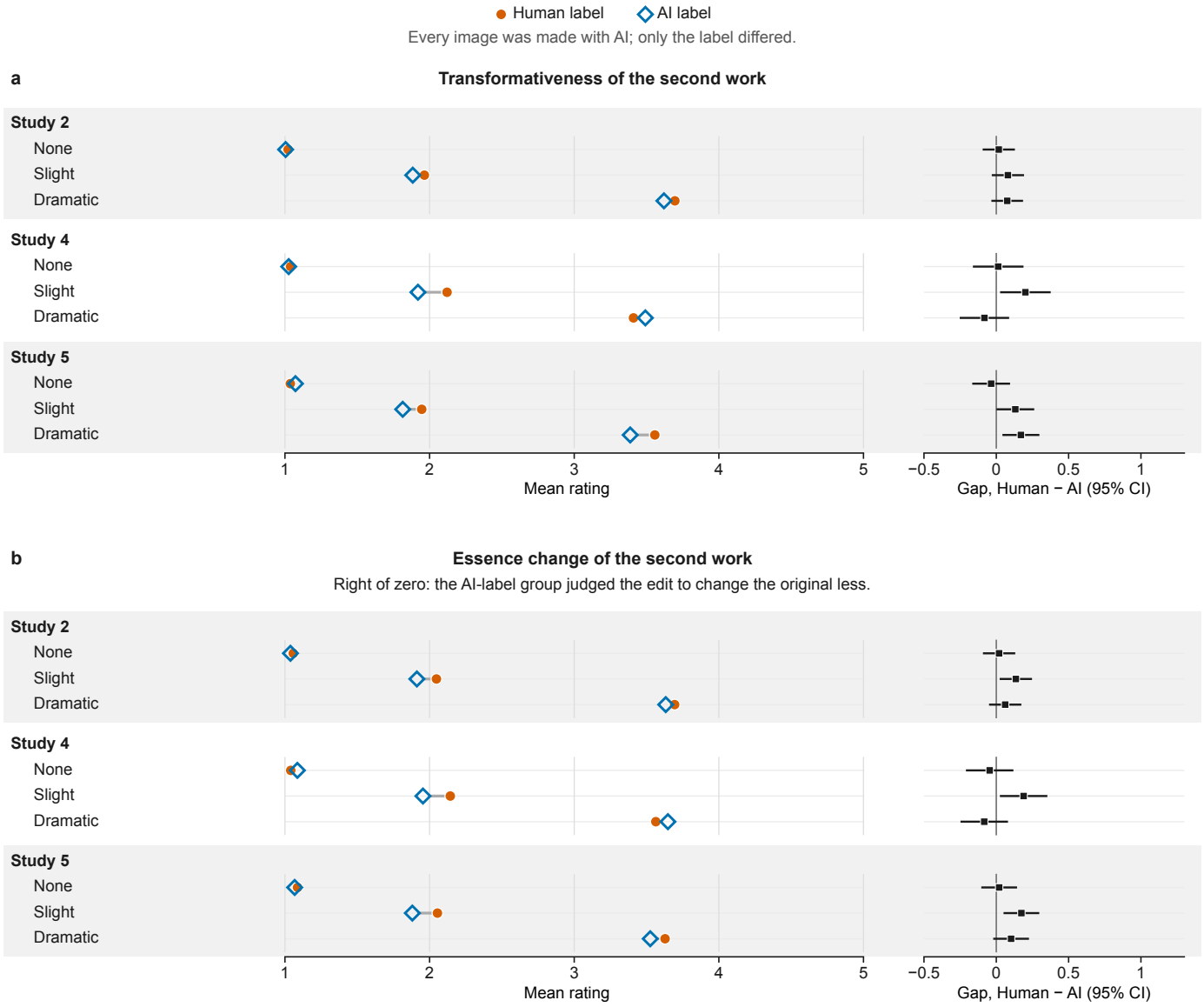
What the breakdown means for the Slight results. Study 5 registered the penalty at each level as secondary tests, judged at .05 without correction. Each was read with an equivalence test, which asks whether the penalty's 90% confidence interval, the range of values the data do not rule out at the 90% level, lies wholly inside ± 0.224 points, half of Study 1's creativity penalty on originals. At Slight the penalty was significant and not shown inside that bound on all three measures; on creativity it was 0.22 points ($p = .001$; Supplementary Table 2). That estimate averages over the three editing times; taken one time at a time, the Slight creativity penalty was 0.06, 0.48 and 0.11 at under 1, 10 and 100 hours (exploratory), so the residue at Slight, what the label carried into ratings of slight edits, rests largely on the 10-hour groups.

Readings the data cannot separate. Each combination of label and editing time was a separate group of 35 to 38 people. One unusual 10-hour group, AI-label raters rating low or Human-label raters rating high, would produce this pattern on all three measures, and these data cannot rule that out; if so, it also inflates Study 5's Slight estimates in Supplementary Table 2. The corrected p sits just under .05, and none of the accounts considered in the paper predicts a peak at 10 hours, so the interaction may be a chance result that survived the correction.

Supplementary Note 6: The pooled analysis of second works

This note reports in full the exploratory analysis that pooled Studies 2, 4 and 5, which the main text summarizes in its Results section on the residue at slight edits. It gives the complete estimates, the rival accounts and the open explanations of the peak at Slight. It also reports two registered results from the separate studies that bear on how the penalty changes across levels: Study 2's test of dilution and Study 4's tests at each level.

Design and measures. Studies 2, 4 and 5, with 235, 117 and 223 participants, asked people to rate second works that Alex was said to have made with his design and sketching tools from original images, all of which had been generated with an AI image generator. A label, varied between participants, said that Alex had made each original either with design and sketching tools (the Human label) or by prompting an AI system (the AI label). Each original came with three second works, and every participant saw all three: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy one. Participants rated each second work's transformativeness (how much it adds new expression, meaning or message to the original), its essence change (how much the essence of the original changed) and the creativity involved in its production, each on a 1-to-5 scale. The label's penalty is the mean rating under the Human label minus the mean under the AI label, in points on that scale; a positive penalty means that the AI label lowered the rating. We call what the label carried into ratings of second works the residue. Before any pooling, each study's mean ratings under the two labels at each level, and the penalty between them, are plotted in main-text Fig. 2b for creativity and in Supplementary Fig. 3 for the other two measures. Study 5's tables give 90% intervals; the 95% intervals plotted for it are computed from the same estimates and standard errors (a standard error is the expected size of an estimate's error).



Supplementary Figure 3: Transformativeness and essence change of the second work under the two labels. Marks and columns are as in main-text Fig. 2b: the two label groups' mean ratings, as predicted by each study's statistical model, and the gap between them (the Human-label mean minus the AI-label mean) with its 95% confidence interval. The label described only the original: both groups read that Alex made the second works with his design and sketching tools. **a**, Transformativeness, how much the second work adds new expression, meaning or message to the original (a lay rating, not a fair-use judgment). **b**, Essence change, how much the original's essence changed (defined in Studies 4 and 5 as its core sense and feeling). None is the original shown again unchanged. As in main-text Fig. 2b, each person saw every level but only one label, so the rise across levels and the gap cannot be compared in size. The gaps are exploratory in Study 2 and registered secondary tests in Studies 4 and 5, each judged at .05 without correction; they average over the hours stated for the original (Study 4) or over stated editing hours (Study 5). Supplementary Table 2 lists the registered tests.

Status and model. This analysis is exploratory. It was specified on 9 October 2026, after every registered result was known; its tests are not corrected for multiple comparisons, and it changes no registered verdict. A p-value is the chance of a difference at least this large if there were none, and a result is called significant when p is below .05. Where this note cites a study's own registered tests, each keeps the status its registration gave it: 'registered confirmatory' marks a test that the registration names as confirmatory, and 'registered secondary' marks one it named outside that set and judged at .05 without correction. A pooled model fitted Studies 2, 4 and 5 together, one measure at a time. It was a linear mixed model, a regression that gives each participant, each original and each second work its own baseline, and it estimated the label's penalty at each level with each study weighted equally, whatever its size. It left out Study 4's stated hours for the original and Study 5's stated editing hours, so each study's penalty averages over them. It also treats each study's participants as different people, so a person who took part in Study 5 and in Study 2 or 4 counts as two (Section 9). Its fits converged, meaning that the fitting procedure settled on a solution, but they are singular: one variance, the spread between originals once each second work had its own baseline, was estimated at zero.

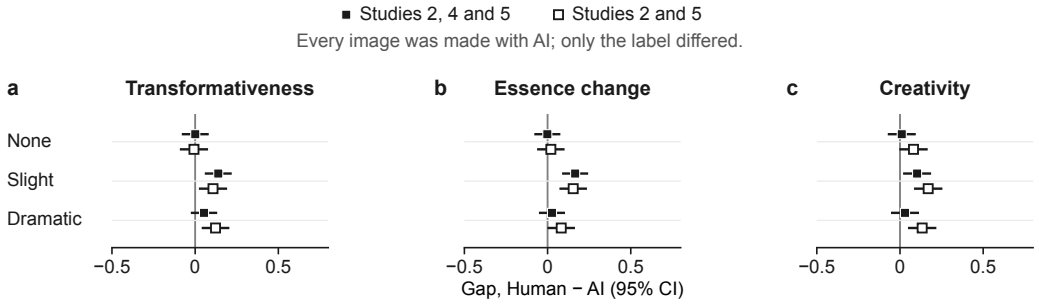
The pooled penalties. On all three measures the pooled penalty was largest at Slight (Supplementary Fig. 4), though not in every study. On creativity it was 0.01, 0.10 and 0.03 points at None, Slight and Dramatic; on transformativeness 0.00, 0.14 and 0.05; and on essence change 0.00, 0.17 and 0.03. Only the Slight penalties differed from zero. Their 95% confidence intervals (CIs), the ranges of values the data do not rule out at the 95% level, ran from 0.02 to 0.19 on creativity, 0.06 to 0.22 on transformativeness and 0.09 to 0.24 on essence change. Ratings at None sat near the floor of the scale, where a penalty has little room to show, so a penalty near zero there says little. Studies 2, 4 and 5 registered a bound of ± 0.224 points for the penalty on second works, half of Study 1's creativity penalty on originals. An equivalence test asks whether a penalty's 90% CI lies wholly inside that bound; a pass means that the penalty is smaller than the bound, not that it is zero. Read this way, the pooled penalty lay inside the bound at every level on creativity and transformativeness, and at None and Dramatic on essence change; at Slight the essence-change interval (0.10 to 0.23) reached past it. For reference, the creativity penalty on originals was 0.45 points in Study 1 (95% CI 0.19 to 0.70; registered confirmatory) and 0.35 in Study 3 averaged over stated time (0.10 to 0.61; an exploratory recomputation from Study 3's registered model). Each pooled creativity penalty on second works was smaller than both, a reading across studies, not a test.

Three rival accounts. Three accounts predict different shapes across the levels. Under taint, the AI origin passes to anything built from it, so the penalty is the same at every level. Under dilution, the penalty is largest at None and shrinks as the edit grows. Under severance, the label does not reach second works at all, so there is no penalty at any level. Two quantities summarize the shape. The slope is the change in penalty per step from None to Dramatic (half of the Dramatic penalty minus the None penalty), which dilution predicts to be negative. The curvature is the Slight penalty minus the average of the None and Dramatic penalties, which is zero for any straight line. The pooled slope did not differ from zero on any measure: 0.01 points on creativity (95% CI -0.02 to 0.04), 0.03 on transformativeness (-0.01 to 0.06) and 0.01 on essence change (-0.02 to 0.05). The fall from None that dilution predicts is therefore not shown, though the floor at None weakens this test. Instead, on all three measures the penalty at Slight lay above the average of the penalties at None and Dramatic: by 0.08 points on creativity (0.02 to 0.14), 0.11 on transformativeness (0.05 to 0.17) and 0.15 on essence change (0.10 to 0.21).

Registered tests in the separate studies. Two registered results from the separate studies bear on how the penalty changes across levels. Study 2 tested dilution directly as its third hypothesis (H3, registered confirmatory): the penalty at None minus the penalty at Dramatic,

which dilution predicts to be positive. The test was judged after Holm's correction across Study 2's confirmatory tests, a correction[4] that raises each p-value in a set of tests so that the chance of calling any result in the set significant by mistake stays at .05. It was not significant on any measure, and every estimate ran against dilution: −0.06 points on transformativeness (95% CI −0.15 to 0.03; Holm $p = .385$), −0.04 on essence change (−0.13 to 0.04; Holm $p = .385$) and −0.08 on creativity (−0.17 to 0.00; Holm $p = .153$). Study 2's ordinal check (registered secondary), a model that treats the five answers as ordered categories rather than evenly spaced numbers, leaned toward dilution on transformativeness and essence change, not significantly (Holm $p > .999$ on both). On creativity its fit did not converge, so that check is not completed. With ratings at None near the floor, this test says little either way.

Study 4 registered the penalty at each level as secondary tests. At Slight, as in Study 5 (main text), they found a penalty that was significant without correction and not shown inside the bound on transformativeness (0.20 points, $p = .024$) and essence change (0.19, $p = .024$), but not on creativity, where the Slight penalty, −0.03 points, lay inside the bound. The essence-change difference at Slight had been reported before registration: from our February 2025 analysis of these data, the CogSci 2025 paper reported a label difference in essence change at Slight and at no other level.[1]



Supplementary Figure 4: Exploratory: the gap on second works, pooled over studies. A pooled statistical model fitted Studies 2, 4 and 5 together, each study weighted equally, and estimated the gap (the Human-label mean minus the AI-label mean) at each edit level for **a**, transformativeness, **b**, essence change and **c**, creativity of the second work. Filled squares pool all three studies; open squares leave out Study 4, whose creativity estimates lay below zero at every level, none significantly. Lines are 95% confidence intervals, not corrected for multiple comparisons. Right of zero, the AI label lowered the rating; for essence change, the AI-label group judged the edit to change the original less. None is the original shown again unchanged, and ratings there sit near the floor of the scale. The fits converged but are singular: one source of variation that the model allows for was estimated at zero. The analysis was specified on 9 October 2026, after every registered result was known, and it changes no registered verdict.

The studies disagree. The pooled curve averages over studies that disagree on creativity (main-text Fig. 2b; Supplementary Table 2). In each study's own model, Study 2's penalty grew from None to Dramatic (0.07, 95% CI −0.04 to 0.18; 0.12, 0.01 to 0.23; and 0.15, 0.04 to 0.27; exploratory and uncorrected, since Study 2 registered no test at each level). Study 4's lay below zero at every level (−0.13, −0.03 and −0.17; registered secondary, none significant), and Study 5's peaked at Slight (0.09, 0.22 and 0.12; registered secondary; at Dramatic, $p = .080$). In the pooled model, the creativity curvature was larger in Studies 4 and 5 (0.12, uncorrected $p = .064$, and 0.12, $p = .016$) than in Study 2 (0.01). Leaving out one study at a time shows how much the pooled readings rest on each. Without Study 4, the pooled creativity penalty at None, Slight and Dramatic was 0.08, 0.17 and 0.13. At Slight its 90% CI (0.10 to 0.24) extended past the bound, where with all three studies it had stayed inside (0.03 to 0.17), and at Dramatic its

90% CI reached 0.20. The transformativeness reading at Slight is as fragile: without Study 2 or without Study 5, the upper end of its 90% CI was 0.26 or 0.23, past the bound. These fits were also singular but converged.

Comparing the accounts as models. Four models were fitted to the same ratings, each freer than the one before: severance (no label effect), taint (one penalty per study, the same at every level), dilution (each study's penalty changing in a straight line across the levels) and a free shape (any penalty at each level, as in the pooled model). Two standard scores for comparing models, AIC (Akaike's information criterion) and BIC (the Bayesian information criterion), reward a model for fitting the ratings and charge it for each added quantity it estimates; BIC charges more. On every measure, AIC favored the free shape and BIC favored severance, the model with no penalty at all. Expressed as weights, each model's share of support among the four, AIC gave the free shape 0.62 on creativity, 0.98 on transformativeness and 1.00 on essence change, and BIC gave severance 1.00, 1.00 and 1.00 after rounding. On creativity, a likelihood-ratio test, which asks whether a model's added quantities improve its fit, found that giving each study one constant penalty fitted better than no penalty ($p = .007$). Study 4's constant penalty ran below zero, however, so this does not show one penalty shared by the three studies. An ordinal check of the pooled creativity penalty did not converge and is not completed, so it adds no evidence either way.

Explanations of the peak at Slight. The cause of the peak at Slight is unknown, and no argument in the paper rests on it. Five explanations remain open. A slight edit may leave Alex's share of the result unclear, so the label may matter more there. Of the ten Slight versions, 7 were also cropped or resized, so Slight is not a pure small edit. Ratings at None sit near the floor, which leaves little room for a penalty to show there. Under the Human label the original was Alex's own earlier work, which may earn the second work a bonus, and no condition with a neutral original was run to check this. The peak may also be chance. Nor is the residue confined to Slight: Study 2's creativity penalty (0.15, uncorrected $p = .007$; exploratory) and Study 5's transformativeness penalty (0.17, $p = .009$; registered secondary, uncorrected) were significant at Dramatic.

Supplementary Note 7: What the explanations referred to

This note gives in full the analyses of participants' written explanations that the main text summarizes in its Results section of the same name.

How the explanations were read. In all five studies, a required text box for each rating asked participants to explain their reasoning, and every box came after the ratings it explained. Eight fixed word lists (for example, words for visible detail, time, skill, and prompting or machines) flagged what each explanation mentioned. We wrote the lists by reading Study 1's explanations with their label and rating hidden, and fixed them before relating them to either, apart from four parts settled after results by label had been read (Section 10). A script then applied the lists without seeing either, and no person or language model classified any single explanation. Every explanation was written after its rating, so these results describe what the words referred to, not why anyone rated as they did. Everything in this note is exploratory, meaning that a registration named it as exploratory or did not name it at all, or descriptive, meaning a share or a count with no test. The analyses of second-work explanations in Studies 2, 4 and 5, of talk of origin and of how far explanations moved, were specified on 9 October 2026, after every registered result was known, and they change no registered verdict. A difference between two shares is given in percentage points, the plain difference between two percentages.

Originals. On originals, what 'effort' referred to depended on the label. In Study 1, 111 participants rated ten AI-generated images on creativity and on effort, told either that Alex made them with design and sketching tools (the Human label) or that he made them by prompting an AI system (the AI label). Study 1's registration named five word-meaning tests as one exploratory family and set the Benjamini–Hochberg correction for them. That correction limits the expected share of false positives among results called significant;[7] a result is called significant when its p-value, the chance of a difference at least this large if there were none, is below .05. The corrected results are descriptive readings, not verdicts. Each test compared how often two kinds of mention occurred, allowing for differences between participants and between images. Under the Human label, 62.5% of effort explanations named visible detail, 10.0% named time and 5.2% named skill. Detail was named more often than time and more often than skill (Benjamini–Hochberg $p < .001$ against time and $p < .001$ against skill). Under the AI label, 50.5% of effort explanations mentioned prompting or machines, against 15.4% under the Human label (descriptive). The difference held among explanations given with the same effort rating of 3 (41.8% against 20.5%; descriptive), so it does not arise only because AI-label raters gave lower ratings. In a third test of the family, AI-label participants mentioned prompting or machines more often when they explained effort than when they explained creativity (Benjamini–Hochberg $p < .001$). The family's last two tests concerned Human-label creativity explanations. These named the idea behind the work (26.8%) more often than its execution (12.1%; Benjamini–Hochberg $p < .001$), but not detectably more often than whether it looked new or familiar (24.5%; $p = .350$). How the word list for new or familiar handles negations was settled after results by label had been read. Because these two tests used Human-label explanations only, they say nothing about what AI-label creativity ratings rested on.

Study 3, in which 85 participants rated nine of the originals under the Human or the AI label, each with a stated time of under 1, 10 or 100 hours, repeated the pattern on effort. Each difference below is AI minus Human, in percentage points, with a 95% confidence interval (CI), the range of differences the data do not rule out at the 95% level. AI-label effort explanations named detail less often and prompting more often (-15.6 for detail, 95% CI -28.5 to -2.9; 19.4 for prompting, 8.3 to 31.7). AI-label creativity explanations also mentioned prompting more often (12.3 points, 3.8 to 21.8). These intervals come from resampling participants, and none of Study 3's word-list comparisons was corrected for multiple comparisons, so an interval that only just excludes zero is tentative. Rating the same pictures, the two label groups wrote about different acts when they explained effort. This does not show that detail drove effort ratings or that 'effort' meant labor.

Second works: talk of origin. On second works, the label left a faint trace in what people wrote. In Studies 2, 4 and 5, participants explained their ratings of second works that Alex was said to have made from each original with his design and sketching tools: None, the original shown again unchanged; Slight, a light edit; and Dramatic, a heavy one. Both label groups were told that Alex made every second work this way; only the description of the original differed. Pooled over the three studies, with each study weighted equally, AI-label explanations mentioned prompting or a machine more often than Human-label ones at every level (3.6%, 3.7% and 5.9% at None, Slight and Dramatic, against 0.5%, 0.6% and 2.8% under the Human label). These shares count explanations of all three ratings from people who explained every level of the same original, so each level holds the same people and images. Such mentions rose from None to Dramatic under both labels, and the two rises did not differ detectably (difference 0.0 percentage points, 95% CI -1.3 to 1.2, from resampling participants). In Study 2, however, AI-label creativity explanations went from 4.2% at None to 8.4% at Dramatic, against 1.2% and 2.6% under the Human label, so their rise was 2.9 percentage points larger (95% CI 0.3 to 5.8; uncorrected).

Second works: how far explanations moved. Under taint, the account on which the AI origin passes to anything built from it, AI-label raters should have kept writing about the original, so their explanations should have changed less from None to Dramatic. That change was measured in two ways: with sentence embeddings, which turn each explanation into a list of numbers whose distances reflect differences in meaning, and with word overlap, the share of content words two explanations have in common.[8,9] For each person and image, the measure began with how far that person's Dramatic explanation lay from their None explanation. From this it subtracted how far their None explanation lay from other participants' Dramatic explanations of the same image, which removes what any two people writing about the same picture share. The label groups were then compared with explanation length held constant; a negative difference, AI minus Human, means that AI-label explanations moved less. Pooled over studies, no difference in change was shown between the label groups. On creativity the difference was −0.006 with sentence embeddings (95% CI −0.023 to 0.011) and −0.001 with word overlap (−0.015 to 0.013). The pooled estimates all lay on the side taint predicts (6 of 6, two distances on each of three measures), but none was clear of zero. Of 24 uncorrected comparisons, pooled and study by study, 1 fell clear of zero on that side (Study 5's transformativeness explanations, word overlap), about what chance alone would produce. No difference shown is not the same as no difference. No study asked participants to recall the label (a manipulation check), so none of this shows that the label was in mind when people rated.

3 Supplementary Methods

The main text's Methods give the design and analysis of the five studies in outline. This section gives the detail behind them, in the same order. It does not repeat what other sections hold. Each study's section quotes that study's materials and gives its exclusions step by step and its sample. Section 4 gives the registrations, what we had seen of each study's data before filing them, how the images were made, and the corrections to earlier reports. Section 10 describes the word lists, and the Supplementary Notes report the results. A registration here is a dated record of a study's hypotheses and analysis plan, not a copyright registration.

Participants

Sample sizes. No power analysis, a calculation of how many participants a study needs to detect an effect of a given size, was run for any study. Each survey set a quota, a planned number of participants: 60 for each label group in Study 1, 125 in Study 2, 50 in Study 3 and 62 in Study 4, and 40 for each of Study 5's six combinations of label and editing time. No quota sent anyone to the other label. Study 2's planning document, which was never submitted, had set a target of 210 participants and a stop at 250. Study 1's registration adds simulations of what its design could detect, run after its data had been collected (Supplementary Note 1); they did not set its sample size. In Studies 2 to 5, a claim that an effect is small rests on the equivalence tests described below.

Exclusions. Each registration states its exclusion rules and their order. The rules were written after the data had been collected, and earlier analyses of Studies 3 and 4 had kept some records that these rules remove. Every study first removed survey test runs and unfinished sessions. The closing screen asked 'Did you complete this experiment seriously throughout (without responding randomly)?', and every study then removed anyone who gave any answer but 'Yes, I did', or no answer. Studies 2 to 5 next removed wrong answers to an arithmetic question, 'What does 8+7 equal?'. Last, Studies 2 and 5 removed records with no recorded condition (there were none in Study 2), and Study 4 removed records that were not valid completions. A valid completion showed exactly six originals, each once, two at each stated time. In Study 1 the sequence also

includes the question about the task, which removed no one, because the survey let everyone continue after one retry. Each study's section gives the number removed at every step.

Participations and people. The five analyzed samples add up to 771 participations, where a participation is one person's session in one study. We counted the people behind them by matching the subject-pool participant code, the code that the survey returned at its end to the pool's page for granting credit and that is the same for a person in every study. The 771 participations came from 747 people. Of these, 24 took part twice, each in one earlier study and then in Study 5. Studies 1 to 4 ran under one subject-pool listing and Study 5 under a listing of its own, which would let earlier participants sign up again. A person in Study 5 who had taken part before had already seen some of the same originals and read the debriefing, which told them that anything produced with AI is insufficiently creative (below). No registered analysis removes them, no analysis checks whether their first session changed their ratings, and the exploratory pooled model of Studies 2, 4 and 5 counts twice those who had taken part in Study 2 or 4. This count, like the age and gender summaries below, was specified on 19 October 2026, after the registered results were known; it is descriptive and tests nothing.

Age, gender and other background questions. The closing screen asked for race, gender, age, native language and formal artistic training. Age was typed as free text and read as the first number in the answer. Gender was chosen from four answers: male, female, other and prefer not to say. No question asked about sex. The table below gives, for each analyzed sample, the number of participants under each label and a summary of age and gender. 'Other or none' joins three answers: other, prefer not to say and no answer.

Study	*N*	Human	AI	Mean age (SD)	Age range	Female	Male	Other or none
Study 1	111	56	55	20.08 (1.69)	18–29	93	17	1
Study 2	235	122	113	19.96 (1.82)	18–34	192	38	5
Study 3	85	43	42	20.16 (1.48)	18–26	60	23	2
Study 4	117	55	62	20.52 (3.07)	18–42	96	20	1
Study 5	223	112	111	20.22 (2.47)	18–45	179	37	7

Human and AI give the number of participants under each label. SD is the standard deviation, a measure of how far ages typically lie from the mean. Supplementary Table 7 gives Study 1's answers on race and artistic training as well; the answers of Studies 2 to 5 to those questions are not summarized here. No analysis compared genders or other background groups. Age, gender and training entered only some of Study 1's robustness refits, as covariates, variables added to a model to adjust for them (below).

Materials and procedure

The label. Every version of the label opened 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks.' The AI label went on, in every study: 'Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.' The Human label went on, in Studies 1, 4 and 5: 'Alex uses his graphic design skills and digital sketching tools.' In Study 2 it went on 'Alex only uses graphic design software and digital sketching tools', and in Study 3 'Alex uses only his graphic design skills and digital sketching tools such as iPad apps and Adobe Photoshop.' Before every original, a short description restated how Alex had made that image. These descriptions differed slightly across studies: in Study 2, the AI version said that Alex prompted the AI system without saying that he revised his prompts, and in Studies 3 to 5 both versions stated a time. In Study 5 the description of each second work stated the editing time in the same words at every level, so at None it said that Alex had spent that time on an unchanged copy. Each study's section quotes every description that participants read.

The images. Section 4 (How the images were made) describes how the ten originals were generated, the provenance records that eight of their files carry, the prompts and the hand edits. Supplementary Note 4 identifies the 7 Slight versions that were also cropped or resized. For that check, the 30 images were extracted, without being re-encoded, from a printout of the Study 5 survey: a PDF file printed from the survey software, which holds each image as the survey served it.

Screens and order. Each survey opened with a welcome screen and then the consent form. The instructions said that participants would be asked to explain their answers. A question about the task followed ('What is your task in this experiment?'); a wrong or missing answer brought back the instructions and the question once, after which every participant went on. Each original stayed on screen alone for at least 8 seconds. In Studies 2, 4 and 5, each second work was introduced and then shown beside its original, headed 'FIRST WORK' and 'SECOND WORK', for at least 10 seconds. Each rating screen gave all of an image's ratings in a fixed order (creativity, then effort; or transformativeness, essence change, then creativity) and below them a required box for explaining each rating. Copying and pasting were blocked in none of the 20 explanation boxes of each of Studies 1 and 3, and in 21 of the 90 boxes of each of Studies 2, 4 and 5.

The originals came in a random order, except that in Studies 4 and 5 the three with a figure came together, as did the three without, the two groups in random order. The three second works of each original also came in a random order. The surveys did not record the order of the originals in Studies 1 and 2, or the order of the second works in any study. In Study 5, the survey script meant to record each original's source time looked for it under a field name that the survey no longer used, so that record is blank for every participant. The arithmetic question came at a random point among the originals in Study 2 and after the last rating in Studies 3 to 5; Study 1 had none. The closing questions followed in this order: the seriousness question, a question about problems with the procedure, the background questions and a box for any other comments.

The debriefing and the law. The last screen, the same in all five studies, read: 'We are studying how humans perceive the copyrightability of artificial intelligence art. Current legal standards state that anything produced with artificial intelligence is insufficiently creative, even if you use AI as a base against which you create new work. There is ongoing litigation against the US Copyright Office with an analogous set of facts that you have seen here.' A final sentence invited questions. The second sentence overstates the law. The US Copyright Office refuses to register material whose expressive elements an AI system determined. It does register a person's own creative selection or arrangement of that material, or a modification of it creative enough to be original, with the AI-generated material left out of the claim.[10,11] The Office treats such a modification by analogy to a derivative work, a work based on an earlier work, whose copyright covers only the material its author contributed (17 U.S.C. § 103(b)). It is only an analogy, because in the Office's view AI output is not itself a work of authorship.[11] The debriefing instead stated as law the categorical view that this paper calls the original sin: that AI involvement leaves even work built on it insufficiently creative. The paper tests that view as a hypothesis about lay judgment, not as a legal rule. The debriefing also did not tell participants that every image had been made with AI.

Measures

Each study's section quotes its rating items and answer labels. Studies 4 and 5 defined essence as 'the core sense and feeling that one has while observing a work of art that makes that work unique', and defined the creativity of a second work as for originals. Study 2 defined neither. In each of Studies 2, 4 and 5, two transformativeness items had a sixth answer point, which puts

them on a different scale. In Study 2 these two items also asked 'How transformative do you consider the second work compared to the first work?', with no definition. As registered, their ratings are left out of the primary analyses: 470 in Study 2, 140 in Study 4 and 274 in Study 5. Study 5's registered second analysis puts them back on a common scale (below).

Registrations

Section 4 lists the six registration documents and, study by study, what we had seen of each study's data before filing. Two points from the Methods are given here in full. First, for Studies 2 to 5 the code reported here was run before filing only with the condition labels shuffled at random across participants, so that every model and table could be checked while no number described the real conditions. In Study 5's shuffled runs, and in one early shuffled run of Study 3's, the ratings were not shuffled, so the text analyses that relate the words to the rating alone used the real ratings. Those outputs were set aside, and none is reported. Second, the CogSci 2025 paper,[1] which first reported Studies 2 and 4, makes several statements that this paper corrects. It called its first study, this paper's Study 2, pre-registered, but no study was registered anywhere before its data were collected. It attributed the prompts to a validated set published by Orwig and colleagues,[12] but we wrote the prompts ourselves. It described the Slight versions as adjustments of hue, coloration and contrast, without the cropping and resizing. It said that a power analysis had set its sample size, but none was run. It gave Study 4's sample as 119, the count before the registered rule that removes 2 invalid completions. Section 4 sets out each correction, with the others.

The registered models

What every model shares. Each registered analysis is a linear mixed model, a regression that also gives each participant and each original its own baseline, called a random intercept. One model was fitted for each rating question, to every individual rating rather than to participants' averages. The models used 2,220 ratings in Study 1, 20,659 in Study 2, 1,525 in Study 3, 6,178 in Study 4 and 11,768 in Study 5. Missing ratings were left out, not filled in. In Study 2, 7 participants lack one trial. Study 3's records lack one rating for each of 3 participants, and 2 other ratings carry an answer code that the final survey did not offer, so they are treated as missing. Models were fitted by restricted maximum likelihood, the usual way of estimating such models, with the R packages lme4 and lmerTest. Studies 2 to 5 coded each factor so that its effects are measured from the average over its levels. Study 1 coded the label and the kind of image against reference levels (Human, and no figure), so its penalty is the averaged difference described next, not a single model coefficient.

Each reported difference is a contrast between estimated marginal means: the mean ratings that the model predicts for two conditions, each averaged with equal weight over the other factors. F tests asked whether a factor changed the mean rating, and t tests whether a single difference departed from zero. Their degrees of freedom, which set how large a statistic must be to count as significant, came from Satterthwaite's approximation, computed for every rating, including in the larger studies, where the software would otherwise have switched to a large-sample approximation. Study 1's contrasts used the Kenward–Roger approximation, as its long registration specifies; it gives the same value, 109, for the penalty. A p-value is the chance of a result at least this far from no effect if there were none. Tests were two-sided at .05, and a 95% confidence interval (CI) is the range of values that the data do not rule out at that level.

The model of each study. In the formulas below, a star joins factors whose effects and interactions (changes in one factor's effect across the levels of another) are all estimated, and a term such as (1 | participant) gives each participant a baseline of its own.

- Study 1, per rating: rating $\sim$ label $*$ kind of image $+ (1 \mid \text{participant}) + (1 \mid \text{original})$. H1 holds if the penalty on both creativity and effort is positive and survives Holm's correction. H2 is tested in one joint model of both ratings, rating $\sim$ label $*$ rating question $*$ kind of image $+ (1 + \text{rating question} \mid \text{participant}) + (1 \mid \text{original:rating question})$, which lets each participant's gap between the two ratings differ. Its corrected test is the F test of the label by rating question interaction, and an ordinal version of the same model must agree in sign and significance. H3 is the label by kind of image interaction. It is judged by a two-sided test together with an equivalence test against ± 0.20, through a four-cell decision table in the long registration. A failed equivalence test is uninformative if the test's power, its chance of passing when the true difference is zero, is below 80%.

- Study 2, per measure: rating $\sim$ label $*$ modification level $*$ kind of image $+ (1 \mid \text{participant}) + (1 \mid \text{original})$. Four tests per measure enter the corrected set of 12: Slight minus None and Dramatic minus Slight, which together decide H1; the penalty averaged over levels, tested for equivalence (H2); and the penalty at None minus the penalty at Dramatic (H3).

- Study 3, per rating: rating $\sim$ label $*$ stated time $+ (1 \mid \text{participant}) + (1 \mid \text{original})$. The corrected set holds the F tests for label, time and their interaction on each rating. The registered decision calls the gap closed by time only if the interaction survives Holm's correction, the gap is smaller at 100 hours than at under 1 hour, and the gap at 100 hours passes the equivalence test. It calls the gap persistent if that equivalence test fails and the gap's 95% CI at 100 hours lies above zero, and otherwise inconclusive.

- Study 4, per measure: rating $\sim$ label $*$ source time $*$ modification level $+ (1 \mid \text{participant}) + (1 \mid \text{original})$. Five tests per measure enter the corrected set of 15: the two steps of H1, the equivalence test of H2, and the F tests for source time and for source time by modification level (H3). Registered secondary tests, judged at .05 without correction, give the penalty at each level, the label by source time interaction, and an equivalence test of 100 hours minus under 1 hour against ± 0.224.

- Study 5, per measure: rating $\sim$ label $*$ editing time $*$ modification level + kind of image $+ (1 \mid \text{participant}) + (1 \mid \text{original})$, with kind of image added on its own. The corrected set holds the F tests for editing time and for label by editing time on each measure. Registered secondary equivalence tests, not corrected, ask whether 100 hours minus under 1 hour of editing time lies inside ± 0.219 and whether the penalty at each level lies inside ± 0.224; the verdict on editing time also reads its corrected F test.

H4. H4 asked whether the AI label spreads ratings more between participants and less within each participant, across images. Under each label it estimated two variances, a variance being a measure of spread: the variance between participants, and the variance within each participant across images. The first came from a model that lets the variance between participants differ by label, and the second from a model that lets the remaining variance differ by label. It compared the labels by the ratio of each variance, AI over Human, with 95% intervals from 1,000 resamples of participants within each label, so its result is a pair of ratios, not a difference between means. H4 holds on a rating only if the ratio between participants lies above 1 and the ratio within participants below 1, each with its whole interval on that side, and it must hold on both ratings. Its entry in Study 1's corrected set is a likelihood-ratio test, which asks whether a model that lets the variance between participants differ by label fits better than one that does not; the rating with the larger p-value enters. Supplementary Note 1 reports the result.

Equivalence tests and bounds. An equivalence test is two one-sided tests, each at .05, of whether an effect lies above a lower bound and below an upper bound.[2,3] Its p-value is the larger of the two one-sided p-values, and judged alone it passes when the effect's 90% CI lies wholly inside the bounds. In Studies 2 and 4, that p-value entered Holm's correction with the study's other confirmatory tests, so an interval can lie inside the bound while the corrected test fails. Study 1's bound of ± 0.20 was set after its analyses had been run, on the ground that a smaller difference is finer than the resolution at which participants use a five-point scale. The bound of ± 0.224 is half of Study 1's registered creativity penalty of 0.448 points, used for all three second-work measures for lack of another benchmark. Study 5's registration fixed the rule for its editing-time bound before any Study 5 model was fitted. Its code read the value, Study 3's registered creativity contrast between 100 hours and under 1 hour (0.219), from the output of Study 3's run with the real labels, and would have stopped had that output come from a shuffled run.

Effect sizes. Study 1's registered standardized effect size is Cohen's d: the difference between the label groups' mean ratings divided by the spread of participants' mean ratings, with a 95% interval from 10,000 resamples. A model-based version divides the penalty by the square root of the summed variances of the model, so that it can be compared with other studies. Section 5 reports both.

Ordinal checks and fitting status

Ordinal checks. As registered checks, the confirmatory models were refitted as cumulative-link mixed models, which treat the five answers as ordered categories rather than evenly spaced numbers and report estimates on the log-odds scale, the logarithm of the odds of a higher answer. They used the R package ordinal, with a logit link and the Laplace approximation, and their tests are Wald tests, which compare an estimate with its standard error. In Study 1, agreement of the ordinal model in sign and significance was part of the registered rule for H2. In Study 2, the nine ordinal p-values for the level steps and the dilution test replaced the linear ones in a corrected set of twelve. The three equivalence tests stayed as they were, because their bound is in rating points and has no ordinal counterpart. Study 4 did the same with its twelve tests of differences in its set of fifteen, and Studies 3 and 5 corrected the ordinal test of each model term over the same six tests. Elsewhere than Study 1's H2, the registrations call the ordinal model a check, so a disagreement is reported beside the verdict rather than replacing it.

Fitting status. Every fitted model carries a status. A linear mixed model converged if its optimizer, the numerical search for the best-fitting values, reported success and raised no warning about convergence. An ordinal model converged if its optimizer reported success and raised no warning about convergence or about the shape of its fit. A model that did not converge stopped without settling on a solution. It yields no p-value, interval or verdict, and its check is reported as not completed. Study 1's registration added this rule in October 2026, after its September estimates had been read, and every study's code applies it. The fits that did not converge were the ordinal checks of Study 2's creativity model and of the pooled creativity model, and the structural topic model of the essence-change explanations. A singular fit converged but put a variance, or a correlation between baselines, at the edge of its possible range, such as a variance of zero, a sign that the data cannot support every quantity the model asks for. Such a fit keeps its numbers and is reported as singular. The singular fits include Study 1's most complex refits, Study 3's refits with random slopes and every pooled linear mixed model of the exploratory analyses.

Random slopes. Two registrations asked that a verdict also hold in a model with random slopes, which let an effect differ from one participant, or one original, to the next. Study 1's

asked this of H3 and Study 3's of its equivalence test. Each let the simpler model decide if the slope model was singular, and Study 3's also if that model stopped with an error or did not converge. Study 1's registration does not list this rule among the three elements fixed before its analyses were run. In both studies the slope model was singular, so the verdict rests on the simpler model.

Study 1's robustness checks and the other registered refits

Study 1. The registered robustness checks refit the penalty in several ways (Supplementary Note 2). A specification curve refit it under every combination of a set of defensible analysis choices. The choices were four. The model had baselines only, or the most complex structure of random slopes. It was a linear mixed model, an ordinal model, or generalized estimating equations, which estimate an average effect while allowing ratings from one participant to be related. It had no covariates, background covariates, or those plus the length of the explanations. And it was fitted to every rating, or to each participant's mean for each kind of image, with standard errors that allow for related ratings. The combinations reduce to 18 distinct models for each rating. Other refits added background measures as covariates, left out one participant or one original at a time, dropped one effort item whose answer labels differed from the rest, or used Bayesian models.

Bayesian models combine the data with stated assumptions about plausible values, called priors. The four Bayesian refits, a linear and an ordinal version for each rating, ran in brms. Their priors were normal(0, 1) on the effects, a bell curve centered on zero with a standard deviation of 1 point, and Student-t(3, 0, 2.5) on the intercept and the standard deviations, a wider curve with heavier tails. Each model drew its estimates in four independent runs, called chains, of 4,000 iterations each. The first 2,000 of each run were discarded as warm-up, while the sampler tuned itself. The sampler took small steps (an adapt_delta setting of 0.95) to avoid failed steps, called divergences. A fit counted as converged only if, for every quantity, the chains agreed (R-hat below 1.01), the draws were worth more than 400 independent ones, and none diverged.

The belief refits and the rescaled analysis. A registered refit of Study 1 left out the Human-label participants who wrote at least once that a work is or looks machine-made, by the word list for a statement that a machine made the work. As exploratory description, 2,000 random groups of Human-label participants of the same size and mean effort rating were removed in the same way for comparison (Supplementary Note 2). Study 4's registration added a refit without the Human-label participants who wrote that a machine made the work, which repeats all fifteen corrected tests (Section 8). Study 5's registration added a second analysis that rescales every item to run from 0 to 1, as (rating − 1) divided by (number of answer points − 1), so that the six-point items can enter. It repeats the six corrected tests and runs no equivalence test, because both bounds are in rating points.

The text measures

Section 10 describes the eight word lists, how they were written and applied, and the two registered checks. The lists flag visible detail, time, skill, concept (the idea behind the work), prompting or machines, a claim that the work looks familiar, the scale's own answer labels, and a statement that a machine made the work. The script that applies them received only identifiers and text, and stopped if handed the label, the rating or any background measure; the label and the rating were joined to the flags afterwards. The first registered check asks how well the words predict the rating and the label for participants left out of the fitting, using ridge regression, a regression that shrinks each word's weight toward zero. The second asks whether participants under the AI label reuse their own words across explanations more than

those under the Human label, relative to other participants writing about the same picture. Study 1's registration also names five tests of what its effort and creativity explanations referred to. They form an exploratory set corrected by the Benjamini–Hochberg method, which limits the expected share of false positives among the results called significant, and their corrected results are descriptive readings, not verdicts.[7] All other analyses of the explanations are exploratory, among them scores of word tone and concreteness from published norms, average ratings of single words collected by other researchers.[13,14]

The exploratory analyses

The analyses in this subsection were specified on 9 and 10 October 2026, after every registered result was known. They are exploratory, their tests are not corrected as a set, and none changes a registered verdict.

The pooled model of second works. A pooled model fitted the second-work ratings of Studies 2, 4 and 5 together, one model per measure: rating $\sim$ label $*$ modification level $*$ study + (1 | study:participant) + (1 | original) + (1 | original:level). It averaged the label's penalty at each level over the three studies with equal weight, whatever their sizes, and averaged Studies 4 and 5 over their time factors, which it leaves out (Supplementary Note 6). It compared four accounts of how the penalty might change with the level: severance (no penalty), taint (the same penalty at every level), dilution (a penalty that changes in a straight line across the levels) and a free shape. Each account was fitted by maximum likelihood, so that their fits can be compared. The comparison used AIC and BIC, two standard scores for choosing between models that reward fit and charge for each added quantity, BIC more heavily. The pooled model treats each study's participants as different people, so anyone who took part in Study 5 and in Study 2 or 4 counts twice. A breakdown of Study 5's registered interaction gave the penalty for each combination of editing time and modification level (Supplementary Note 5).

The explanations of second works. Two analyses described the second-work explanations (Supplementary Note 7): how often they mentioned prompting or machines, by the word lists, and how far each participant's explanations moved from None to Dramatic. The second was measured by word overlap and by a sentence-embedding model (all-mpnet-base-v2), which turns each explanation into a list of numbers whose distances reflect differences in meaning.[8,9] Each person's distance was taken relative to other participants writing about the same picture, and the labels were compared against 200 shuffles of the label within each study. A structural topic model, which finds groups of words that occur together and relates how often they occur to the conditions, described the pooled explanations (Section 11).

The image check. An image check measured each edited image's distance from its original with two image-recognition networks, DINOv2 and CLIP,[5,6] and with a color measure, the difference between the two images' color histograms. It asked whether the versions of each artwork are arranged as the design assumes, by rules fixed in its code before any distance was computed (Supplementary Note 4). Neither network was built to judge art, so neither distance shows how different two images looked to participants.

Software, random seeds and run records

The models ran in R 4.5.2 (lme4 2.0-1, lmerTest 3.2-1, emmeans 2.0.3, ordinal 2025.12-29, brms 2.23.0 with rstan 2.32.7, glmnet 5.0, stm 1.3.8). The text measures and the image check ran in Python 3.12.3 (pandas 2.2.2, NumPy 1.26.4, scikit-learn 1.5.2, SciPy 1.13.0, sentence-transformers 3.3.1, PyTorch 2.5.1 and, for the image check, transformers 4.46.2). Each run

records the versions it used; only the image check also fixes them in a file of requirements. Refitting the four Bayesian models needs a C++ compiler.

Random steps used fixed seeds. A seed is the starting number of a random-number generator, so a rerun with the same seed repeats the same random draws. The R code used 20260421, as did the text code of Studies 2 and 3. Study 1's text code used the text module's defaults, and the text code of Studies 4 and 5 used 20260903, as their registrations name. The image networks were run at fixed versions (DINOv2 revision f9e44c8, CLIP revision 3d74acf). The sentence-embedding model was loaded only from the copy stored on the analysis computer, revision e8c3b32, the only copy there; the run records do not name the revision.

Each analysis writes a run record that lists a checksum, a fingerprint of a file's exact bytes, for every input and output, and later analyses check the files they read against those records. Every result, and every count of participants, records or ratings, printed in the paper is read from the tables that the analysis writes. The script that reads them records the table, row and column behind each number, with the checksum of the table it came from.

4 Registrations

A registration is a dated record of a study's hypotheses and analysis plan. It has nothing to do with copyright registration. This section lists the six registration documents of the five studies and what each covers. It says what we had seen of each study's data when we filed them, defines the status labels that every result in the paper carries, states how the images were made, and corrects statements made in earlier reports of these studies.

The six documents

All six documents were filed together on 8 October 2026 on the Open Science Framework (OSF), an online archive for study plans, as one registration in the project at `https://osf.io/u3d69/overview?view_only=be795b275c3548428d74457df31a4159`. Each study has a short registration that answers the nine questions of the form used by AsPredicted, another registration service: whether data had been collected, the hypotheses, the measures, the conditions, the analyses, the exclusions, the sample size, anything else to record, and a title. Study 1 also has a long registration, which sets out its full analysis plan, the legal reasoning behind its predictions and the rules for its text measures. Where Study 1's short registration is silent, the long one governs. Supplementary Table 4 lists the six documents, the study each covers and what each registers. Copies of the six files are shared with the data (main text, Data availability).

Every document was filed after data collection had ended in every study. The study numbers follow the argument of the paper, not the order in which the studies were run. Study 2 ran first, from 23 November to 5 December 2024, then Study 4, Study 1 and Study 3 in early 2025, and Study 5 from 21 October to 2 November 2025. In this paper, 'registered' therefore means written into one of these documents. It does not mean fixed before the data were collected, and no study in this paper was registered before its data were collected.

For Studies 2 to 5, each registration was written before the analysis code reported here was run on the real data. Until the filing, that code was run only blind: every model and table was produced with the condition labels shuffled at random across participants, so that the code could be checked while no number described the real conditions. The runs with the real labels followed on 8 October 2026, after the filing. One exception concerns the written explanations. In Study 5's blind runs, and in one early blind run of Study 3's, the ratings were not shuffled, so the text analyses that relate the words to the rating alone used the real ratings. Their outputs were set aside, and no estimate from them is reported anywhere. Study 1's registration is different: it was written after its analyses had been run, which makes it a postregistration.

Supplementary Table 4: The six registration documents. All six were filed together on the Open Science Framework (OSF) on 8 October 2026, after every study's data had been collected. Studies are listed in the paper's order; the order in which they were run was 2, 4, 1, 3, 5. *Confirmatory*: the small set of tests that a registration names as the study's confirmatory tests, judged together after Holm's correction. Bounds are in rating points on 1-to-5 scales.

Document	Study, data collection and analyzed sample	What it registers
Study 1, short registration	Study 1; 25 February to 3 March 2025; $N = 111$	H1, the AI label lowers both creativity and effort; H2, the drop is larger on effort; H3, the drop is the same for images with and without a human figure; H4, under the AI label ratings vary more between people and less within a person. Four confirmatory tests (two for H1, one each for H2 and H4); H3 decided by its own rule against ± 0.20. Five word-meaning tests on the explanations as an exploratory set, and the two text checks.
Study 1, long registration	Study 1	The full plan, which governs where the short registration is silent: why the predictions matter for the law (section 2), design, exclusions, every model and decision rule, the robustness checks, the word lists and text analyses (section 9), which decisions were taken with the estimates already known (section 8), and the materials (section 10).
Study 2	Study 2; 23 November to 5 December 2024; $N = 235$	H1, ratings of the second work rise from None to Slight to Dramatic; H2, the label's effect, averaged over levels, lies inside ± 0.224 on each measure; H3, whether the label's effect shrinks from None to Dramatic (no direction predicted). Twelve confirmatory tests, four per measure; ordinal checks; the two text checks on creativity explanations.
Study 3	Study 3; 4 to 9 March 2025; $N = 85$	The label, the stated time and their interaction on creativity and on effort (six confirmatory F tests); an equivalence test of the Human-minus-AI gap at 100 hours against ± 0.224; a rule deciding whether time closes the gap, the gap persists, or the result is inconclusive; ordinal checks; the two text checks on both ratings.
Study 4	Study 4; 27 January to 2 February 2025; $N = 117$	H1 and H2 as in Study 2; H3, whether the hours stated for the original change ratings of the second work, alone or with the level (no direction predicted). Fifteen confirmatory tests, five per measure. Secondary: the label's effect at each level, its change with stated hours, an equivalence test of 100 hours against under 1 hour, and a refit without Human-label participants who wrote that a machine made the work; the two text checks on creativity explanations.
Study 5	Study 5; 21 October to 2 November 2025; $N = 223$	The effect of stated editing hours, and its interaction with the label, on each measure (six confirmatory F tests); an equivalence test of editing time against ± 0.219; the label's effect at each level against ± 0.224, nine tests each judged at .05 without correction; ordinal checks; a second analysis with every item rescaled from 0 to 1; the two text checks on creativity explanations.

What we had seen before filing

The first question of each registration asks whether data had been collected, and each answer lists what we had already done with them. The account below follows those answers.

Study 1. In April 2025 we compared the two label groups' mean ratings with Welch t tests, which compare two means without assuming that the groups vary equally, and with analyses of variance by label and kind of image, on the 111 participants that the registered exclusions keep. The first author showed these results at a lab meeting on 18 April 2025. A public talk by the three authors on 26 September 2025 showed the same t tests, with effect sizes, for both ratings. Both drops on the originals, and the larger drop on effort, had therefore been seen before Study 1 was registered. In November 2025 we ran topic models and sentiment scores on the explanations. By April 2026 a language model had coded the explanations with the label, and for sentiment the rating, in its instructions, and a manuscript draft reported the means, effect sizes and robustness checks along with those codes. None of that text work is used here. Before 7 September 2026, other exploratory work had predicted the ratings from the words. In September and October 2026 we ran the registered code itself, and a code review on 8 September 2026 summarized its results for the first author. Section 8 of the long registration names only three elements fixed before those analyses: the model for each rating, the sequence of exclusions and the convention for effect sizes. It lists the decisions taken with the estimates in view. They include the joint model that tests H2, H3's bound and decision table, the set of tests corrected together, and the quantity that decides H4; the rule for fits that did not converge was added in October 2026. Study 1's H1 and H2 are therefore registered tests of patterns already seen, not predictions made in advance.

Study 2. By February 2025 we had fitted an analysis of variance to the 235 participants that the registered exclusions keep. The CogSci 2025 paper reported it as its first study,[1] and the September 2025 talk showed it with plots of every condition. Ratings rose steeply with the level of modification. The label made a difference only to creativity, and only averaged over levels, not at any single level. No interaction was significant. In April 2025 we ran keyword, topic and sentiment analyses of the explanations. After that, only counts and a few pooled answer frequencies were read from the data. Study 2's registered test of the label's effect on creativity therefore re-tests an effect already known.

Study 3. In April 2025 we ran a mixed analysis of variance, one with factors that vary both between and within participants, on the finished records, including some that the registered exclusions later removed, and we saw its mean for each label and stated time. The September 2025 talk reported effects of the label and of stated time on both ratings, and a change in the label's effect with time on effort but not on creativity; it called the label's effect significant at some stated times and not at others. No analysis of the explanations had been run. We wrote the hypotheses and the plan knowing these results.

Study 4. In February 2025 we fitted mixed analyses of variance to these data. The CogSci 2025 paper reported them as its second study, and the September 2025 talk reported them as well. They showed a large effect of modification, no significant main effect of the label, an effect of the hours stated for the original on creativity only, and an effect of the label on essence change that varied with the level of modification. We found no analysis of the explanations. For the registration we read counts, screening answers, image slots and answer codes, and summarized no rating by condition. We wrote the hypotheses knowing the earlier results.

Study 5. Study 5 was run after the CogSci 2025 paper and the talk, and neither reports it. A November 2025 manuscript draft reported a sample size that the registered exclusions do not

reproduce, with age and gender, and no rating result. In 2026, coding sessions in which an AI assistant worked for the first author displayed the exclusion counts and cell sizes (7 September), one participant's three ratings and explanations for one trial (22 September), and a pooled count of answers in the extra category of two six-point items (6 October). No rating had been summarized by condition, and no model of the ratings had been fitted. Study 5 is therefore the only study registered before any of its ratings had been averaged or compared by label. Both of its equivalence bounds were fixed before any Study 5 model was fitted; the bound on editing time is read from Study 3's registered result.

What each status label means

Every result in the paper carries one of the statuses defined below. Supplementary Table 1 gives the status and verdict of each registered test.

Registered confirmatory: one of the small set of tests that a registration names as the study's confirmatory tests. It is judged by the registration's own rule, usually after Holm's correction.[4] That correction raises each p-value in the set, so that the chance of calling any result in the set significant by mistake stays at .05. A p-value is the chance of a result at least this far from no effect if there were no effect. Words such as 'supported', 'holds', 'not supported' and 'inconclusive' are verdicts under that rule.

Registered secondary: an analysis that a registration names but places outside the confirmatory set, such as the robustness checks, the two text checks run in every study, and, in Studies 4 and 5, the label's effect at each level of modification. These are judged without correction unless the registration says otherwise. Study 1's H3 is a registered hypothesis decided outside the confirmatory set by its own rule, a significance test read together with an equivalence test; Supplementary Table 1 labels its status 'own rule'. An equivalence test asks whether an effect is shown to lie inside a stated bound. It passes when the effect's 90% confidence interval, the range of values that the data do not rule out at the 90% level, lies wholly inside the bound, which shows that the effect is small, not that it is zero.[2,3]

Exploratory: an analysis that a registration names as exploratory, or that no registration names. It carries no verdict. Where a registration fixes a correction for an exploratory set, as Study 1's does for its five word-meaning tests, the corrected results are descriptive readings, not verdicts. The analyses that pool Studies 2, 4 and 5, the image check, the breakdown of Study 5's interaction, and the analyses of talk of origin and of how far explanations moved were specified on 9 and 10 October 2026, after every registered result was known; all are exploratory.

Descriptive: a count, a mean, a share or a fitting status, with no test. A *registered description* is a descriptive result that a registration asks us to report.

Two terms describe how a model fitted. A model that did not converge stopped without settling on a solution; it yields no p-value or verdict, and its check is reported as not completed. A singular fit converged but estimated a variance or a correlation at the edge of its possible range, a sign that the data cannot support that many quantities; it keeps its numbers and is reported as singular.

How the images were made

The ten original images were generated with OpenAI's DALL·E, used through ChatGPT. The long Study 1 registration (section 10) and the CogSci 2025 paper both name DALL·E as the generator, and both are right.[1] Eight of the ten original files carry a provenance record: a statement of where the image came from, which the service that made the image wrote into the file and signed digitally, so that a later change to the record would show. These records follow C2PA, an industry standard for such records set by the Coalition for Content Provenance and Authenticity. Each of the eight records was signed by OpenAI. Each names DALL·E as the software that created the image, and its two entries were written by OpenAI's programming

interface and by ChatGPT. Each also classes the image as made by a trained algorithm, the term these records use for content that an AI system generated. They are dated 6 to 8 November 2024, before the first study began. The other two original files carry no provenance record, so for those two the generator rests on our account. The image check (Supplementary Note 4) extracted the images that the surveys displayed, and the original files match them: eight pixel for pixel, among them the two files without a record, and two with only slight differences in pixel values.

We wrote the prompts ourselves; no file records them.

We made the Slight and Dramatic versions of each original by hand, in Adobe Photoshop, an image-editing program that we operated manually. One Dramatic version was instead exported from Dzine, an online image editor that we used only as a manual editor, and the Dzine logo on the exported file was removed before the surveys showed it. No generative or other AI editing tool changed any image. Of the edited files that we kept, seven of the ten Slight files record the program that saved them, and each names Adobe Photoshop; the other three, and all ten Dramatic files, record none. For those files the program rests on our account, as the generator does for the two originals without a provenance record. The None second work was the original's own image file, shown again unchanged. Every original, in both label groups and in every study, was therefore generated by an AI system, so the Human label, which said that Alex made the originals with design and sketching tools, was a cover story.

Corrections to earlier reports

The statements below, made in earlier reports of these studies, are wrong, and this paper does not repeat them. Most come from the CogSci 2025 paper,[1] our proceedings paper that reported Studies 2 and 4 as its Study 1 and Study 2.

The source of the prompts. The CogSci 2025 paper says that the originals were generated from established prompts in a validated set developed by Orwig and colleagues.[12] That is wrong: we wrote the prompts ourselves. The long Study 1 registration (section 10) gives the correct account, that our approach drew on that study but that the prompts were our own.

Power analysis. The CogSci 2025 paper says that the sample size of its first study, this paper's Study 2, was set by a power analysis run in the program G*Power, and that its second study, this paper's Study 4, followed the same criteria. A power analysis calculates how many participants a study needs in order to detect an effect of a given size. No power analysis was run for any of the five studies. The planned sample sizes were the quotas set in each survey, a number of participants for each label group or, in Study 5, for each combination of label and editing time; each study's section gives them. The simulations of what Study 1's design could detect (Supplementary Note 1) were computed with its registered analyses, after its data had been collected, and did not set its sample size.

Pre-registration. The CogSci 2025 paper described its first study, this paper's Study 2, as pre-registered, and it linked to a page that it said held all the pre-registrations. Neither Study 2 nor Study 4 was registered with any registry before its data were collected. Their only planning document was an OSF preregistration template filled in with Google Docs. It was begun on 5 November 2024, before Study 2 ran, last edited on 29 January 2025, while Study 4's data were being collected, and never submitted. Its plan also differed from the registered analyses: it planned analyses of variance rather than mixed models, had no rule excluding participants who said they had not taken part seriously, and for Study 4 planned a measure of perceived effort that the survey does not contain. The Study 4 registration says that the CogSci 2025 submission called Study 4 pre-registered too; the published paper makes the claim of its first study.

The item wording in Study 2. The CogSci 2025 paper quotes the essence-change and creativity items of its first study, this paper's Study 2, with definitions, and says that every item used a five-point scale. In the Study 2 survey only the transformativeness item carried a definition, and two transformativeness items had six answer points (Section 6).

The Slight edits. The CogSci 2025 paper described the Slight versions as adjustments of hue, coloration and contrast, and the registrations of Studies 2, 4 and 5 describe them as adjustments of color and contrast. Of the ten Slight versions, 7 were also cropped or resized (Supplementary Note 4).

Study 4's sample. The CogSci 2025 paper gives the sample of its second study, this paper's Study 4, as 119 participants. That is the count left after the first four registered exclusion steps. The export held 140 records, and those steps removed 7 survey test runs, 8 unfinished records and 6 participants who did not answer that they had taken part seriously; the fourth step, a wrong answer to an arithmetic question, removed nobody. The registered exclusions then remove 2 more records that are not valid completions: one participant rated all ten originals rather than six, and another's record names one original in two of its six image slots. That leaves 117. The CogSci 2025 paper's own degrees of freedom for this study fit the smaller number, which is also the number of participants that the Study 4 registration gives for the February 2025 analyses, all but one of whom are in the registered sample.

Viewing time. The CogSci 2025 paper says that each pair of an original and its second work stayed on screen for at least 8 seconds. In the surveys of Studies 2 and 4, as in Study 5, each such pair stayed on screen for at least 10 seconds; the 8-second minimum applied to an original shown alone.

The law. The CogSci 2025 paper described the US Copyright Office's current policy as treating AI assistance as an 'original sin', and it read its ratings as judgments of the qualities that copyright protection requires. This paper does neither. The Office refuses copyright registration for material whose expressive elements an AI system determined. It does register a person's own creative selection or arrangement of that material, or a modification of it creative enough to meet copyright's standard of originality, with the AI-generated material left out of the claim.[10,11] In this paper, 'original sin' names a hypothesis about lay judgment, that an AI origin passes to whatever is built from it; it is not a legal rule. The ratings are lay judgments of creativity, effort, transformativeness and essence change, not judgments of authorship or of whether a work can be protected.

Reading the CogSci 2025 paper beside this one. Four further points are not errors, but a reader comparing the two papers needs them. First, the names differ: the CogSci 2025 paper's Study 1 is this paper's Study 2, and its Study 2 is this paper's Study 4. Its creator-type factor is the label, and its effort-level factor is the number of hours stated for the original. Second, in Study 2 both papers analyze the same 235 participants, left after the same two exclusions: 31 unfinished records and 9 participants who did not answer that they had taken part seriously. Study 4's sample is corrected above. Third, the analyses differ: the CogSci 2025 paper fitted analyses of variance to each participant's average ratings, while the registered analyses fit mixed models to every rating, with a baseline for each participant and each original, and correct their confirmatory tests together. Fourth, one registered result differs: the CogSci 2025 paper reported that the hours stated for the original changed creativity ratings of the second work. Study 4's registered model does not reproduce that effect ($F(2, 1970.2) = 1.47$, $p = .230$ before correction; Holm $p > .999$). Some exploratory comparisons also differ, and the sections on Studies 2 and 4 set out the two analyses side by side.

One number in a registration is also not used. Study 1's registration gives the effect sizes that the September 2025 talk showed. An effect size here is Cohen's d, the difference between the label groups' mean ratings divided by the spread of participants' mean ratings. The talk's value for creativity matches the analysis reported here (0.66), but its value for effort matches no table that the analysis writes, which gives 1.15. The paper uses the values that the analysis gives.

5 Study 1

Study 1 asked whether a label saying that an AI system made an image lowers how creative, and how effortful, people judge that same image to be. Its 111 participants each rated ten images for creativity and for effort. Each participant was told one of two things about the artist, Alex: that he made the images with design and sketching tools (the Human label) or by prompting an AI system (the AI label). The main text reports the registered tests of Study 1's four hypotheses (H1 to H4). Supplementary Note 1 gives H3 and H4 in full, Supplementary Note 2 the robustness checks and the belief check, and Supplementary Note 7 what the written explanations referred to; Supplementary Table 1 lists each verdict. A registration is a dated record of a study's hypotheses and analysis plan, unrelated to copyright registration. Study 1's was filed on the Open Science Framework (OSF), an online archive for study plans, on 8 October 2026, after its data had been collected and analyzed (Section 4). This section gives the materials, the procedure and the sample, defines the terms its tables use, and presents those tables.

Materials

Every participant saw the same ten images. All ten were generated with OpenAI's DALL·E, an AI image generator, used through ChatGPT and given prompts that we wrote (Section 4), so the Human label was a cover story. Five images show a human or humanoid figure and five do not; the analysis calls this the kind of image. The same ten images were the originals of Studies 2 to 5.

Participants read the label once, at the start. The Human label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex uses his graphic design skills and digital sketching tools.' The AI label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.' The AI label is one sentence longer.

Before each image, the Human group read 'Alex had a new idea for a digital artwork. Alex worked with his graphic design and digital sketching tools, revising as his work as he went, which created a digital artwork.' The AI group read 'Alex had a new idea for a work of digital artwork. Alex developed and input prompts into the AI image generator, revising each prompt as he went, which created a digital artwork.' Both went on: 'Click the arrow to see the digital artwork created. Note that you will be required to stay on the page for at least 8 seconds before proceeding. Please pay close attention to the artwork.' These quotations reproduce the survey exactly, slips included.

Each image was rated twice, on five-point scales scored 1 to 5 by the position of the answer chosen. The creativity item read 'How much creativity was involved in the production of this work? Creativity is defined as the presence of at least a minimal original, intellectual contribution to the creation of the work.' Its answers ran from 'No creativity involved' through minimal, moderate and significant creativity to 'A great deal of creativity involved'. The effort item read 'How much effort was involved in the production of this work?' and carried no definition. Its answers ran from 'No effort involved' through minimal, moderate and significant effort to 'A great deal of effort involved'. One effort item, for the first image with a figure, labeled its top

two answers differently: 'A lot of effort involved' and 'Significant effort involved'. It is scored by position like the others, and a registered check drops it (Supplementary Note 2). After the two ratings of an image, two required text boxes asked 'Please explain your reasoning for your rating on creativity:' and 'Please explain your reasoning for your rating on effort:'. Copying and pasting into these boxes were not blocked.

Procedure

By our account, participants were students in the UCLA Psychology subject pool who took part for course credit. Every analyzed record carries the code that the survey sent, at its end, to the pool's page for granting credit. The consent form, the same in all five studies, also offered hourly pay to people who signed up outside the pool, and no record shows that anyone took that route. After the consent form and the instructions, a task check asked 'What is your task in this experiment?' with five answers, the correct one being 'Viewing images for a set amount of time and answering questions'. An answer was not required. A wrong answer brought the instructions back once, and the participant then went on whatever the second answer.

A randomizer in the survey software, Qualtrics, then assigned each participant to one of the two labels, presenting each equally often. The survey also set a cap of 60 participants per label, but the cap never changed which label anyone saw. The ten images followed in a random order that the survey did not record. For each image, the description above came first and then the image, which stayed on screen for at least 8 seconds. A separate screen, without the image, then asked for the creativity rating and the effort rating, followed by the explanation of each, in that order. Every explanation was therefore written after both ratings of its image. A final block asked 'Did you complete this experiment seriously throughout (without responding randomly)?', with the answers 'No, I didn't' and 'Yes, I did', then whether any part of the procedure had been unclear, and then race, gender, age, native language and formal artistic training. No question asked whether participants noticed or believed the label. The debriefing, the same in all five studies, told participants that 'Current legal standards state that anything produced with artificial intelligence is insufficiently creative, even if you use AI as a base against which you create new work.' That sentence overstates the law. The US Copyright Office refuses copyright registration for material whose expressive elements an AI system determined, but it does register a person's own creative selection or arrangement of that material, or a modification of it creative enough to meet copyright's standard of originality, with the AI-generated material left out of the claim.[10,11] Participants read the debriefing only after their last rating.

Exclusions and sample

The survey export held 119 records. The registered exclusions removed, in order, 2 survey previews (test runs by the researchers), 3 unfinished records, and 3 records with any answer but 'Yes, I did' to the seriousness question, a blank included. The task check removed 0 records, because it let everyone continue. That left 111 participants, 56 under the Human label and 55 under the AI label. Each gave all 20 ratings and 20 explanations, 2,220 of each in all. The data were collected from 25 February to 3 March 2025. Supplementary Table 5 gives the flow, and Supplementary Table 6 gives the removals by label, with Fisher's exact test, a test for counts in small tables, of whether the share removed differed between labels.

The participants' mean age was 20.08 years (standard deviation, SD, a measure of spread: 1.69; range 18 to 29). Of the 111, 93 answered the gender question Female, 17 Male and 1 Other. Supplementary Table 7 gives race and formal artistic training as well. Supplementary Table 8 compares the two label groups on these background measures, as a check that random assignment produced similar groups.

Supplementary Table 5: Study 1: exclusions. The registered exclusion steps in the order they were applied, with the records each step removed and the records left after it. A record is one session in the survey export.

Step	Removed	Records left
Records in the survey export, test runs included	–	119
Survey previews (test runs by the researchers) removed	2	117
Unfinished records removed	3	114
Any answer but 'Yes, I did' to the seriousness question, a blank included, removed	3	111
Wrong answer to the task check removed	0	111

The task check asked 'What is your task in this experiment?'. A wrong answer brought the instructions back once and then let the participant continue, so this step removed no one. The seriousness question asked 'Did you complete this experiment seriously throughout (without responding randomly)?'.

Supplementary Table 6: Study 1: removals by label. The records left once survey previews had been removed, by the label each participant had been shown: those analyzed and those removed at each later step (descriptive).

Records	Human label	AI label	Total
Analyzed	56	55	111
Removed as unfinished	1	2	3
Removed by the seriousness question	2	1	3

Fisher's exact test, a test for counts in small tables, compared the two labels. Removed at any step against analyzed: $p > .999$. The whole table of steps by label: $p > .999$. Every record carried a label.

Supplementary Table 7: Study 1: the analyzed participants. Their ages, their answers to the background questions at the end of the survey, and the label each was shown (descriptive).

Question	Answer	Participants (%)
Age in years	Mean 20.08 (SD 1.69); range 18 to 29	111 (100.0%)
Gender	Female	93 (83.8%)
	Male	17 (15.3%)
	Other	1 (0.9%)
Race	Asian	35 (31.5%)
	White or Caucasian	35 (31.5%)
	Other	22 (19.8%)
	White or Caucasian; Asian	6 (5.4%)
	Black or African American	4 (3.6%)
	White or Caucasian; Black or African American	3 (2.7%)
	American Indian/Native American or Alaska Native	1 (0.9%)
	Asian; Native Hawaiian or Other Pacific Islander	1 (0.9%)
	Black or African American; Other	1 (0.9%)
	Prefer not to say	1 (0.9%)
	White or Caucasian; Asian; Native Hawaiian or Other Pacific Islander	1 (0.9%)
	White or Caucasian; Other	1 (0.9%)
Formal artistic training	No formal training or experience	43 (38.7%)
	A little formal training or experience	41 (36.9%)
	Moderate formal training or experience	21 (18.9%)
	Extensive formal training or experience	5 (4.5%)
	Very extensive formal training or experience	1 (0.9%)
Label shown	Human label	56 (50.5%)
	AI label	55 (49.5%)

SD, standard deviation, a measure of spread. Percentages are of all analyzed participants. Answers are in the survey's own words. The gender question offered Male, Female, Other and Prefer not to say. Participants could choose more than one race; a row that names several answers, separated by semicolons, counts those who chose exactly that combination. Race was self-reported, by choosing one or more of the survey's categories; it is reported to describe the sample and to check that the label groups were alike (Supplementary Table 8), and no analysis compared racial groups. The training question asked for the level of 'formal artistic training or experience, such as painting, drawing, sculpture, ceramics, photography, and/or film-video'.

Supplementary Table 8: Study 1: did the two label groups differ in background? Tests comparing the participants shown the Human label with those shown the AI label, as a check that random assignment produced similar groups (descriptive; the four tests are not corrected).

Background measure	Test	Statistic	p
Age	Welch's t test	$t(92.1) = 1.88$	$p = .064$
Gender (female, male, other)	Fisher's exact test	–	$p = .519$
Formal artistic training (five ordered answers)	Wilcoxon rank-sum test	$W = 1{,}579$	$p = .809$
Race (the four most common answers; all others pooled)	Fisher's exact test	–	$p = .992$

Welch's t test compares two means without assuming that the two groups vary equally; its degrees of freedom are in parentheses. Fisher's exact test is a test for counts in small tables; for race, its p-value was simulated from 10,000 random tables. The Wilcoxon rank-sum test compares two groups' ordered answers by their ranks; W is its statistic. Every analyzed participant enters every test.

Terms used in the tables

The tables below use the paper's own terms, and each table's note defines its abbreviations. A few terms recur. Most tests rest on a linear mixed model, a regression that predicts each rating from the label, the kind of image and the two together, while giving each participant and each image a baseline of its own. A random slope lets an effect also vary from one participant, or one image, to the next; a 'maximal' structure includes every random slope the design allows. A specification curve refits the same quantity under every combination of a set of reasonable analysis choices. An ordinal model treats the five answers as ordered categories rather than as evenly spaced numbers, and it reports its estimates in log-odds, the logarithm of the odds of a higher answer.

The H1 contrast is the label's penalty: the Human-label mean minus the AI-label mean, averaged with equal weight over the two kinds of image. The primary model's AI-label coefficient is a different quantity. It is the AI-minus-Human difference for images without a figure only, so its sign runs the other way.

A p-value is the chance of a result at least this far from no effect if there were no effect. A 95% confidence interval (CI) is the range of values the data do not rule out at the 95% level. A fit converged when it settled on a solution, and it is singular when it estimated a variance or a correlation at the edge of its possible range. File names ending in .csv, such as `study1_specification_curve.csv`, name the full tables that the analysis code writes to Study 1's tables folder.

Results not reported elsewhere

The confirmatory tests and the models behind them. Study 1's registration named four confirmatory tests and corrected them together by Holm's method,[4] which raises each p-value so that the chance of calling any of the four significant by mistake stays at .05. Supplementary Table 9 lists the four with their p-values before and after the correction. Its H4 row is the test of spread between people on creativity alone, and passing it does not mean that H4 passed (Supplementary Note 1). Supplementary Table 10 gives every estimate of the primary model of each rating. Supplementary Table 11 gives H2 in its two registered versions, which agree.

Supplementary Table 9: Study 1: the four confirmatory tests and Holm's correction. The registration named these four as its confirmatory tests and corrected them together by Holm's method (registered confirmatory). All four pass after the correction. H1 and H2 are supported; H4 is not, because it needed more than this one test (note a).

Test	Estimate (SE)	Statistic	p	Holm p
H1, creativity: the label's penalty	0.45 (0.13)	$t(109) = 3.49$	$p < .001$	$p = .001$
H1, effort: the label's penalty	0.92 (0.15)	$t(109) = 6.08$	$p < .001$	$p < .001$
H2: the penalty on effort minus the penalty on creativity	0.47 (0.10)	$F(1, 109) = 23.02$	$p < .001$	$p < .001$
H4, between-person part, creativity: AI-to-Human ratio of the variance between people[a]	2.04	$\chi^2(1) = 4.98$	$p = .026$	$p = .026$

The penalty is the Human-label mean rating minus the AI-label mean rating, averaged with equal weight over images with and without a human figure, in points on the 1-to-5 scale. SE, standard error. Degrees of freedom are in parentheses. Holm's method orders the four p-values from smallest to largest and compares each with a threshold that starts strict and relaxes as tests pass, so that the chance of calling any of the four significant by mistake stays at .05; the Holm p is each p-value adjusted in that way. The H4 row is a likelihood-ratio test, which compares a model in which the variance between people differs by label with one in which it does not. It is shown for creativity, the weaker of the two ratings, and its estimate has no standard error. [a]This row is only the between-person part of H4. H4 as registered also required a smaller variance within people under the AI label, with both ratios' resampled intervals on the predicted side of 1, on both ratings. That held on effort and not on creativity, so H4 was not supported (Supplementary Note 1).

Supplementary Table 10: Study 1: every estimate of the primary model of each rating. The registered model, fitted to each rating separately; its coefficients are descriptive. The registered H1 test is not the AI-label row here, which is the effect for images without a figure only and runs AI minus Human, but the penalty averaged over both kinds of image (Supplementary Table 9).

Term	Estimate (SE)	95% CI	p
Creativity			
Baseline: Human label, images without a figure	3.37 (0.22)	2.88 to 3.86	$p < .001$
AI label, images without a figure (AI minus Human)	−0.41 (0.14)	−0.68 to −0.14	$p = .003$
Images with a figure, Human label (with minus without)	0.35 (0.29)	−0.32 to 1.02	$p = .268$
Label by kind of image (the H3 quantity)	−0.08 (0.09)	−0.27 to 0.11	$p = .393$
Effort			
Baseline: Human label, images without a figure	3.72 (0.18)	3.34 to 4.10	$p < .001$
AI label, images without a figure (AI minus Human)	−0.91 (0.16)	−1.22 to −0.60	$p < .001$
Images with a figure, Human label (with minus without)	0.09 (0.21)	−0.39 to 0.57	$p = .685$
Label by kind of image (the H3 quantity)	−0.02 (0.10)	−0.21 to 0.17	$p = .833$

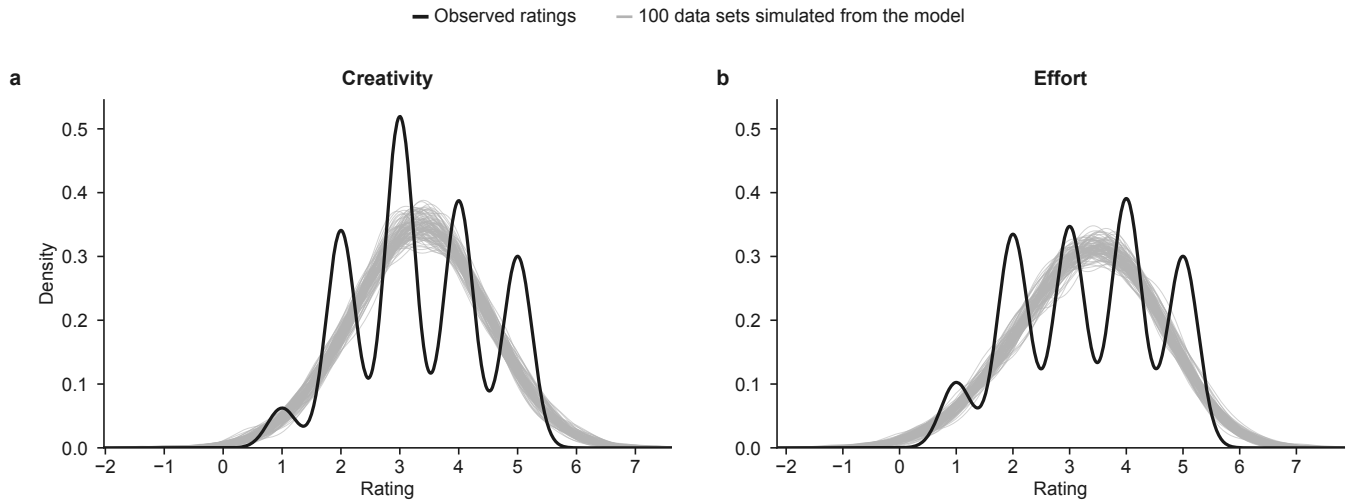
Estimates are in points on the 1-to-5 scale. The model is a linear mixed model, which gives each participant and each image a baseline of its own; the Human label and images without a human figure are its reference levels. The label-by-kind-of-image row is how much the AI label's effect changes when the image has a figure, the quantity H3 tests. SE, standard error; CI, confidence interval. The p-values use Satterthwaite's approximation to the degrees of freedom.

Supplementary Table 11: Study 1's H2 in its two registered versions. H2 predicted that the AI label lowers effort ratings more than creativity ratings. The quantity is the label's penalty on effort minus its penalty on creativity, averaged over the two kinds of image, where a penalty is the Human-label mean minus the AI-label mean. Both versions fit both ratings together, with a baseline and an effort-minus-creativity difference for each participant and a baseline for each image on each rating. The linear model treats the five answers as evenly spaced numbers, so its estimate is in rating points. The ordinal model treats them as ordered categories, so its estimate is on that model's log-odds scale and cannot be compared in size with the first. The registration requires the two to agree in sign and significance; they do, so H2 is supported (registered confirmatory). Neither fit was singular. CI, confidence interval.

Version	Estimate	95% CI	SE	Test statistic	p
Linear mixed model (rating points)	0.47	0.28 to 0.67	0.10	$t(109) = 4.80$	$p < .001$; Holm $p < .001$
Ordinal mixed model (log-odds)	1.22	–	0.26	$z = 4.72$	$p < .001$

H3 and H4. Supplementary Note 1 reports both hypotheses, with their estimates, intervals, power and verdicts. It does not print H4's likelihood-ratio tests. Each compares a model in which one spread differs between the label groups with a model in which it does not, and its statistic is referred to a chi-square (χ^2) distribution with one degree of freedom. For the spread between people, the tests gave $\chi^2(1) = 4.98$ on creativity ($p = .026$) and $\chi^2(1) = 18.15$ on effort ($p < .001$); for the spread within people, $p = .025$ and $p < .001$. H4's verdict does not rest on these tests. It rests on the two ratios of spreads and their resampled intervals, and as registered H4 was not supported (Supplementary Note 1). In simulated samples of this size, the estimated ratio of spreads between people ranged across its middle 95% from 1.12 to 3.99 on creativity and from 2.08 to 7.42 on effort, when the true ratios equal those observed (registered description of what the design can resolve). The analysis tables `study1_h3_equivalence.csv`, `study1_h3_equivalence_power.csv`, `study1_h4_variance_test.csv` and `study1_h4_resolution.csv` give every row.

Robustness of the penalty. Supplementary Note 2 describes these registered checks and their results. Supplementary Table 12 gathers several of them; its specification-curve row summarizes the 36 refits of the specification curve, each of which is in the analysis table `study1_specification_curve.csv`. The table's Bayesian rows report the primary model's AI-label coefficient, not the H1 contrast. The Bayesian refits estimate the model by combining the data with prior assumptions and summarize the result as a distribution. In them, the H1 contrast was 0.44 points on creativity (95% credible interval, the range holding 95% of that distribution, 0.18 to 0.68) and 0.91 on effort (0.61 to 1.21). Supplementary Fig. 5 checks the linear Bayesian refit, which treats ratings as continuous numbers as the primary model does, against the data. The registered maximal structure lets the label's penalty vary from image to image and the effect of the kind of image vary from person to person. With it, the H1 contrast was 0.45 on creativity ($p < .001$) and 0.92 on effort ($p < .001$). Both fits were singular, so these values are reported with that status (analysis table `study1_lmm_fully_maximal.csv`).



Supplementary Figure 5: Study 1: the linear Bayesian refit checked against the data. A posterior predictive check compares ratings simulated from a fitted model with the ratings observed: **a**, creativity; **b**, effort. The horizontal axis is the rating; the height of a curve shows how common each rating value is, smoothed across neighboring values. The black line is the observed ratings; the gray lines are 100 sets of ratings simulated from the saved model, with the draws fixed by a seed. The model treats ratings as continuous numbers, so its simulated ratings are smooth and run past 1 and 5, while the observed ratings take only the five scale values. This mismatch is why the registration also asks for the ordinal model, which treats the answers as ordered categories. The Bayesian refit is a registered secondary check.

Supplementary Table 12: Study 1: robustness checks on the label's penalty. Each row refits or re-tests the H1 contrast, or the related quantity the row names, under another analysis that the registration lists. The status of each row is in parentheses; none of these checks carries a verdict of its own.

Check (status)	Creativity	Effort
The H1 contrast, primary model *(registered confirmatory)*	0.45 (SE 0.13); $t(109) = 3.49$, $p < .001$; Holm $p = .001$	0.92 (SE 0.15); $t(109) = 6.08$, $p < .001$; Holm $p < .001$
Specification curve: the H1 contrast refitted under every combination of the registered analysis choices *(registered secondary; the registration calls it descriptive)*	18 of 18 specifications estimated, 18 recorded as converged; 18 positive and 0 negative; 18 with p below .05; median $p < .001$	18 of 18 specifications estimated, 18 recorded as converged; 18 positive and 0 negative; 18 with p below .05; median $p < .001$
Bayesian refit, linear: the AI-label effect for images without a figure (AI minus Human) *(registered secondary)*	Posterior mean −0.40 (95% credible interval −0.66 to −0.13); 0.01 points from the primary model's estimate	Posterior mean −0.90 (95% credible interval −1.21 to −0.59); 0.01 points from the primary model's estimate
Bayesian refit, ordinal: the same effect, in log-odds *(registered secondary)*	Posterior mean −0.93 (95% credible interval −1.57 to −0.29)	Posterior mean −1.92 (95% credible interval −2.66 to −1.15)
Equivalence at wider bounds: the label-by-kind-of-image interaction, the H3 quantity *(descriptive; no verdict on H3)*	−0.08 (90% CI −0.24 to 0.08); inside ±0.30 ($p = .010$) and inside ±0.50 ($p < .001$)	−0.02 (90% CI −0.18 to 0.14); inside ±0.30 ($p = .002$) and inside ±0.50 ($p < .001$)

The H1 contrast is the Human-label mean rating minus the AI-label mean rating, averaged with equal weight over images with and without a human figure, in points on the 1-to-5 scale. SE, standard error; CI, confidence interval; ±, plus or minus, in rating points. The specification curve crosses the registered choices of random effects, model family, covariates and level of aggregation; its estimates are on each family's own scale, so only their signs are counted. A Bayesian refit estimates the model by combining the data with prior assumptions and summarizes each quantity as a distribution; the credible interval holds the middle 95% of it. The two Bayesian rows report the AI-label coefficient of Supplementary Table 10, not the H1 contrast; the text gives the H1 contrast from the same refits. An ordinal model treats the five answers as ordered categories, and log-odds are the logarithm of the odds of a higher answer. An equivalence test asks whether an effect is shown to lie inside a bound; its p is the larger of two one-sided tests. The last row applies such tests to the H3 quantity at wider bounds than the registered one, as description, and carries no verdict on H3; the registered test of H3, at ±0.20, is in Supplementary Note 1.

Supplementary Table 13 gives the largest and the average change in the penalty when the model is refitted once without each participant and once without each image. Supplementary Table 14 gives H1 on effort and H2 with the differently labeled effort item set aside, and H2 with that image set aside from both ratings. The registration names this check as a sensitivity analysis, settled after the estimates had been read (registered secondary).

Supplementary Table 13: Study 1: does any one participant or image carry the penalty? The primary model was refitted once without each participant and once without each image (registered secondary). The table gives the largest and the average size of the resulting change in the H1 contrast. Every refit kept the penalty above zero.

Left out in turn	Rating	Refits	Usable	Largest change	Average change
Each participant	Creativity	111	111	0.03	0.01
	Effort	111	111	0.04	0.01
Each image	Creativity	10	10	0.04	0.01
	Effort	10	10	0.03	0.01

The H1 contrast is the Human-label mean rating minus the AI-label mean rating, averaged with equal weight over images with and without a human figure. Changes are in points on the 1-to-5 scale, without regard to sign. A usable refit is one that converged without error. The analysis tables `study1_influence_lopo.csv` and `study1_influence_loso.csv` give the estimate from every refit.

Supplementary Table 14: Study 1: the differently worded effort item set aside. One effort item, for the first image with a figure, labeled its top two answers differently from the other nine. The table gives H1 on effort and H2 recomputed without that item's ratings, and H2 without that image on either rating. The registration names this as a sensitivity check, settled after the September 2026 estimates had been read (registered secondary; no verdict).

Ratings left out	Estimate (SE)	95% CI
H1: the penalty on effort		
None	0.92 (0.15)	0.62 to 1.22
The effort rating of the first image with a figure	0.93 (0.15)	0.63 to 1.22
H2: the penalty on effort minus the penalty on creativity		
None	0.47 (0.10)	0.28 to 0.67
The effort rating of the first image with a figure	0.48 (0.10)	0.29 to 0.67
The first image with a figure, on both ratings	0.47 (0.10)	0.27 to 0.66

Estimates are in points on the 1-to-5 scale, averaged with equal weight over images with and without a human figure; the penalty is the Human-label mean minus the AI-label mean. SE, standard error; CI, confidence interval. On that item the top two answers read 'A lot of effort involved' and 'Significant effort involved'; on the other nine, 'Significant effort involved' and 'A great deal of effort involved'. Every item is scored 1 to 5 by the position of the answer chosen.

Supplementary Table 15 gives the registered effect sizes. Cohen's d on participants' mean ratings was 0.66 on creativity (95% CI 0.30 to 1.06) and 1.15 on effort (0.77 to 1.64). Averaging each person's ten ratings first removes much of the variation, so these values cannot be read against the usual small, medium and large benchmarks. The model-based values, 0.41 and 0.80, put the penalty on the scale of a single rating and are the ones to compare with other studies.

Supplementary Table 15: Study 1: effect sizes of the label's penalty. The registered effect sizes (registered secondary; no verdict). Averaging each person's ten ratings first removes much of the variation, so d on participants' mean ratings is larger than the model-based d by construction and cannot be read against the usual small, medium and large benchmarks. The model-based d is the one to compare with other studies.

Rating	d on participants' means	95% CI	Model-based d
Creativity	0.66	0.30 to 1.06	0.41
Effort	1.15	0.77 to 1.64	0.80

Cohen's d here is the Human-label mean minus the AI-label mean, divided by the pooled standard deviation of participants' mean ratings (56 Human-label and 55 AI-label participants). Its 95% confidence interval (CI) comes from 10,000 resamples of participants drawn within each label group. The model-based d divides the H1 contrast by the square root of the primary model's summed variances (between participants, between images, and the rest), which puts it on the scale of a single rating.

The written explanations. Section 10 describes the eight word lists that measured the explanations and how a script applied them without seeing the label or the rating. Every explanation was written after its rating, so none of these results shows why anyone rated as they did. Supplementary Table 16 gives how often each list flagged the Human group's explanations, and Supplementary Table 17 how often the effort explanations of each group mentioned prompting or machines; Supplementary Note 7 reads both. Supplementary Table 18 gives the five registered word-meaning tests, an exploratory set whose corrected results are descriptive readings, not verdicts. Supplementary Table 19 refits them with each test's difference free to vary from person to person, a check that the registration names and that changes no reading: every estimate kept its sign, and concept against prior exposure stayed short of significance.

Supplementary Table 16: Study 1: what the Human-label group's explanations mentioned. The share of that group's explanations of each rating that each word list flagged (descriptive). One explanation can mention several things, so the shares need not sum to 100%.

Word list	Share	95% CI
Effort explanations		
Visible detail in the picture	62.5%	56.4% to 68.8%
Time and industriousness	10.0%	6.6% to 13.8%
Skill and training	5.2%	2.1% to 9.1%
None of these three	32.0%	26.8% to 37.3%
Creativity explanations		
The idea behind the work	26.8%	22.0% to 31.8%
Says the work looks familiar	17.0%	13.0% to 21.6%
Says the work does not look familiar	7.5%	5.0% to 10.2%
Raises familiarity either way	24.5%	18.9% to 30.4%
Visible detail and rendering	12.1%	9.1% to 15.5%
None of the idea, familiarity or detail	47.1%	40.7% to 53.9%

Each share is of the 560 explanations that the 56 Human-label participants wrote for that rating. CI, confidence interval, from 2,000 resamples of participants. A word list is a fixed list of words and phrases, applied by a script that saw neither the label nor the rating (Section 10).

Supplementary Table 17: Study 1: effort explanations that mention prompting or machines, by label (descriptive). Under the AI label such talk restates what participants were told. Under the Human label a mention is not a rejection of the account, since it can compare, deny or wonder.

Label	Share of effort explanations	95% CI
Human label	15.4%	10.4% to 20.2%
AI label	50.5%	40.7% to 60.3%

The share of each group's effort explanations that the prompting word list flagged: any mention of a machine, a prompt or a generated image. CI, confidence interval, from 2,000 resamples of participants. The test of prompting talk on effort against creativity, under the AI label, is in Supplementary Table 18.

Supplementary Table 18: Study 1's five word-meaning tests. Each test asks whether one word list flagged explanations more often than another, in the explanations named, or, in the last row, whether one list flagged effort explanations more often than creativity explanations. Each is a mixed logistic regression, a model for a yes-or-no outcome (whether a list flagged an explanation) with a baseline for each participant and each image, so that one person's twenty explanations do not count as twenty independent observations. The estimate is the difference in the log-odds that the first list flags an explanation rather than the second; a positive value is the predicted direction, and all five are positive. The registration names the five as one exploratory set corrected by the Benjamini–Hochberg method, which limits the expected share of false positives among results called significant; the corrected results are descriptive readings, not verdicts. The detail-against-skill fit converged only when restarted from where its first attempt stopped; the prompting fit was singular, with no variation between images. Supplementary Note 7 gives the shares of explanations behind each test.

Explanations	Comparison	Estimate	SE	Uncorrected	Corrected
Effort, Human label	Visible detail against time	3.08	0.19	$p < .001$	$p < .001$
Effort, Human label	Visible detail against skill	3.87	0.24	$p < .001$	$p < .001$
Creativity, Human label	Concept against visible detail (execution)	1.03	0.16	$p < .001$	$p < .001$
Creativity, Human label	Concept against prior exposure (novelty)	0.13	0.14	$p = .350$	$p = .350$
AI label, effort against creativity	Prompting or machines	1.78	0.18	$p < .001$	$p < .001$

Supplementary Table 19: Study 1: the five word-meaning tests with each person's difference free to vary. The registered model gives each participant a baseline of their own but one difference between the two lists that all participants share; the refit lets that difference vary from person to person. The registration names this refit as a sensitivity check that cannot change a reading (exploratory; its p-values are uncorrected). Every estimate kept its sign.

Explanations	Comparison	Registered	Refit	z	p
Effort, Human label	Visible detail against time	3.08 (0.19)	3.57 (0.37)	9.60	$p < .001$
Effort, Human label	Visible detail against skill	3.87 (0.24)	5.66 (0.95)	5.98	$p < .001$
Creativity, Human label	Concept against visible detail (execution)	1.03 (0.16)	1.03 (0.22)	4.63	$p < .001$
Creativity, Human label	Concept against prior exposure (novelty)	0.13 (0.14)	0.28 (0.21)	1.30	$p = .195$
AI label, effort against creativity	Prompting or machines	1.78 (0.18)	2.11 (0.46)	4.58	$p < .001$

Each cell gives an estimate with its standard error (SE) in parentheses, from the registered model and from the refit. Estimates are differences in log-odds, the logarithm of the odds that the first list flags an explanation rather than the second; a positive value is the predicted direction. z and p test the refit's estimate against zero. Every refit converged and none was singular (a variance or correlation estimated at the edge of its range). Supplementary Table 18 gives the registered tests with their corrected p-values.

The registration also asks whether a participant rated lower, on creativity, the particular images that they called familiar (registered secondary, description only). Within a person, an image whose explanation called it familiar received a creativity rating that differed from that person's other images by −0.56 points under the Human label (95% CI −0.77 to −0.35) and by −0.65 under the AI label (−0.85 to −0.45); a negative value means a lower rating. The registration names effort as the point of comparison; there the values were −0.49 (−1.06 to 0.09) and −0.44 (−0.83 to −0.05). The explanation followed the rating, so these associations do not show that calling an image familiar lowered its rating. The analysis table `study1_text_novelty_within_between.csv` gives every term.

The two checks run in every study (Section 10) are registered secondary. In the first, the words of explanations from participants left out of the fit predicted the rating, with correlations of 0.49 on creativity (95% CI 0.44 to 0.55) and 0.63 on effort (0.57 to 0.68). Both lie above the 95th percentile of the correlations obtained after each person's ratings were shuffled among that person's own explanations (0.16 and 0.39). They also told the label groups apart. The AUC, the chance that a randomly chosen AI-label explanation scores above a randomly chosen Human-label one, so that 0.5 is chance, was 0.59 on creativity (0.53 to 0.64) and 0.72 on effort (0.66 to 0.78). Added to the rating, the words raised the AUC by 0.036 (0.003 to 0.067) and 0.051 (0.020 to 0.083), so by the registered reading the words add to the rating in telling the groups apart. Recovery of the rating was already partly known from earlier exploratory work (Section 4). In the second check, AI-label participants reused their own words across images more than Human-label participants on effort (difference 0.057, 95% CI 0.021 to 0.104; Holm $p = .002$), as predicted, but not detectably on creativity (0.025, −0.003 to 0.059; Holm $p = .092$). The registered reading is 'supported' on effort and 'inconclusive' on creativity, which is not evidence that the groups reuse their words equally. Supplementary Tables 20 and 21 give every row.

Supplementary Table 20: Study 1: can the rating and the label be recovered from the words? The first of the two checks run in every study (registered secondary). Each model was fitted to some participants' explanations and scored on the explanations of participants left out of the fit. By the registered reading, both the rating and the label are recovered on both ratings, and the words add to the rating in telling the two label groups apart.

What is predicted	Statistic	Estimate	95% CI	Shuffled baseline
Creativity explanations				
The rating, from the words	r	0.49	0.44 to 0.55	0.16
The label, from the words	AUC	0.59	0.53 to 0.64	0.51
The label, from the rating	AUC	0.60	0.53 to 0.66	0.51
The label, from the rating and the words	AUC	0.63	0.57 to 0.69	–
What the words add: rating and words, minus rating alone	AUC difference	0.036	0.003 to 0.067	–
Effort explanations				
The rating, from the words	r	0.63	0.57 to 0.68	0.39
The label, from the words	AUC	0.72	0.66 to 0.78	0.50
The label, from the rating	AUC	0.70	0.63 to 0.77	0.53
The label, from the rating and the words	AUC	0.75	0.68 to 0.81	–
What the words add: rating and words, minus rating alone	AUC difference	0.051	0.020 to 0.083	–

r, the correlation between predicted and observed ratings (0, no relation; 1, a perfect one). AUC, the chance that a randomly chosen AI-label explanation is scored as more AI-like than a randomly chosen Human-label one (0.5 is chance; 1, perfect separation). The models are ridge regressions, which shrink their many word weights toward zero, on counts of each explanation's content words. Participants were split into 10 groups, and each group in turn was left out of the fit and scored, so that no participant's explanations were both fitted and scored. CI, confidence interval, from 2,000 resamples of participants. The shuffled baseline is the 95th percentile of the statistic over 200 reruns with the outcome shuffled: the rating among each participant's own explanations, the label across participants. An estimate above it is read as signal. Every explanation was written after its rating, so recovering the rating is not evidence that the words produced it.

Supplementary Table 21: Study 1: how much each person reused their own words, by label. The second of the two checks run in every study (registered secondary). It predicted that participants shown the AI label would reuse their own words more. By the registered reading the prediction is supported on effort and inconclusive on creativity, which is not evidence that the two groups reuse their words equally.

Measure	Human label (95% CI)	AI label (95% CI)
Creativity explanations		
Overlap among a person's own explanations	0.069 (0.055 to 0.083)	0.094 (0.070 to 0.124)
Overlap with others' explanations of the same image	0.027 (0.024 to 0.031)	0.029 (0.026 to 0.033)
Own overlap minus others' overlap	0.041 (0.029 to 0.054)	0.065 (0.041 to 0.094)
Content words per explanation	8.03 (6.80 to 9.40)	9.17 (7.88 to 10.55)
Registered comparison (AI minus Human in own minus others' overlap, at equal length): 0.025 (95% CI −0.003 to 0.059); $p = .092$; Holm $p = .092$; inconclusive.		
Effort explanations		
Overlap among a person's own explanations	0.099 (0.082 to 0.118)	0.157 (0.121 to 0.201)
Overlap with others' explanations of the same image	0.036 (0.032 to 0.041)	0.036 (0.030 to 0.042)
Own overlap minus others' overlap	0.063 (0.046 to 0.081)	0.121 (0.088 to 0.163)
Content words per explanation	7.99 (6.63 to 9.53)	7.89 (6.76 to 9.16)
Registered comparison (AI minus Human in own minus others' overlap, at equal length): 0.057 (95% CI 0.021 to 0.104); $p < .001$; Holm $p = .002$; supported.		

Overlap is the share of content words that two explanations have in common: the words in both divided by the words in either (0, none in common; 1, the same words). A person's own overlap is averaged over all pairs of that person's explanations of one rating; others' overlap pairs each of those explanations with other participants' explanations of the same image and rating. The registered comparison is a regression of own minus others' overlap on the label, holding the mean number of content words constant, because overlap falls as texts get longer. CI, confidence interval, from 2,000 resamples of participants within each label group, with others' overlap rebuilt each time. Holm's correction runs over the two ratings. Supported means that the corrected test rejected no difference, in the predicted direction; inconclusive, that it did not.

The registration names the general description of the explanations as exploratory, and none of it carries a verdict. The tone and concreteness of the words, scored from published norms, did not differ between the label groups on either rating after a Benjamini–Hochberg correction, which limits the expected share of false positives among results called significant (analysis table `study1_text_affect_by_arm.csv`). A topic model, a method that finds groups of words that tend to occur together without being told the label or the rating, chose the number of topics for each rating by a stability rule (Supplementary Table 22). After correction, AI-label explanations gave more weight on both ratings to a topic led by 'ai', with 'created' and 'prompt' among its leading words on creativity and 'generated' and 'generator' on effort. On effort they gave less weight to a topic led by 'details', 'small' and 'many' and to one led by 'looks', 'generated', 'good' and 'real' (analysis table `study1_text_topics_by_arm.csv`). The topics agreed only weakly with groups formed from sentence embeddings, lists of numbers that a pretrained model assigns to each explanation (analysis table `text_topics_embedding_agreement.csv`). A lasso, a regression that sets most word weights to exactly zero, kept few words for telling the label groups apart once the rating was known, led by 'due', 'prompt' and 'ai' (analysis table `study1_text_prediction_weights.csv`).

Supplementary Table 22: Study 1: how the number of topics was chosen. For each number of topics from 3 to 10, how stable the topic model of each rating's explanations was (exploratory). The most stable number was used: 6 for both ratings (in bold).

Number of topics	Creativity: stability (SD)	Effort: stability (SD)
3	0.83 (0.13)	0.86 (0.15)
4	0.89 (0.08)	0.87 (0.14)
5	0.88 (0.07)	0.94 (0.08)
6	**0.92 (0.05)**	**0.98 (0.01)**
7	0.84 (0.07)	0.95 (0.05)
8	0.82 (0.08)	0.91 (0.06)
9	0.84 (0.06)	0.87 (0.06)
10	0.87 (0.07)	0.86 (0.06)

A topic model (here non-negative matrix factorization) finds groups of words that tend to occur together. At each number of topics it was refitted ten times, each time on a random 80% of the participants. The topics of two refits were matched one to one, and their agreement is the mean cosine similarity of the matched topics' word weights (1 when two topics weight the same words in the same proportions, 0 when they share no word). Stability is that agreement averaged over every pair of refits; SD, its standard deviation. A tie at three decimals goes to the smaller number.

6 Study 2

Study 2 asked whether the AI label on an original reaches ratings of a second work that Alex was said to have made from it with his design and sketching tools. It was the first of the five studies to be run, from 23 November to 5 December 2024, and our CogSci 2025 paper,[1] a peer-reviewed proceedings paper, reported it as its Study 1. A registration is a dated record of a study's hypotheses and analysis plan, unrelated to copyright registration. Study 2's was filed on the Open Science Framework (OSF) on 8 October 2026, after its data had been collected and analyzed. Its registered verdicts are in Supplementary Table 1, the label's effect averaged over levels is in Supplementary Table 2, and the main text reports both. Supplementary Note 6 gives its registered test of dilution (H3), and Supplementary Note 7 the analyses of its explanations pooled with those of Studies 4 and 5. This section gives the materials and procedure, the sample, how the registered analysis differs from the earlier one, and the results that no other part of the paper prints.

Materials

Every participant saw the same ten original images. All ten were generated with OpenAI's DALL·E, an AI image generator, so the Human label described below was a cover story (Section 4 gives the evidence for the generator). Five originals show a human or humanoid figure and five do not, and the registered model includes this image kind as a factor. Each original came with three second works: None, the original's own image file shown again unchanged; Slight, described in the registration as adjustments of color and contrast; and Dramatic, substantial changes. We made the Slight and Dramatic versions by hand in image-editing software, and no AI tool changed any image (Section 4). Of the ten Slight versions, 7 were also cropped or resized (Supplementary Note 4), although the CogSci 2025 paper describes the Slight edits only as adjustments of hue, coloration and contrast. Studies 2, 4 and 5 showed the same 30 images.

Participants read the label once, at the start. The Human label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex only uses graphic design software and digital sketching tools.' The AI label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.'

Before each original, the Human group read 'Alex had a new idea for a digital artwork. Alex created a new piece of digital artwork only using graphic design software and digital sketching tools.' The AI group read 'Alex had a new idea for a work of digital artwork. Alex prompted the artificial intelligence (AI) system for a new piece of digital artwork, which created a new digital artwork.' Both went on: 'Click the arrow to see the digital artwork created. Note that you will be required to stay on the page for at least 8 seconds before proceeding.'

Both groups read the same description of each second work. Before the first of the three, it was 'Alex looked at the first artwork, and decided to produced a secondary digital artwork. Using the first work as a base, he produced the second digital artwork with his graphic design software and digital sketching tools.' Before each of the other two, it was 'Later, Alex looked at the very first artwork again. Alex decided to produce a different secondary digital artwork. Using the very first artwork as a base, he produced a different secondary digital artwork with his graphic design software and digital sketching tools.' Each ended 'Click the arrow to see the digital artworks:'. These quotations reproduce the survey exactly, slips included. In both groups, then, Alex made every second work himself; only the account of the original differed. No description said how much had been changed or how long anything took.

Three features of this wording belong to Study 2 alone. Its Human label says that Alex 'only uses graphic design software', where Studies 1, 4 and 5 say that he 'uses his graphic design skills' and Study 3 names the tools. Its AI vignette before each original says nothing about Alex revising his prompts, unlike the other four studies. And its second-work descriptions name Alex's 'graphic design software', where Studies 4 and 5 name 'his own graphic design skills'.

On the screen after each pair of original and second work, which showed neither image, participants gave three ratings on five-point scales, always in the order below, each scored 1 to 5 by the position of the answer chosen. The transformativeness item read 'How transformative is the second work compared to the first work? Transformativeness is the degree the second work adds new expression, meaning, or message to the first work.' Its answers ran from 'Not at all transformative' through slightly, moderately and very transformative to 'Highly transformative'. The essence item read 'How much has the essence of the first work changed in the second work?', with answers from 'No change in essence' through very little, moderate and significant change to 'Complete change in essence'. The creativity item read 'How much creativity was involved in the production of the second work?', with answers from 'No creativity involved' through minimal, moderate and significant creativity to 'A great deal of creativity involved'. Below the three ratings, a required text box for each asked 'Please explain your reasoning for your rating on transformativeness for your responses:' (or on essence, or on creativity). Each explanation was therefore written after its rating had been given, although a rating could still be changed before the participant left the screen. A script blocked copying and pasting in 21 of the 90 explanation boxes and not in the others.

Study 2 is the only study whose essence and creativity items carried no definition. Studies 4 and 5 defined essence as 'the core sense and feeling that one has while observing a work of art that makes that work unique', and the other four studies defined creativity as 'the presence of at least a minimal original, intellectual contribution to the creation of the work'. The CogSci 2025 paper quotes all three items with definitions, but the Study 2 survey file shows a definition on the transformativeness item only. Comparisons of creativity ratings between Study 2 and the other studies are therefore exploratory, as Study 2's registration states. Two transformativeness items, both for one original (at None and at Slight), had six answer points, a reworded question and no definition. As registered, their 470 ratings and explanations were left out. One creativity item asked about 'creating the second work', and three essence items had slips in their answer labels; scoring by answer position is unaffected.

Procedure

Participants took part online in a Qualtrics survey. By our account, participants were students in the UCLA Psychology subject pool who took part for course credit. Every analyzed record carries the code that the survey sent, at its end, to the pool's page for granting credit. The consent form, the same in all five studies, also offered hourly pay to people who signed up outside the pool, and no record shows that anyone took that route. The survey assigned each participant at random to one label for the whole session, presenting each label equally often. It also set a quota of 125 participants for each label group, a planned number that neither stopped anyone nor moved anyone to the other label. Each participant saw all ten originals in random order. For each, the vignette came first, then the original alone for at least 8 seconds, then the three second works. Each second work was introduced by its description and then shown beside the original, under the headings 'FIRST WORK' and 'SECOND WORK', for at least 10 seconds; its ratings and explanations followed on the next screen. The first of the three levels was drawn at random and the other two followed in random order; the order was not recorded. Each participant thus rated 30 second works, apart from 7 whose records lack the Slight trial of one original.

A question about the task ('What is your task in this experiment?') came before the label and allowed one retry, so it ended no session. An arithmetic question ('What does 8+7 equal?') appeared at a varying point among the originals. After the last rating, a closing screen asked first whether the participant had completed the experiment seriously, without responding randomly, then whether any part of the procedure had been unclear, and then about race, gender, age, native language and formal artistic training. No question asked participants to recall the label, a check known as a manipulation check. The debriefing was the same as in the other four studies; Section 5 quotes it and corrects its statement of the law. Participants read it only after their last rating.

Exclusions and sample

The registered exclusion rules, applied in order, left 235 of the 275 records in the survey export. There were no survey test runs. Then 31 unfinished sessions were removed, and 9 participants who gave any answer but 'Yes, I did' to the seriousness question, a blank included. No one gave a wrong answer to the arithmetic question, and every record had a label. Of the 235 participants analyzed, 122 saw the Human label and 113 the AI label. They gave 20,659 analyzed ratings, each with its explanation. The same unfinished and non-serious records were removed in the CogSci 2025 paper, which analyzed the same participants.

The participants' mean age was 19.96 years (standard deviation 1.82; range 18 to 34). Of the 235, 192 described their gender as female, 38 as male, and 5 chose 'Other' or 'Prefer not to say' or gave no answer. The answers on race, native language and formal artistic training are not summarized here. The CogSci 2025 paper gives age and gender figures for this study, but their counts add up to the number of finished sessions, not to the analyzed sample, so they are not reused.

How the registered analysis differs from the CogSci 2025 analysis

The CogSci 2025 paper analyzed these data with mixed analyses of variance (ANOVAs). A mixed ANOVA compares condition means computed from each participant's average rating in each condition; here the label varied between participants and the level within them. Pairwise t tests, comparisons of two means, followed between levels, and the two labels were compared at each level. The registered analysis differs in four ways. First, it fits a linear mixed model to every rating rather than to each participant's averages. This is a regression that gives each participant and each original its own baseline, so that differences between originals are not mistaken for differences between conditions, and it includes image kind as a factor. Second, it

fixes twelve confirmatory tests and corrects them together with Holm's method, which raises each p-value so that the chance of calling any of the twelve significant by mistake stays at .05. A p-value is the chance of a difference at least this large if there were no effect. Third, it asks about the label with an equivalence test: whether the label's effect, averaged over levels, is shown to lie inside ± 0.224 points, half of Study 1's creativity penalty on originals, rather than whether it differs from zero. Fourth, it adds a direct test of dilution (H3; Supplementary Note 6), the account on which the label's effect is largest at None and shrinks as the edit grows, and an ordinal check.

The two analyses agree that ratings rose steeply with the level of modification (Supplementary Table 23b). On the label, the CogSci 2025 paper found a small creativity difference averaged over levels and none at any single level. In the registered model, the ordinary two-sided test, which asks whether that averaged creativity difference is zero, is also significant ($p = .026$), but it is description only. The registered verdict rests on the equivalence test instead, and it reads 'not shown to be small': the test fails after Holm's correction, although its uncorrected interval lies inside the bound (Supplementary Table 2). Because the averaged creativity difference had been reported before registration, this registered test re-analyzes an effect already known. The registered model's comparisons at each level are exploratory and uncorrected (Supplementary Table 23a). Two of the three creativity comparisons fall below .05, where the earlier analysis, after its own correction, found none. They are not evidence of a penalty at any particular level.

The CogSci 2025 paper also calls this experiment pre-registered. It was not registered with any registry before data collection. Its only planning document, a template begun in November 2024 and never submitted, was last edited after collection had ended (Study 2's registration, question 1).

Results not reported elsewhere

Each result below carries its status. A registered confirmatory test is one that the registration names as confirmatory, judged by its rule after Holm's correction; a result is called significant when its p-value is below .05. A registered secondary analysis is one the registration names outside that set. An exploratory analysis was named as exploratory or not named at all, and carries no verdict. A descriptive result, such as a mean, carries no test.

Registered tests and predicted means. Supplementary Table 23 gives the model's predicted mean for each label at each level, that is, the mean rating the model estimates for that condition with the other factors averaged (descriptive); the label difference at each level (exploratory); and each registered test of H1 with its ordinal check. The ordinal check is a cumulative-link mixed model, which treats the five answers as ordered categories rather than evenly spaced numbers. On transformativeness and essence change it reached the same verdict as the linear model on every test it repeated. On creativity its fit did not converge, meaning that the fitting procedure stopped without settling on a solution, so the check is not completed and carries no verdict. All three registered linear models converged, and none was singular, that is, none estimated a source of variation at zero. Supplementary Note 6 gives the registered test of dilution (H3) with its ordinal check, and Supplementary Table 2 the test of H2.

Registered text checks. Two checks on the creativity explanations were registered for every study, and Study 2 ran both (registered secondary; Section 10 describes both). In the first, ridge regression, a regression that shrinks each word's weight toward zero, predicted each rating and each participant's label from the words of explanations written by participants left out of the fit. The words predicted the rating well: predicted and actual ratings correlated at 0.66 (95% confidence interval, CI, the range of values the data do not rule out at the 95% level: 0.63 to 0.68). In runs with the ratings shuffled, 95% of correlations stayed below 0.03. The words did not show the label. The AUC, the chance that a randomly chosen AI-label explanation scores

Supplementary Table 23: Study 2: predicted mean ratings, and the registered tests of modification (H1). Ratings are on 1-to-5 scales. **a**, The linear mixed model's predicted mean for each label at each level (descriptive), and the Human-label mean minus the AI-label mean with its 95% confidence interval (CI) and two-sided test. Study 2 registered no test at each level, so these differences are exploratory and uncorrected. **b**, The registered confirmatory tests of H1, corrected together with H2 and H3 by Holm's method across twelve tests, beside the registered ordinal check. Supplementary Note 6 gives the test of H3 and Supplementary Table 2 the test of H2.

a Predicted means and the label difference

Level	Human label	AI label	Human minus AI	95% CI	Two-sided test
Transformativeness					
None	1.02	1.00	0.02	−0.09 to 0.13	$p = .762$
Slight	1.96	1.88	0.08	−0.03 to 0.19	$p = .157$
Dramatic	3.70	3.62	0.08	−0.03 to 0.18	$p = .175$
Averaged over levels[a]	2.23	2.17	0.06	−0.04 to 0.16	$p = .249$
Essence change					
None	1.06	1.04	0.02	−0.09 to 0.13	$p = .734$
Slight	2.05	1.91	0.14	0.02 to 0.25	$p = .018$
Dramatic	3.69	3.63	0.06	−0.05 to 0.17	$p = .279$
Averaged over levels[a]	2.27	2.19	0.07	−0.03 to 0.17	$p = .161$
Creativity					
None	1.16	1.09	0.07	−0.04 to 0.18	$p = .220$
Slight	1.88	1.76	0.12	0.01 to 0.23	$p = .034$
Dramatic	3.56	3.41	0.15	0.04 to 0.27	$p = .007$
Averaged over levels[a]	2.20	2.08	0.11	0.01 to 0.22	$p = .026$

b Registered confirmatory tests of H1, with the ordinal check

					Ordinal check	
Step	Estimate	95% CI	t (df)	After Holm	Log-odds	After Holm
Transformativeness						
Slight minus None	0.91	0.87 to 0.96	39.49 (6319.7)	$p < .001$	4.95	$p < .001$
Dramatic minus Slight	1.73	1.69 to 1.78	75.38 (6281.9)	$p < .001$	3.81	$p < .001$
Essence change						
Slight minus None	0.93	0.89 to 0.97	43.88 (6,790)	$p < .001$	4.90	$p < .001$
Dramatic minus Slight	1.68	1.64 to 1.73	79.26 (6,790)	$p < .001$	3.75	$p < .001$
Creativity						
Slight minus None	0.70	0.65 to 0.74	32.49 (6,790)	$p < .001$	not completed[b]	
Dramatic minus Slight	1.67	1.63 to 1.71	77.98 (6,790)	$p < .001$		

Predicted means and differences come from the registered model, averaged equally over image kind (originals with and without a human figure). A difference is computed before rounding, so it can differ in its last digit from the difference of the rounded means. df: degrees of freedom by Satterthwaite's approximation. The ordinal check is a cumulative-link mixed model, which treats the five answers as ordered categories. Its estimates are differences in the log-odds of a higher answer, and its Holm family is the nine ordinal p-values with the three equivalence p-values. [a]The ordinary two-sided test of the averaged difference is description only; the registered H2 verdict rests on the equivalence test in Supplementary Table 2. [b]The ordinal model of creativity did not converge, so under the registered rule it has no p-value and no verdict.

above a randomly chosen Human-label one, so that 0.5 is chance, was 0.53 (0.50 to 0.56), an interval that reaches chance; the rating alone gave 0.51. Adding the words to the rating raised the AUC by 0.025 (0.001 to 0.050), which by the registered reading means that the words add to the rating. The second check predicted that AI-label participants would reuse their own words across explanations more than Human-label participants. It measured each person's word overlap among their own explanations, the share of words two explanations have in common, minus their overlap with other people's explanations of the same second work, with explanation length held constant. The difference, AI minus Human, was 0.006 (-0.015 to 0.026; Holm $p = .562$), so the prediction is inconclusive, which is not evidence that the groups reuse words equally. Every explanation was written after its rating, so these checks show what the words track, not why anyone rated as they did.

Exploratory text analyses. The registration names the remaining text analyses as exploratory, and none changes a verdict. Run on the transformativeness explanations, the first check told the label groups apart, though its interval cleared chance only barely; on the essence-change explanations it did not. Word tone and concreteness, scored from published norms, were compared between labels with a Benjamini–Hochberg correction, which limits the expected share of false positives among results called significant. A few small transformativeness terms survived it, among them slightly less pleasant words in AI-label explanations at the average rating; no creativity or essence-change term did. Topics found without labels, groups of words that tend to occur together, showed one transformativeness topic, led by the words 'look', 'identical' and 'exact', with more weight in AI-label explanations after correction, and no creativity or essence-change topic that differed. A lasso, a regression that sets most word weights to exactly zero, kept a single word for telling the labels apart from creativity explanations: 'ai'. Their estimates are in the Study 2 tables that the analysis code writes (main text, Code availability). The analyses pooled over Studies 2, 4 and 5 are in Supplementary Notes 6 and 7 and in Section 11.

7 Study 3

Study 3 asked whether stating how long Alex spent on an original closes the gap between the ratings given under the two labels. Its data were collected from 4 to 9 March 2025. Its registration, a dated record of its hypotheses and analysis plan, was filed on the Open Science Framework (OSF) on 8 October 2026, after an April 2025 analysis of the same data and a September 2025 talk (Supplementary Table 1). Supplementary Note 3 gives every registered test, the gap at each stated time and the predicted means. This section gives the materials, procedure and sample, and the checks and text analyses that Note 3 does not.

Materials

Study 3 used Study 1's ten originals. All ten were generated with OpenAI's DALL·E, an AI image generator (Section 4), so the Human label was a cover story, and five show a human or humanoid figure. Each participant saw nine of the ten, drawn at random, and each was shown alone for at least 8 seconds. There were no second works.

The Human label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex uses only his graphic design skills and digital sketching tools such as iPad apps and Adobe Photoshop.' The AI label, the same in all five studies, read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.'

Before each original, the survey inserted a stated time, 'less than 1 hour', '10 hours' or '100 hours', where *time* appears below. The Human group read 'Alex had a new idea for a digital artwork. Alex spent *time* working only with his graphic design and digital sketching tools, revising as his work as he goes, which creates a digital artwork.' The AI group read 'Alex had a new idea for a work of digital artwork. Alex spent *time* developing and inputting prompts into the AI image generator, revising each prompt as he goes, which creates a digital artwork.' Both went on: 'Click the arrow to see the digital artwork created. Note that you will be required to stay on the page for at least 8 seconds before proceeding.' The quotations reproduce the survey exactly, slips included.

Two features of this wording belong to Study 3 alone. Its Human label is the only one that names the tools. And of the two studies that asked about originals, it is the only one whose vignettes state a time. Under the Human label, the hours were spent drawing; under the AI label, they were spent revising prompts, so there more time also meant more control over the image, and the two cannot be told apart.

On the screen after each original, which no longer showed the image, participants rated creativity and then effort, on five-point scales scored 1 to 5 by the position of the answer chosen. The creativity item read 'How much creativity was involved in the production of this work? Creativity is defined as the presence of at least a minimal original, intellectual contribution to the creation of the work.' Its answers ran from 'No creativity involved' through minimal, moderate and significant creativity to 'A great deal of creativity involved'. The effort item read 'How much effort was involved in the production of this work?', with no definition, and its answers ran from 'No effort involved' through minimal, moderate and significant effort to 'A great deal of effort involved'. Their wording matches Study 1's, except that one of Study 1's effort items had different answer labels; in Study 3 the effort answers were the same for every image. Below the two ratings, a required text box for each asked 'Please explain your reasoning for your rating on creativity:' (or 'on effort:'). Each explanation was therefore written after its rating had been given, although a rating could still be changed before the participant left the screen. No box blocked copying or pasting.

Procedure

Participants took part online in a Qualtrics survey. By our account, participants were students in the UCLA Psychology subject pool who took part for course credit. Every analyzed record carries the code that the survey sent, at its end, to the pool's page for granting credit. The consent form, the same in all five studies, also offered hourly pay to people who signed up outside the pool, and no record shows that anyone took that route. The survey assigned each participant at random to one label for the whole session, presenting each label equally often. It also set a quota of 50 participants for each label group, a planned number that neither stopped anyone nor moved anyone to the other label. A script attached a stated time to each trial, drawing at random among the times not yet used three times, so each participant saw three originals at each time in a random order. On each trial, participants read the vignette with its stated time and saw the original for at least 8 seconds; on the next screen they gave both ratings and then wrote both explanations: 18 ratings and 18 explanations in all.

A question about the task ('What is your task in this experiment?') came before the label and allowed one retry, so it ended no session. An arithmetic question ('What does 8+7 equal?') came after the trials. A closing screen then asked first whether the participant had completed the experiment seriously, without responding randomly, then whether any part of the procedure had been unclear, and then about race, gender, age, native language and formal artistic training. No question asked participants to recall the label (a manipulation check) or whether they believed it. The debriefing was the same as in the other four studies; Section 5 quotes it and corrects its statement of the law. Participants read it only after their last rating.

Exclusions and sample

The registered exclusion rules, applied in order, left 85 of the 100 records in the survey export: 4 survey test runs were removed, then 2 unfinished records, 7 participants who gave any answer but 'Yes, I did' to the seriousness question, a blank included, and 2 who answered the arithmetic question wrongly. Of the 85 participants analyzed, 43 saw the Human label and 42 the AI label. Of their 1,530 explanations, 1,525 came with a scored rating. The few remaining ratings were blank or carried an answer code that the final survey did not offer, and they were treated as missing.

The participants' mean age was 20.16 years (standard deviation 1.48; range 18 to 26). Of the 85, 60 described their gender as female, 23 as male, and 2 chose 'Other' or 'Prefer not to say' or gave no answer. The answers on race, native language and formal artistic training are not summarized here.

Results not reported elsewhere

Each result below carries its status. A registered confirmatory test is one that the registration names as confirmatory, judged by its rule after Holm's correction, which raises each p-value in a set of tests so that the chance of calling any of them significant by mistake stays at .05. A p-value is the chance of a result at least this large if there were no effect, and a result is called significant when p is below .05. A registered secondary analysis is one the registration names outside that set. An exploratory analysis was named as exploratory or not named at all, and carries no verdict.

The ordinal check. The registration named a cumulative-link mixed model as a check on each confirmatory verdict (registered secondary). This model treats the five answers as ordered categories rather than evenly spaced numbers, and it tests each term with a chi-square statistic, corrected by Holm's method over the same six tests as the linear model. On creativity, the label gave $\chi^2 = 7.66$ (Holm $p = .023$), stated time 11.65 (Holm $p = .015$), and their interaction

0.07 (Holm $p = .966$). On effort, the label gave 7.62 (Holm $p = .023$) and stated time 89.01 (Holm $p < .001$); Supplementary Note 3 gives the effort interaction, which came nearest to the threshold. Every verdict matched the linear model's. Both registered linear models and both ordinal models converged, meaning that the fitting procedure settled on a solution, and none of the four was singular, that is, none estimated a source of variation at zero. The registered refit that also lets the time effect vary from person to person and the label's effect vary from image to image was singular on both ratings, so, as registered, its requirement was waived (Supplementary Note 3).

The contrast that set Study 5's bound. Averaged over labels, the creativity mean at 100 hours minus the mean at under 1 hour was 0.219 points before rounding (registered secondary; Supplementary Note 3). Before fitting any of its own models, Study 5 took this value as its bound for stated editing time, ±0.219 points.

Registered text checks. Two checks on the explanations were registered for every study, and Study 3 ran both on creativity and on effort (registered secondary; Section 10 describes both). In the first, ridge regression, a regression that shrinks each word's weight toward zero, predicted each rating and each participant's label from the words of explanations written by participants left out of the fit. The words predicted the rating: predicted and actual ratings correlated at 0.30 on creativity (95% confidence interval, CI, the range of values the data do not rule out at the 95% level: 0.24 to 0.36) and 0.58 on effort (0.52 to 0.63). In runs with the ratings shuffled, 95% of correlations stayed below 0.03 and 0.27 respectively. The label was shown by the effort explanations but not by the creativity explanations. The AUC, the chance that a randomly chosen AI-label explanation scores above a randomly chosen Human-label one, so that 0.5 is chance, was 0.65 on effort (0.58 to 0.72) and 0.51 on creativity (0.45 to 0.57, an interval that reaches chance). The rating alone gave 0.59 and 0.57. Adding the words to the rating changed the AUC by 0.057 on effort (0.000 to 0.110; the lower end lies just below zero before rounding) and −0.012 on creativity (−0.053 to 0.024), so on both the registered reading is inconclusive. The second check predicted that AI-label participants would reuse their own words across explanations more than Human-label participants, measured as a person's word overlap among their own explanations minus their overlap with other people's explanations of the same image, with explanation length held constant, and Holm's correction ran across the two ratings. The difference, AI minus Human, was 0.009 on creativity (−0.037 to 0.053; Holm $p > .999$) and −0.003 on effort (−0.043 to 0.036; Holm $p > .999$), so the prediction is inconclusive on both, which is not evidence that the groups reuse words equally. Every explanation was written after its rating, so these checks show what the words track, not why anyone rated as they did.

Exploratory text analyses. Supplementary Note 7 gives the word-list comparisons between labels. Compared across stated times instead, without correction, effort explanations at 100 hours mentioned time more often, and prompting and machines less often, than at under 1 hour, and creativity explanations at 100 hours mentioned visible detail more often. Participants often restated the stated time, so a mention of time may echo the vignette. Word tone and concreteness, scored from published norms, did not differ by label after a Benjamini–Hochberg correction, which limits the expected share of false positives among results called significant. Two topics found without labels, groups of words that tend to occur together, leaned toward the AI label before correction: a creativity topic led by 'ai', 'created' and 'generated', and an effort topic led by 'hours', 'spent' and 'prompt'. Neither survived correction. The words that carried the prediction of the rating mostly echoed the answer labels, such as 'minimal' and 'great'. Their estimates are in the Study 3 tables that the analysis code writes (main text, Code availability).

8 Study 4

Study 4 asked whether two things said about an original, its label and the hours Alex spent on it, carry over to ratings of the second works he was said to have made from it with his design and sketching tools. Its data were collected from 27 January to 2 February 2025, and our CogSci 2025 paper[1] reported it as its Study 2. Its registration, a dated record of its hypotheses and analysis plan, was filed on the Open Science Framework (OSF) on 8 October 2026, after analyses of the same data had been reported in that paper and in a September 2025 talk (Supplementary Table 1). The main text and Supplementary Table 2 report the label's effect, and Supplementary Note 6 discusses its registered tests at each level. This section gives the materials, procedure and sample, explains why the registered sample is smaller than the one the CogSci 2025 paper gives, sets out how the two analyses differ, and reports the results that no other part of the paper prints.

Materials

Study 4 used the 30 images of Study 2: ten originals, all generated with OpenAI's DALL·E, an AI image generator, each with a Slight and a Dramatic version that we made by hand and a None second work, the original shown again unchanged (Section 4; Supplementary Note 4). The Human label was therefore a cover story. Each participant saw six of the ten originals, three with a human or humanoid figure and three without, each with a stated time and each followed by its three second works.

The Human label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex uses his graphic design skills and digital sketching tools.' The AI label, the same in all five studies, read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.'

Before each original, the survey inserted a stated time, 'less than 1 hour', '10 hours' or '100 hours', where *time* appears below. The Human group read 'Alex had a new idea for a digital artwork. Alex spent *time* working with his graphic design and digital sketching tools, revising as his work as he goes, which creates a digital artwork.' The AI group read 'Alex had a new idea for a work of digital artwork. Alex spent *time* developing and inputting prompts into the AI image generator, revising each prompt as he goes, which creates a digital artwork.' Both went on: 'Click the arrow to see the digital artwork created. Note that you will be required to stay on the page for at least 8 seconds before proceeding.'

Both groups read the same description of each second work. Before the first of the three, it was 'Alex looked at the first artwork. He decided to produced a secondary digital artwork. Using the first artwork as a base, he produced a second digital artwork with his own graphic design skills and digital sketching tools.' Before each of the other two, it was 'Later, Alex looked at the very first artwork again. Alex decided to produce a different secondary digital artwork. Using the very first artwork as a base, he produced a different secondary digital artwork with his own graphic design skills and digital sketching tools.' Each ended 'Click the arrow to see the digital artworks:'. The quotations reproduce the survey exactly, slips included. No time was stated for the second work.

The three ratings were those of Study 2, with two changes: the essence and creativity items now carried definitions. The essence item read 'How much has the essence of the first work changed in the second work? Essence refers to the core sense and feeling that one has while observing a work of art that makes that work unique.' The creativity item read 'How much creativity was involved in the production of the second work? Creativity is defined as the presence of at least a minimal original, intellectual contribution to the creation of the work.' The transformativeness item and every answer scale were as in Study 2, scored 1 to 5 by answer position. The rating screen was also laid out as in Study 2, with the three ratings first and a

required explanation box for each below them, and a script again blocked copying and pasting in 21 of the survey's 90 explanation boxes and not in the others. Two transformativeness items, both for one original, again had six answer points; in Study 4 they kept the usual question and definition. As registered, their 140 ratings were left out of every model. The slips in three essence answer labels and one creativity question recurred; scoring by answer position is unaffected.

Study 4 thus differs from Study 2 in its design and in its wording. In design, it showed six originals per person rather than ten, each with a stated time. In wording, its essence and creativity items carried definitions, its Human label and vignettes followed Study 1's, its AI vignette mentioned revised prompts, and its second works were said to be made with Alex's 'own graphic design skills'.

Procedure

Participants took part online in a Qualtrics survey. By our account, participants were students in the UCLA Psychology subject pool who took part for course credit. Every analyzed record carries the code that the survey sent, at its end, to the pool's page for granting credit. The consent form, the same in all five studies, also offered hourly pay to people who signed up outside the pool, and no record shows that anyone took that route. The survey assigned each participant at random to one label for the whole session, presenting each label equally often. It also set a quota of 62 participants for each label group, a planned number that neither stopped anyone nor moved anyone to the other label. A script attached a stated time to each original, drawing at random among the times not yet used twice, so each participant saw two originals at each time in a random order. Each original was shown alone for at least 8 seconds, and each second work beside it for at least 10 seconds. The order of the three levels was random and was not recorded. Each participant rated 18 second works, giving 54 ratings and 54 explanations.

The task question, the arithmetic question (after the trials) and the closing screen, which asked the seriousness question before the background questions, were as in Study 2. No question asked participants to recall the label (a manipulation check). The debriefing was the same as in the other four studies; Section 5 quotes it and corrects its statement of the law. Participants read it only after their last rating.

Exclusions and sample

The registered exclusion rules, applied in order, left 117 of the 140 records in the survey export. First, 7 preview records, which are test runs, were removed. Then 8 unfinished records were removed, and 6 participants who gave any answer but 'Yes, I did' to the seriousness question, a blank included. No one answered the arithmetic question wrongly. Last, the registration defines a valid completion as one that rates exactly six originals, each named once in the survey's six image slots, two at each stated time. Of the remaining records, 2 failed this rule: one had rated all ten originals, and one named the same original in two slots, so its stated times were not split two, two and two. Of the 117 participants analyzed, 55 saw the Human label and 62 the AI label. They gave 6,178 analyzed ratings and 6,318 explanations.

The CogSci 2025 paper gives this study's sample as 119 participants. Its count before the seriousness question, and the number it removed at that question, both match the pipeline's, so its reported sample is the count left before the validity rule, and the difference from the registered sample is the 2 records that the rule removes. The paper starts from a smaller count of recruited participants than the export's records that are not test runs, and it removes correspondingly fewer unfinished sessions. Study 4's registration adds that the analyses of variance behind the CogSci 2025 paper were fitted to as many participants as the registered sample holds, all but one of them in that sample. The degrees of freedom that the CogSci 2025 paper gives for this study fit that number, not the sample it states.

The participants' mean age was 20.52 years (standard deviation 3.07; range 18 to 42). Of the 117, 96 described their gender as female, 20 as male, and 1 chose 'Other' or 'Prefer not to say' or gave no answer. The answers on race, native language and formal artistic training are not summarized here. For this study, the CogSci 2025 paper gives age and gender figures for the participants it recruited, not for the analyzed sample.

How the registered analysis differs from the CogSci 2025 analysis

The CogSci 2025 paper analyzed these data with mixed analyses of variance (ANOVAs). A mixed ANOVA compares condition means computed from each participant's average rating in each condition; here the label varied between participants, and the stated time and the level within them. The registered analysis instead fits a linear mixed model to every rating, a regression that gives each participant and each original its own baseline. It fixes fifteen confirmatory tests, five per measure: two steps of modification (H1), an equivalence test of the label (H2) and two F tests of the stated time (H3). It corrects the fifteen together with Holm's method, which raises each p-value so that the chance of calling any of them significant by mistake stays at .05. A p-value is the chance of a result at least this large if there were no effect. An equivalence test asks whether an effect is shown to lie inside a stated bound, here ± 0.224 points, half of Study 1's creativity penalty on originals. An F test asks whether a factor changes the mean rating.

The two analyses agree that ratings rose steeply with the level and that the label's overall effect was not significant; in the registered model that F test lies outside the registered set and is context only (Supplementary Table 2). They part on the hours stated for the original. The CogSci 2025 paper found an effect of those hours on creativity, with the second works of 100-hour originals rated lowest. In the registered model the F test of stated hours on creativity was not significant (uncorrected $p = .230$, Holm $p > .999$), the ordinal check agreed, and the predicted creativity mean at 100 hours minus the mean at under 1 hour was 0.03 points, inside the bound (registered secondary; Supplementary Table 24c). The analyses also differ in what they hold constant. Each participant saw a random six of the ten originals, each at a randomly assigned time, so a participant's average at one stated time also reflects which originals happened to receive it. A baseline for each original, as in the registered model, removes that source of variation; an ANOVA on participants' averages does not. Whether this difference accounts for the different results was not tested.

The CogSci 2025 paper also found that the label's effect on essence change differed by level, with a difference at Slight only. The registered secondary tests at each level likewise found differences favoring the Human label at Slight, on transformativeness and essence change, without correction (Supplementary Table 2). Outside the registered set of tests, the model also has a term that lets the label's effect differ by level. Examined as an exploratory, uncorrected test, it was significant on transformativeness ($F(2, 1823.5) = 5.93$, $p = .003$) and essence change ($F(2, 1963.6) = 7.37$, $p < .001$), and not on creativity ($p = .171$).

Study 4 was not registered with any registry before data collection. Its only planning document was a form that was never submitted and was last edited during collection. That form planned analyses of variance and a measure of perceived effort that the survey did not contain, and it predicted that stated time would raise ratings most under the AI label. These data do not show that pattern: the registered secondary test of whether the stated hours changed the label's effect was not significant on any measure (Supplementary Table 24c).

Results not reported elsewhere

Each result below carries its status. A registered confirmatory test is one that the registration names as confirmatory, judged by its rule after Holm's correction; a result is called significant when its p-value is below .05. A registered secondary analysis is one the registration names

outside that set, judged without correction. An exploratory analysis was named as exploratory or not named at all, and carries no verdict.

Registered tests and predicted means. Supplementary Table 24 gives every registered test of H1 and H3 with its ordinal check, the registered secondary tests of the stated hours, and the predicted means. The ordinal check is a cumulative-link mixed model, which treats the five answers as ordered categories rather than evenly spaced numbers; it reached the same verdict as the linear model on all twelve tests it repeated. All three registered linear models and all three ordinal models converged, meaning that the fitting procedure settled on a solution, and none was singular, that is, none estimated a source of variation at zero.

Participants who wrote that a machine made the work. A fixed word list, run without access to labels or ratings, flagged explanations stating that a machine made the work. Of the 55 Human-label participants, 7 wrote this at least once, in 12 of their 2,970 explanations. This is a count of statements, not of disbelief. Without them, the registered secondary refit used 110 participants and changed 1 of the 15 registered verdicts, the essence-change equivalence test (main text). Every refit converged.

Registered text checks. Two checks on the creativity explanations were registered for every study, and Study 4 ran both (registered secondary; Section 10 describes both). In the first, ridge regression, a regression that shrinks each word's weight toward zero, predicted each rating and each participant's label from the words of explanations written by participants left out of the fit. The words predicted the rating: predicted and actual ratings correlated at 0.58 (95% confidence interval, CI, the range of values the data do not rule out at the 95% level: 0.53 to 0.63), and in runs with the ratings shuffled, 95% of correlations stayed below 0.01. The words did not show the label. The AUC, the chance that a randomly chosen AI-label explanation scores above a randomly chosen Human-label one, so that 0.5 is chance, was 0.55 (0.50 to 0.60), an interval that reaches chance; the rating alone gave 0.51. Adding the words to the rating changed the AUC by 0.032 (-0.016 to 0.081), which is inconclusive. The second check predicted that AI-label participants would reuse their own words across explanations more than Human-label participants, measured as a person's word overlap among their own explanations minus their overlap with other people's explanations of the same second work, with explanation length held constant. The difference, AI minus Human, was 0.004 (-0.019 to 0.026; Holm $p = .765$), so the prediction is inconclusive, which is not evidence that the groups reuse words equally. Every explanation was written after its rating, so these checks show what the words track, not why anyone rated as they did.

Exploratory text analyses. The registration names the remaining text analyses as exploratory. No topic found without labels, a group of words that tend to occur together, carried more weight under one label than the other after a Benjamini–Hochberg correction, which limits the expected share of false positives among results called significant. Word tone and concreteness, scored from published norms, did not differ by label after the same correction. In predicting the rating, words such as 'completely', 'different' and 'style' went with higher ratings and words such as 'exact', 'identical' and 'copy' with lower ones, on all three measures. Their estimates are in the Study 4 tables that the analysis code writes (main text, Code availability). Section 11 reports the label's effect at each stated time and level.

Supplementary Table 24: Study 4: the registered tests of modification (H1) and of the hours stated for the original (H3), and the predicted means. Ratings are on 1-to-5 scales. **a**, H1, registered confirmatory: the difference between predicted means at adjacent levels, averaged over label and stated hours. **b**, H3, registered confirmatory: F tests of whether the hours stated for the original, alone or by level, change the mean rating. Panels a and b are corrected together with H2 by Holm's method across fifteen tests. **c**, Registered secondary tests, each uncorrected: whether the stated hours change the label's effect, and 100 hours minus under 1 hour, read with an equivalence test against ±0.224 points. **d**, Predicted means (descriptive).

a H1: ratings rise with the level

					Ordinal check	
Step	Estimate	SE	t (df)	After Holm	Log-odds	After Holm
Transformativeness						
Slight minus None	0.99	0.04	23.15 (1822.8)	$p < .001$	4.89	$p < .001$
Dramatic minus Slight	1.43	0.04	33.77 (1814.1)	$p < .001$	3.18	$p < .001$
Essence change						
Slight minus None	0.99	0.04	25.64 (1963.6)	$p < .001$	4.69	$p < .001$
Dramatic minus Slight	1.56	0.04	40.51 (1963.6)	$p < .001$	3.51	$p < .001$
Creativity						
Slight minus None	0.71	0.04	18.22 (1962.7)	$p < .001$	3.08	$p < .001$
Dramatic minus Slight	1.41	0.04	36.43 (1962.7)	$p < .001$	3.33	$p < .001$

b H3: the hours stated for the original

				Ordinal check	
Term	F (df)	Uncorrected	After Holm	χ^2 (df)	After Holm
Transformativeness					
Stated hours	0.17 (2, 1836.1)	$p = .843$	$p > .999$	0.33 (2)	$p > .999$
Stated hours by level	0.97 (4, 1825.2)	$p = .423$	$p > .999$	2.63 (4)	$p > .999$
Essence change					
Stated hours	0.04 (2, 1969.4)	$p = .960$	$p > .999$	0.81 (2)	$p > .999$
Stated hours by level	1.41 (4, 1963.6)	$p = .229$	$p > .999$	5.12 (4)	$p > .999$
Creativity					
Stated hours	1.47 (2, 1970.2)	$p = .230$	$p > .999$	0.81 (2)	$p > .999$
Stated hours by level	0.38 (4, 1962.7)	$p = .825$	$p > .999$	0.79 (4)	$p > .999$

c Registered secondary tests of the hours stated for the original

	Label by stated hours		100 hours minus under 1 hour		
Measure	F	Test	Estimate	90% CI	Equivalence test
Transformativeness	0.68	$p = .509$	−0.02	−0.09 to 0.05	$p < .001$
Essence change	1.24	$p = .289$	0.01	−0.05 to 0.07	$p < .001$
Creativity	0.37	$p = .691$	0.03	−0.03 to 0.10	$p < .001$

d Predicted means

	By level			By label	
Measure	None	Slight	Dramatic	Human label	AI label
Transformativeness	1.03	2.02	3.45	2.19	2.15
Essence change	1.06	2.05	3.61	2.25	2.23
Creativity	1.22	1.93	3.34	2.11	2.22

Estimates come from the registered linear mixed model, which gives each participant and each original its own baseline. SE: standard error. df: degrees of freedom by Satterthwaite's approximation. The ordinal check is a cumulative-link mixed model, which treats the five answers as ordered categories; it repeats the H1 and H3 tests as Wald tests and, having no rating-point scale, does not repeat the equivalence tests. Its H1 estimates are differences in the log-odds of a higher answer. The label-by-hours F tests have two numerator degrees of freedom. Panel d averages each level over label and stated hours, and each label over level and stated hours. Supplementary Table 2 gives the label's effect averaged over levels and at each level.

9 Study 5

Study 5 asked whether the hours Alex was said to have spent making a second work change its ratings, or change the AI label's effect on them. Its data were collected from 21 October to 2 November 2025, after the CogSci 2025 paper and the September 2025 talk, neither of which reports it. Its registration, a dated record of its hypotheses and analysis plan, was filed on the Open Science Framework (OSF) on 8 October 2026. That was after data collection but before any rating had been averaged or compared by label (Supplementary Table 1). The main text reports the registered tests; Supplementary Note 5 and Supplementary Fig. 2 break down the one confirmatory test that was significant after Holm's correction; and Supplementary Table 2 gives the label's effect at each level. This section gives the materials, procedure and sample, and the results that no other part of the paper prints.

Materials

Study 5 used the 30 images of Studies 2 and 4: ten originals, all generated with OpenAI's DALL·E, an AI image generator, each with a Slight and a Dramatic version that we made by hand and a None second work, the original shown again unchanged (Section 4; Supplementary Note 4). The Human label was therefore a cover story. Each participant saw six of the ten originals, three with a human or humanoid figure and three without, each followed by its three second works.

The labels and the vignette before each original were worded as in Study 4. The Human label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex uses his graphic design skills and digital sketching tools.' The AI label read: 'There is a digital artist named Alex. Alex is a skilled artist who creates digital artworks. Alex develops and inputs prompts into an artificial intelligence (AI) system. The AI system generates digital artworks based on those prompts.' Before each original, the Human group read 'Alex had a new idea for a digital artwork. Alex spent *time* working with his graphic design and digital sketching tools, revising as his work as he goes, which creates a digital artwork.' The AI group read 'Alex had a new idea for a work of digital artwork. Alex spent *time* developing and inputting prompts into the AI image generator, revising each prompt as he goes, which creates a digital artwork.' Here *time* was 'less than 1 hour', '10 hours' or '100 hours', two originals at each. The survey showed this time on every trial but did not record it, because the field that should have stored it had been renamed in the survey. It therefore enters no analysis.

The description of each second work now stated the hours Alex spent making it. Before the first of the three, both groups read 'Alex looked at the first artwork. He decided to produced a secondary digital artwork. Using the first artwork as a base, he spent *editing time* producing a second digital artwork with his own graphic design skills and digital sketching tools.' Before each of the other two, they read 'Later, Alex looked at the very first artwork again. Alex decided to produce a different secondary digital artwork. Using the very first artwork as a base, he spent *editing time* producing a different secondary digital artwork with his own graphic design skills and digital sketching tools.' Each ended 'Click the arrow to see the digital artworks:'. The editing time was 'less than 1 hour', '10 hours' or '100 hours', and it was the same for a given participant throughout. Because the sentence was the same at every level, at None it said that Alex had spent the stated hours on an unchanged copy (Supplementary Note 5). The quotations reproduce the survey exactly, slips included.

The three ratings, their definitions, their answer scales, the layout of the rating screen and the required explanations were as in Study 4, and a script again blocked copying and pasting in 21 of the survey's 90 explanation boxes and not in the others. Two transformativeness items, both for one original, had six answer points. As registered, their 274 ratings were left out of the primary models and entered only the rescaled analysis described below. The slips in three essence answer labels and one creativity question recurred; scoring by answer position is

unaffected.

Study 5 thus differs from Study 4 in two ways. A stated editing time was added, and it varied between participants, whereas Study 4's stated time for the original varied within them. And the stated time for the original, though shown, was not recorded.

Procedure

Participants took part online in a Qualtrics survey. By our account, participants were students in the UCLA Psychology subject pool who took part for course credit. Every analyzed record carries the code that the survey sent, at its end, to the pool's page for granting credit. The consent form, the same in all five studies, also offered hourly pay to people who signed up outside the pool, and no record shows that anyone took that route. The survey assigned each participant at random to one of the six combinations of label and editing time for the whole session, presenting each combination equally often. It also set a quota of 40 participants for each combination, and it never moved anyone from one combination to another. Each original was shown alone for at least 8 seconds, and each second work beside it for at least 10 seconds. The order of the three levels was random and was not recorded. Each participant rated 18 second works, giving 54 ratings and 54 explanations.

The task question, the arithmetic question (after the trials) and the closing screen, which asked the seriousness question before the background questions, were as in Studies 2 to 4. No question asked participants to recall the label (a manipulation check). The debriefing was the same as in the other four studies; Section 5 quotes it and corrects its statement of the law. Participants read it only after their last rating.

Exclusions and sample

The registered exclusion rules, applied in order, left 223 of the 259 records in the survey export. There were no survey test runs. Then 17 unfinished records were removed, 16 participants who gave any answer but 'Yes, I did' to the seriousness question, a blank included, 1 who answered the arithmetic question wrongly, and 2 records with no recorded condition, which held no ratings. Of the 223 participants analyzed, 112 saw the Human label and 111 the AI label; Supplementary Note 5 gives the six group sizes. They gave 12,042 ratings, each with its explanation, of which 11,768 entered the primary models once the six-point items were left out.

The participants' mean age was 20.22 years (standard deviation 2.47; range 18 to 45). Of the 223, 179 described their gender as female, 37 as male, and 7 chose 'Other' or 'Prefer not to say' or gave no answer. The answers on race, native language and formal artistic training are not summarized here.

Study 5 had its own listing in the subject pool, separate from the one that Studies 1 to 4 shared, and some of its participants had taken part in one earlier study. Across the five studies, the 771 analyzed records come from 747 people. Of these, 24 took part in two studies, in each case Study 5 and one earlier study, so they had already seen some of the same originals, and read the same debriefing, before. No registered analysis removes them, and no analysis checks whether their first session changed their ratings. The pooled analyses of Supplementary Notes 6 and 7 and Section 11 count a person who took part in Study 5 and in Study 2 or 4 as two participants.

Results not reported elsewhere

Each result below carries its status. A registered confirmatory test is one that the registration names as confirmatory, judged by its rule after Holm's correction, which raises each p-value in a set of tests so that the chance of calling any of them significant by mistake stays at .05. A p-value is the chance of a result at least this large if there were no effect, and a result is called

significant when p is below .05. A registered secondary analysis is one the registration names outside that set. An exploratory analysis was named as exploratory or not named at all, and carries no verdict.

Registered tests and predicted means. Supplementary Table 25 gives all six registered confirmatory tests, the registered secondary equivalence tests of editing time on all three measures, and the predicted means. An F test asks whether a factor changes the mean rating. An equivalence test asks whether an effect is shown to lie inside a stated bound; a pass means that the effect is small, not that it is zero. Two registered checks repeated the six confirmatory tests. The first was a cumulative-link mixed model, which treats the five answers as ordered categories rather than evenly spaced numbers; it reached the same verdict as the linear model on all six (Supplementary Note 5 gives its creativity interaction). The second rescaled every item from 0 to 1, so that the two six-point items could be included, and it too gave the same six verdicts. All registered linear, ordinal and rescaled models converged, meaning that the fitting procedure settled on a solution, and none was singular, that is, none estimated a source of variation at zero.

The label's penalty at each editing time, on every measure. Supplementary Note 5 explains the exploratory breakdown of the creativity interaction. Supplementary Table 26 extends it to transformativeness and essence change, whose registered interactions were not significant. On both, as on creativity, the largest penalties at Slight and Dramatic fell in the 10-hour groups. Because editing time varied between participants, each column of that table compares separate groups, and one unusual 10-hour group could produce this pattern on all three measures.

Other terms of the registered model. The registered model contains terms that the registration names no test of. Examined as exploratory, uncorrected tests, the term that lets the label's effect differ by level was significant on transformativeness and essence change, the label's overall effect was significant on creativity, and the term that lets the effect of editing time differ by level fell just under .05 on creativity. The registration's own test of the label is the set of equivalence tests at each level in Supplementary Table 2, not this overall term. Supplementary Note 5 reports the three-way term on creativity. Their statistics, apart from the three-way term, are in the Study 5 tables that the analysis code writes (main text, Code availability).

Participants who wrote that a machine made the work. A fixed word list, run without access to labels or ratings, flagged explanations stating that a machine made the work (exploratory count). Of the 112 Human-label participants, 18 wrote this at least once, in 38 of their 6,048 explanations. This is a count of statements, not of disbelief, and Study 5 registered no refit without these participants.

Registered text checks. Two checks on the creativity explanations were registered for every study, and Study 5 ran both (registered secondary; Section 10 describes both). In the first, ridge regression, a regression that shrinks each word's weight toward zero, predicted each rating and each participant's label from the words of explanations written by participants left out of the fit. The words predicted the rating: predicted and actual ratings correlated at 0.60 (95% confidence interval, CI, the range of values the data do not rule out at the 95% level: 0.57 to 0.63), and in runs with the ratings shuffled, 95% of correlations stayed below 0.04. The words showed the label, barely. The AUC, the chance that a randomly chosen AI-label explanation scores above a randomly chosen Human-label one, so that 0.5 is chance, was 0.54 (0.50 to 0.57), with a lower end just above chance before rounding; the rating alone gave 0.53. Adding the words to the rating changed the AUC by 0.013 (-0.011 to 0.036), which is inconclusive. The second check predicted that AI-label participants would reuse their own words across explanations more than

Supplementary Table 25: Study 5: the registered tests of stated editing time, and the predicted means. Ratings are on 1-to-5 scales. **a**, The six registered confirmatory F tests, corrected together by Holm's method, beside the registered second analysis with every item rescaled from 0 to 1. **b**, The registered secondary equivalence tests of editing time, each uncorrected: 100 hours minus under 1 hour, averaged over label and level, against a bound of ±0.219 points, the size of Study 3's creativity time effect. **c**, Predicted means (descriptive), each averaged over the factors not named.

a Registered confirmatory tests

				Rescaled analysis	
Term	F (df)	Uncorrected	After Holm	F	After Holm
Transformativeness					
Editing time	0.66 (2, 216.6)	$p = .520$	$p > .999$	0.68	$p > .999$
Label by editing time	1.49 (2, 216.6)	$p = .227$	$p > .999$	1.61	$p > .999$
Essence change					
Editing time	0.54 (2, 216.9)	$p = .586$	$p > .999$	0.54	$p > .999$
Label by editing time	1.61 (2, 216.9)	$p = .202$	$p > .999$	1.61	$p > .999$
Creativity					
Editing time	0.21 (2, 216.8)	$p = .807$	$p > .999$	0.21	$p > .999$
Label by editing time	4.93 (2, 216.8)	$p = .008$	$p = .048$	4.93	$p = .048$

b Registered secondary tests: 100 hours minus under 1 hour of stated editing time

Measure	Estimate	90% CI	Two-sided test	Equivalence test
Transformativeness	−0.03	−0.14 to 0.09	$p = .691$	$p = .003$
Essence change	0.07	−0.04 to 0.18	$p = .303$	$p = .011$
Creativity	0.04	−0.08 to 0.15	$p = .598$	$p = .005$

c Predicted means

	None		Slight		Dramatic		Stated editing time		
Measure	Human	AI	Human	AI	Human	AI	Under 1 hour	10 hours	100 hours
Transformativeness	1.04	1.07	1.95	1.81	3.56	3.39	2.17	2.09	2.14
Essence change	1.09	1.07	2.05	1.88	3.63	3.53	2.17	2.21	2.24
Creativity	1.21	1.12	1.92	1.70	3.32	3.20	2.07	2.06	2.11

Estimates come from the registered linear mixed model, which gives each participant and each original its own baseline. df: degrees of freedom by Satterthwaite's approximation; the rescaled tests have the same numerator degrees of freedom. The rescaled analysis includes the two six-point transformativeness items, which the primary models leave out; essence change and creativity have no six-point items, so their rescaled tests repeat the primary ones on a 0-to-1 scale. An equivalence test passes when the 90% confidence interval (CI) lies wholly inside the bound, which means small, not zero. Human, AI: the label of the original. Supplementary Table 2 gives the label's effect at each level with its registered verdict.

Supplementary Table 26: Exploratory: Study 5's label penalty at each stated editing time and level. Each entry is the predicted mean rating under the Human label minus the predicted mean under the AI label, in points on the 1-to-5 scale, with its two-sided p-value, not corrected for the 27 comparisons. A positive value means that the AI label lowered the rating. Editing time varied between participants, so each column compares two separate groups of 35 to 38 people. The breakdown was specified on 9 October 2026, after the registered results were known, and it changes no registered verdict. Among the registered tests behind it, only the creativity interaction was significant (Supplementary Table 25a).

	Stated editing time		
Level	Under 1 hour	10 hours	100 hours
Transformativeness			
None	−0.10 ($p = .386$)	0.03 ($p = .805$)	−0.03 ($p = .771$)
Slight	0.06 ($p = .598$)	0.27 ($p = .021$)	0.07 ($p = .556$)
Dramatic	0.14 ($p = .234$)	0.38 ($p < .001$)	0.00 ($p = .975$)
Essence change			
None	0.00 ($p = .992$)	0.10 ($p = .372$)	−0.04 ($p = .723$)
Slight	0.15 ($p = .185$)	0.27 ($p = .013$)	0.10 ($p = .343$)
Dramatic	−0.02 ($p = .876$)	0.33 ($p = .002$)	−0.01 ($p = .945$)
Creativity			
None	0.00 ($p = .966$)	0.21 ($p = .070$)	0.07 ($p = .535$)
Slight	0.06 ($p = .598$)	0.48 ($p < .001$)	0.11 ($p = .333$)
Dramatic	0.11 ($p = .360$)	0.49 ($p < .001$)	−0.25 ($p = .031$)

The estimates come from a refit of Study 5's registered model in the cross-study analysis, which checks that, averaged over editing time with equal weights, the refit reproduces Study 5's registered penalty at each level. None is the original shown again unchanged, where ratings sit near the floor of the scale. Supplementary Figure 2 plots the creativity rows.

Human-label participants, measured as a person's word overlap among their own explanations minus their overlap with other people's explanations of the same second work, with explanation length held constant. The difference, AI minus Human, was 0.009 (−0.009 to 0.026; Holm $p = .320$): in the predicted direction but inconclusive, which is not evidence that the groups reuse words equally. Every explanation was written after its rating, so these checks show what the words track, not why anyone rated as they did.

Exploratory text analyses. The registration names the remaining text analyses as exploratory. Word tone and concreteness, scored from published norms, did not differ by label after a Benjamini–Hochberg correction, which limits the expected share of false positives among results called significant. No topic found without labels, a group of words that tend to occur together, differed in weight by label after the same correction. Before correction, AI-label explanations leaned toward topics about the second work looking identical or exact, and Human-label explanations toward a topic about changes in essence. In predicting the rating, words for the size of a change, such as 'dramatically' and 'complete', and words of praise went with higher ratings, and 'identical', 'exact' and 'copy' with lower ones. Their estimates are in the Study 5 tables that the analysis code writes (main text, Code availability).

10 Text measures shared by all five studies

In all five studies, a required text box for each rating asked participants to explain their reasoning for it, and every box came after the ratings it explained. This section describes how those explanations were measured, which was the same in every study: eight fixed word lists, applied by a script that never saw the label or the rating. It then describes the two checks on the explanations that every registration names, and the status of each use of the word lists. A registration is a dated record of a study's hypotheses and analysis plan (Section 4). Supplementary Note 7 gives what the explanations referred to, and each study's section gives its results on the two checks. Every explanation was written after its rating, so no result from the explanations shows why anyone rated as they did.

The eight word lists

A word list here is a rule: a written list of words and phrases, in their ordinary forms, that a computer searches for in every explanation in the same way, after lowering its case. When an explanation contains a phrase on a list, the list flags that explanation. Supplementary Table 27 gives each list, what it flags and examples of what it matches.

A list records a topic, not an endorsement: an explanation saying that a picture has barely any detail is talk about detail. For four lists (visible detail, time, skill and concept), a second reading records whether the matched phrase was asserted or denied. It looks for a negating word, such as 'not', 'never', 'no', 'little' or 'lacks', in the thirty characters before the phrase, cut back to the start of the phrase's clause. The prior-exposure list uses the same window to separate a claim that a work looks familiar from a denial that it does. The machine-as-maker list counts a phrase as not asserted when it is denied or sits inside a supposition, such as one opened by 'if', 'whether' or 'unless'; a hedged impression such as 'looks AI generated' counts as asserted.

The lists, the negating words and every reading are written out in full in the text module that all five studies share (`shared/python/study_text.py` in the analysis code; main text, Code availability). Each study's run also writes the rules in full to that study's `tables/text_rules.txt`, with a SHA-256 checksum of that file, a fingerprint of its exact bytes that changes if one character changes, in `tables/text_rules.sha256`. A third file, `tables/text_rules_grounding.txt`, states what each list is meant to measure and the research it draws on. The five studies' rules files differ only in their title line and in the line that gives the number of explanations. A test in the shared folder (`test_reproduces_study1.py`) runs the shared module on Study 1's explanations and confirms that it reproduces the outputs of Study 1's own text analysis exactly.

How the lists were written and applied

We wrote the lists by reading Study 1's explanations with the label and the rating hidden. The lists were then frozen before any quantity that links them to the label or to the rating was computed, with four exceptions settled after results by label had been read. In September 2026 we settled how the prior-exposure list reads a negation, and wrote the machine-as-maker list itself. In October 2026 we settled that list's reading of whether a claim is asserted, and the rule that reads 'little' as a denial only when it names an amount ('little detail') and not a size ('the little details'). Study 1's long registration records all four, and the paper's results that use those parts should be weighed with that in mind. The registrations of Studies 2 to 5 state that the lists run unchanged, and they did.

The script that applies the lists receives only identifiers, for the participant, the image and the rating asked about, and the text of each explanation. A check in the code stops the script if it receives any column that names the label, the rating or a background measure such as age.

Supplementary Table 27: The eight word lists. Each list is a fixed search pattern that a script applies in the same way to every explanation, after lowering its case, without seeing the label or the rating. The examples are a selection; each study's rules file gives every pattern in full, with the phrases each list sets aside because they mean something else in these explanations. A list records a topic, whatever the writer said about it: 'barely any detail' is detail talk.

List (name in the code)	What it flags	Examples of what it matches	Further reading of a match
Visible detail (`says_detail`)	Detail and how much is in the picture	detail, intricate, complex, texture, shading, brushwork, layers, 'so many different elements'	Asserted or denied
Time (`says_time`)	Time spent, duration or speed	time, hours, minutes, days, tedious, painstaking, spent, 'took hours', quick, instant	Asserted or denied
Skill (`says_skill`)	Skill, craft and training	skill, craft, technique, talent, trained, mastery, expertise, professional, execution, 'by hand'	Asserted or denied
Concept (`says_concept`)	The idea, meaning or thought behind the work	idea, concept, meaning, message, symbol, theme, story, represents, conveys, intention, 'thought behind'	Asserted or denied
Prompting or machines (`says_prompt`)	Prompting, typing or generating, or a machine named in any role	prompt, typed, generated, input, AI, artificial intelligence, computer, algorithm, software, an image generator named by kind or brand	None
Prior exposure (`prior_exposure`)	Says the work looks familiar, already seen, generic or derivative	'seen before', familiar, 'reminds me of', generic, cliché, typical, unoriginal, derivative, 'nothing new'	Three values: familiarity asserted, familiarity denied, or neither
Scale echo (`anchor_echo`)	Repeats the rating scale's own wording	'moderate creativity', 'a great deal of effort', 'effort involved'	None
Machine as maker (`says_machine_made`)	Names a machine, program or image generator as making the work	'AI-generated', 'machine-made', 'created by a computer'; a person using a tool ('Alex used an AI') is set aside	Asserted, or not asserted (denied or only supposed)

The label and the rating are joined to the flags only after the lists have run, and a second check stops the join if either had arrived earlier.

Our judgment entered these measures at one point: we read explanations to write the lists, as described above. No person and no language model labeled, coded or judged any single explanation for these measures, and no one read a flagged explanation to decide what it meant. Every flag and every count comes from the script, so each can be recomputed from the rules and the text alone. Samples of the explanations that each list flagged and did not flag, with the matched phrase marked, are saved with the analysis outputs, to be shared with the data (main text, Data availability), so that a reader can judge each list; nothing reported depends on what that reader concludes. Before Study 1 was registered, a language model had coded its explanations in exploratory work that saw the label (Section 4); that work was retired, and nothing in the paper uses it. One pretrained model does appear: a sentence-embedding model (all-mpnet-base-v2), which turns each explanation into a list of numbers whose distances reflect differences in meaning.[8,9] It is used in two exploratory analyses only, a check on the topic model within each study and the measure of how far explanations moved in Supplementary Note 7, and it gives no explanation a label.

The two checks run in every study

Study 1's long registration fixed two checks to run unchanged in every study where the rating exists, and each later registration names them. Both are registered secondary: named in a registration, but outside the set of tests that it names as confirmatory. In Studies 1 and 3 they ran on the explanations of both ratings; in Studies 2, 4 and 5 they ran on the creativity explanations only, as those registrations specify.

The first check asks how closely the words alone track the rating, and how well they tell the two label groups apart. Each explanation is reduced to counts of its content words, with function words such as 'the' and 'of' removed. Ridge regression, a regression that shrinks each word's weight toward zero so that more words than explanations can be fitted without memorizing them, predicts the rating, and its counterpart for a yes-or-no outcome predicts the label. Every prediction is made for participants left out of the fit: the participants are split into ten groups, and each group's explanations are predicted by a model fitted to the other nine. For the rating, the check reports the correlation between predicted and observed ratings. It sets that correlation beside the same correlation computed 200 times after each person's ratings were shuffled among that person's own explanations, which keeps each person's average and scrambles only which image got which rating.

For the label, the check reports the AUC, the chance that a randomly chosen AI-label explanation scores above a randomly chosen Human-label one, so that 0.5 is chance; its baseline shuffles the label across participants. Recovery is read as present when the estimate exceeds 95% of the shuffled values and its 95% confidence interval (CI), the range of values the data do not rule out at the 95% level, clears chance. Otherwise recovery is not shown, which is not evidence that the words carry nothing. The check also reports how much the words add to the rating in telling the groups apart, as the AUC of the rating and the words together minus that of the rating alone. The words add to the rating when that interval lies wholly above zero, and the result is inconclusive when it includes zero. This first check is a description: it carries no prediction and is corrected with no other test.

The second check is the text counterpart of Study 1's H4, which predicted that under the AI label ratings would spread out more between people and less across one person's ratings of different images. It predicts that AI-label participants reuse their own words across their explanations more than Human-label participants do. For each participant it takes the average overlap among that person's own explanations, the share of content words that two explanations have in common. It subtracts the average overlap between that person's explanations and other participants' explanations of the same image, or of the same second work, which removes what

any two people writing about the same picture share. The two label groups are then compared with explanation length held constant, with an interval from resampling participants within each group 2,000 times. Within each study the ratings checked are corrected together by Holm's method.[4] The reading is 'supported' when the corrected p-value is below .05 with the AI group higher, 'opposite direction' when it is below .05 with the AI group lower, and 'inconclusive' otherwise. A p-value is the chance of a difference at least this large if there were none. An inconclusive reading is not evidence that the groups reuse their words equally.

Across the five studies, the words recovered the rating in every study and on every rating checked. They told the label groups apart in Study 1 on both ratings, in Study 3 on effort and, barely, in Study 5. They were not shown to do so in Studies 2 and 4 or on Study 3's creativity explanations, where the intervals reached chance. The words added to the rating in telling the groups apart in Studies 1 and 2, and the result was inconclusive in Studies 3, 4 and 5. Two of these readings turn on the third decimal place. Study 5's interval for the label begins just above chance, and those of Studies 2 and 4 just below it, although all three print as 0.50 at two decimals. Study 3's interval for what the words add on effort begins just below zero, although it prints as 0.000. Own-word reuse was greater under the AI label, as predicted, only in Study 1's effort explanations; every other reading was inconclusive. Supplementary Table 1 summarizes these readings, and each study's section gives its estimates.

Other uses of the word lists, and their status

Every other use of the lists is listed here with its status, so that no result from the explanations is read above it.

Study 1's five word-meaning tests asked what 'effort' and 'creativity' referred to in the explanations. The registration names them as one exploratory set, corrected by the Benjamini–Hochberg method, which limits the expected share of false positives among results called significant.[7] Their corrected results are descriptive readings, not verdicts (Supplementary Note 7; Supplementary Table 18).

Study 1's belief check counts Human-label participants who mentioned machines or prompting, using the prompting list, and those who wrote that a work is or looks machine-made, using the machine-as-maker list read for assertion. The two counts, and a refit of the penalty without the second group, are registered secondary; the comparison with randomly chosen groups is exploratory description (Supplementary Note 2). Neither count measures disbelief in the label.

Study 1's familiarity association, between calling an image familiar and its creativity rating within a person, is registered secondary description (Section 5).

Study 4's refit without the 7 of 55 Human-label participants whom the machine-as-maker list flagged for writing that a machine made the work is registered secondary (Supplementary Table 1). *Study 5's count* of such participants, 18 of 112, is exploratory.

Comparisons of how often the lists fired under the two labels, in Study 3's explanations of originals and in the second-work explanations of Studies 2, 4 and 5, are exploratory and uncorrected (Supplementary Note 7).

The general description of the explanations, run in every study, uses no word list. It scores the tone and concreteness of the words against published norms, the average ratings that large samples of people gave to single English words.[13,14] It finds topics, groups of words that tend to occur together, with a model given no label and no rating, and it reports which words a lasso, a regression that sets most word weights to exactly zero, keeps when it predicts the label or the rating. Every registration names this description as exploratory, and it carries no verdict.

11 Exploratory cross-study analyses

Every analysis in this section is exploratory. None is named in any registration: the pooled analysis of penalties and the two analyses of explanations were specified on 9 October 2026, and the image check on 10 October 2026, after every registered result was known. Their tests are not corrected for multiple comparisons, and they change no registered verdict. The pooled analyses treat each study's participants as different people, so a person who took part in Study 5 and in Study 2 or 4 counts as two (Section 9). A p-value is the chance of a result at least this large if there were no effect.

Most of these analyses are reported elsewhere. The check of the stimulus images is in Supplementary Note 4. Supplementary Note 6 and Supplementary Fig. 4 give the model that pools the label's penalty on second works over Studies 2, 4 and 5, with the comparison of rival accounts and the results when one study is left out. Supplementary Note 7 gives the analyses of what the explanations referred to, including how often they mentioned prompting or a machine and how far they moved from None to Dramatic. Supplementary Note 5, Supplementary Fig. 2 and Supplementary Table 26 give Study 5's penalty at each stated editing time. This section reports the rest: the shape of the penalty within each study, a check of the pooled model, Study 4's penalty at each stated time, the explanation shift on the two measures other than creativity, and a structural topic model of the explanations.

The shape of the penalty within each study. Here the penalty is the predicted mean rating of a second work under the Human label minus the predicted mean under the AI label. Supplementary Table 28 gives its slope and curvature across the three levels in each study, from the pooled model. The slope is the change in penalty per step from None to Dramatic, which dilution, the account on which the penalty is largest at None and shrinks as the edit grows, predicts to be negative. The curvature measures how far the Slight penalty stands above a straight line through the None and Dramatic penalties. In no study did the penalty fall significantly from None to Dramatic on any measure. It rose significantly before correction in two places: Study 2's creativity penalty (slope 0.04 points per step, $p = .045$) and Study 5's transformativeness penalty (0.10, $p < .001$). The Study 2 creativity slope is Study 2's registered H3 contrast with its sign reversed and halved, refitted in a model that gives each second work its own baseline; the registered test, which was not significant (Holm $p = .153$; Supplementary Note 6), governs. The curvature was positive, a peak at Slight, and clear of zero before correction on essence change in all three studies, on transformativeness in Study 4 only, and on creativity in Study 5 only, with Study 4 close. Of 24 uncorrected tests, about one in twenty can fall below .05 by chance alone.

A check of the pooled model. The pooled model leaves out the time factors of Studies 4 and 5 and adds a baseline for each second work, so its estimate of each study's penalty need not equal that study's registered estimate. A descriptive check compared them. At every level and on every measure, the pooled model's estimate lay within a few thousandths of a point of the registered estimate. Only the point estimates agree this closely; the pooled model's standard errors, and so its p-values, differ from each study's own.

Study 4's penalty at each stated time. A refit of Study 4's registered model estimated the label's penalty at each of the three hours stated for the original and each level, on each measure, without correction. It showed no consistent pattern across the stated hours. Of its 27 cells, 1 had a 95% confidence interval that excluded zero: essence change at Dramatic for 100-hour originals, where second works from AI-labeled originals were rated higher (penalty -0.28 points; 95% confidence interval -0.51 to -0.06). With this many uncorrected intervals, a cell like this can arise by chance. None of Study 4's registered tests of the stated hours was

Supplementary Table 28: Exploratory: the shape of the label's penalty across levels, study by study. The penalty is the predicted mean rating under the Human label minus the predicted mean under the AI label. The slope is the change in penalty per step from None to Dramatic, half of the Dramatic penalty minus the None penalty; dilution predicts it to be negative. The curvature is the Slight penalty minus the average of the None and Dramatic penalties; it is zero for any straight line, and positive when the penalty peaks at Slight. All values come from the pooled model of Studies 2, 4 and 5, specified on 9 October 2026 after the registered results were known. The p-values are two-sided and not corrected for the 24 tests.

	Slope		Curvature	
Scope	Estimate	Test	Estimate	Test
Transformativeness				
Study 2	0.03	$p = .188$	0.03	$p = .376$
Study 4	−0.05	$p = .204$	0.24	$p < .001$
Study 5	0.10	$p < .001$	0.07	$p = .201$
Pooled	0.03	$p = .143$	0.11	$p < .001$
Essence change				
Study 2	0.02	$p = .306$	0.09	$p = .008$
Study 4	−0.02	$p = .589$	0.25	$p < .001$
Study 5	0.04	$p = .131$	0.11	$p = .018$
Pooled	0.01	$p = .410$	0.15	$p < .001$
Creativity				
Study 2	0.04	$p = .045$	0.01	$p = .794$
Study 4	−0.02	$p = .545$	0.12	$p = .064$
Study 5	0.01	$p = .704$	0.12	$p = .016$
Pooled	0.01	$p = .573$	0.08	$p = .006$

The pooled model gives each participant, each original and each second work its own baseline, and it leaves out Study 4's stated hours for the original and Study 5's stated editing hours, so each study's values average over them. Its fits converged but are singular: one source of variation, the spread between originals once each second work has its own baseline, was estimated at zero. Supplementary Note 6 gives the pooled slopes and curvatures with their 95% confidence intervals.

significant (Supplementary Table 24b). The cell estimates are in the cross-study tables that the analysis code writes (main text, Code availability).

How far explanations moved, on transformativeness and essence change. Under taint, the account on which an AI origin passes to anything built from it, AI-label raters should keep writing about the original, so their explanations should change less from None to Dramatic. Supplementary Note 7 describes the measure and gives the creativity estimates. On the other two measures, too, no difference between the label groups was shown. The pooled difference, AI minus Human, was −0.015 on transformativeness with sentence embeddings (95% confidence interval −0.031 to 0.001; permutation $p = .085$) and −0.007 with word overlap (−0.020 to 0.005; $p = .303$). On essence change it was −0.006 with sentence embeddings (−0.023 to 0.011; $p = .458$) and −0.001 with word overlap (−0.015 to 0.013; $p = .930$). A sentence embedding turns an explanation into a list of numbers whose distances reflect differences in meaning; word overlap is the share of content words two explanations have in common. The permutation p-value is the share of label-shuffled refits, shuffled within each study, that gave a difference at least this large. The essence-change model with word overlap was singular, estimating one source of variation at zero, but it converged and keeps its numbers. On creativity, in both label groups, a person's Dramatic explanation lay closer to their own None explanation than other people's Dramatic explanations of the same image did, by similar amounts (descriptive). No difference shown is not the same as no difference.

A structural topic model of the explanations. A topic model finds groups of words that tend to occur together, called topics, and describes each response as a mix of them. A structural topic model also lets the share of each topic depend on recorded features of the response, here the label, the level, their combination, the rating and the study. One model was fitted to each rating's explanations, pooled over Studies 2, 4 and 5. Words describing the task and each study's own item wording were removed, and only words used in at least five responses were kept. The number of topics was chosen between four and twelve by a rule fixed in code: the number whose model best predicted words set aside from fitting. No person or language model named or judged any topic; each is described by its most probable words.

The explanations were short, and responses that kept no word after this pruning were dropped. At None, the AI-label group lost a larger share of its responses this way than the Human-label group, so a label difference at None compares responses selected differently in the two groups.

On creativity, the topic whose most probable words were 'ai', 'similar', 'feel' and 'takes' took a larger share of AI-label explanations than of Human-label ones at None, by 0.038 (95% confidence interval 0.033 to 0.043), and at Slight, by 0.017 (0.012 to 0.022), but not at Dramatic (−0.002; −0.007 to 0.003). A topic share runs from 0 to 1. The difference at Dramatic minus the difference at None was −0.041 (−0.048 to −0.033). Three cautions apply. This topic was poorly recovered when the model was refitted from other starting points, so it may not survive another start. The intervals treat each person's many responses as independent, so they are likely too narrow. And the None comparison is affected by the uneven loss of responses described above. On transformativeness, at None, AI-label explanations gave more share than Human-label ones to a topic led by 'look', 'transformation' and 'identical' and less to a topic led by 'difference', 'see' and 'looks', and both differences shrank toward Dramatic. The model of the essence-change explanations did not converge within its limit of iterations, so it is not completed: its label effects are description only, with no interval, p-value or verdict.

Like every text analysis in the paper, these describe what people wrote after rating; they do not show why anyone rated as they did.

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